

Beyond Variance: Asymmetric Labor Income Risk and Portfolio Entry*

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Abstract

Earnings opportunities have different portfolio implications for households outside and inside the stock market. I measure earnings growth net of industry–education-specific age profiles in the Current Population Survey and relate its distribution to subsequent portfolios in the Survey of Income and Program Participation. At a given variance, a source standard deviation of upside composition predicts 0.88 percentage point more entry into stocks or mutual funds. Using real log earnings, the same composition measure predicts a 0.36-point higher subsequent risky share among initial nonholders and a 1.62-point lower share among holders. The negative holder association is not simply dilution from a growing financial-asset denominator. A model with participation costs and safe nontradable human capital generates the opposite allocation responses at unchanged earnings mean and variance. Earnings-tail composition connects market access and incumbent allocation in ways that variance alone cannot distinguish.

Keywords: labor income risk; household portfolios; tail composition; stock market participation; measurement error

JEL codes: C18, D14, G11, J31

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1 Introduction

An earnings distribution can become more dispersed because adverse outcomes worsen or favorable outcomes improve. Variance combines these changes, although their implications for household portfolios need not be the same. Earnings opportunities can make stock-market access worthwhile for a household that does not invest. For an existing investor, the same opportunities can change the balance between financial wealth and nontradable human capital, lowering the fraction of financial assets held in stocks. This paper studies how the composition of earnings risk relates to these two portfolio decisions.

The central finding is that initial ownership distinguishes the portfolio implications of upside earnings risk. At a given earnings variance, a more upside-oriented historical distribution predicts greater subsequent entry into stocks or mutual funds. Among existing holders, it predicts a lower subsequent risky share. A model combining a participation cost with safe human capital generates this contrast for an explicit change in the earnings distribution that preserves its mean and variance. Thus, favorable earnings opportunities need not produce a uniform increase in financial risk taking.

I estimate earnings distributions using the Current Population Survey (CPS) and assign them to later household portfolios in the Survey of Income and Program Participation (SIPP). Following [Catherine, Sodini, and Zhang \(2024\)](#), I remove a cubic age profile from real log earnings growth separately within industry–education groups. The main measure includes wage, business, and farm earnings. Centered lower and upper second partial moments divide variance into its adverse and favorable components. Their difference measures upside composition; controlling for their sum isolates the orientation of a given variance. The comparison also holds mean growth, dispersion curvature, and observed household resources fixed.

In 27,419 transitions beginning without holdings, a source standard deviation of upside composition is associated with 0.88 percentage point more stock-and-mutual-fund entry. This is approximately one tenth of the sample’s 8.82% entry rate. The estimate is significant with standard errors clustered by the 30 industry–education groups. Using wages on the same earnings histories gives 0.78 point; replacing real log growth with nominal arc growth gives 0.69 and 0.74 point for total earnings and wages. All four age-adjusted entry coefficients are positive and significant. Both earnings endpoints are positive, so the finding concerns the distribution of earnings among workers with positive earnings, not simply the incidence of earnings loss to zero.

The allocation comparison follows 31,569 consecutive person-year pairs and fixes ownership at baseline. With total real log earnings growth, a common source standard deviation of upside composition is associated with a 0.36-percentage-point higher next-year risky share among nonholders and a 1.62-point lower share among holders. Every risk and household-control coefficient varies by initial ownership, and the holder equation controls the starting share. The directly estimated holder-minus-nonholder difference is -1.98 points ($p = 0.0118$). Future exits remain in the holder sample. The result therefore concerns subsequent allocation among initial investors, rather than a sample

selected on continued participation. The nominal-arc companion retains the positive nonholder association but does not precisely estimate the holder contrast.

The economic framework makes initial market access a state variable. A nonholder must pay a fixed cost before investing; an incumbent has already obtained access. With safe future labor income, the household's financial equity share declines with financial wealth because human capital supplies an implicit bond position (Viceira, 2001; Cocco, Gomes, and Maenhout, 2005). I construct two earnings distributions with identical mean and variance for which an upside-oriented rotation raises entry and lowers incumbent shares. This links the participation and allocation margins without assigning different preferences to holders and nonholders. It also connects the earnings-distribution comparison to lottery evidence: windfalls increase participation among prior nonholders (Briggs et al., 2021), while reducing equity shares among prior participants (Briggs et al., 2023).

A balance-sheet decomposition sharpens the interpretation of the allocation evidence. I divide each share change exactly into a risky-balance contribution and an asset-denominator contribution. In the real-log holder comparison, the risky-balance component is negative and the denominator component offsets part of it. The lower risky share is therefore not simply dilution from a growing denominator: it is accounted for by the risky-balance component of portfolio-share changes.

The historical assignment separates measurement from portfolio outcomes. CPS earnings origins run through 2017, with destinations through 2018; portfolio baselines are 2019–2023 and outcomes are 2020–2024. SIPP workers never contribute to their own assigned CPS distributions. This avoids using a household's concurrent earnings realization to construct the regressor for its portfolio, while leaving the economic interpretation conditional on sorting into earnings environments. The main inference treats estimated CPS moments as given. Joint remeasurement of the two surveys produces wider, method-sensitive entry inference. A richer earnings-rank adjustment also attenuates entry estimates without precisely establishing their change. These comparisons, reported in the Online Appendix, delimit the historical-risk interpretation; they are not the earnings-profile adjustment used in the main tables.

The contribution is to connect the composition of labor income risk with portfolio choice across initial ownership states. Catherine (2022) shows that nonlinear earnings risk and its comovement with stock returns alter life-cycle portfolios, including comparisons at fixed variance and persistence. Catherine, Sodini, and Zhang (2024) relates systematic earnings exposure to Swedish household portfolios and tests the human-capital interpretation. I follow their earnings-profile approach but study a different empirical object: the favorable-versus-adverse composition of earnings variation and subsequent U.S. portfolio transitions. The comparison adds an extensive-margin target and an initial-state allocation contrast to models incorporating richer earnings dynamics (Shen, 2024; Gálvez and Paz-Pardo, 2026). It does not require claiming that modern life-cycle models use variance alone, or that partial moments dominate every other shape statistic.

The paper also connects measured earnings environments with evidence on subjective risk. In the Survey of Consumer Expectations (SCE), perceived earnings dispersion is negatively associated with retirement equity shares, while a centered upside tilt has a weaker association with reported

reallocation. These respondent-level comparisons complement research using elicited earnings beliefs (Guiso, Jappelli, and Terlizzese, 1996; Wang, 2023; Caplin et al., 2026; Arellano et al., 2026). They concern a different information object and portfolio account from the CPS-to-SIPP assignment; they do not establish that assigned tails equal perceived tails. More broadly, the distinction between favorable and adverse uncertainty connects household portfolios to the economic role of two-sided uncertainty (Segal, Shaliastovich, and Yaron, 2015).

Section 2 develops the model and its predictions. Section 3 describes the data and empirical design. Section 4 presents entry, allocation by initial ownership, and the main construction comparisons. Section 5 examines balance-sheet channels and perceived risk. Section 6 concludes. The Online Appendix contains proofs, complete specification families, and additional measurement and inference exercises.

2 Earnings Composition and Portfolio Allocation

Why can earnings opportunities encourage stock-market entry while lowering the financial risky share of households that already invest? Entry depends on whether access is worth its cost; allocation depends on the composition of resources after access is obtained. This section shows how the same change in an earnings distribution can move these margins in opposite directions, even with unchanged mean and variance.

2.1 Portfolio choice and initial market access

A household begins with financial wealth w . A transitory earnings innovation Y is realized before its next portfolio decision, giving financial resources $w_1 = w + Y > 0$. Remaining nontradable labor income is a safe terminal payoff $h > 0$. Utility is logarithmic, the safe net return is zero, and independent risky excess returns equal $\bar{r} > 0$ with probability $p \in (1/2, 1)$ and $-\bar{r}$ otherwise. Investors cannot borrow or short stocks. Saving is fixed: the innovation enters financial resources before allocation.

A baseline holder has already paid a one-time market-access cost. Maximizing expected utility over risky dollars $a \in [0, w_1]$ gives, for a feasible interior optimum,

$$a^*(w_1) = \eta(w_1 + h), \quad \alpha^*(w_1) = \frac{a^*(w_1)}{w_1} = \eta \left(1 + \frac{h}{w_1} \right), \quad \eta = \frac{2p - 1}{\bar{r}}. \quad (1)$$

Safe human capital acts like an implicit bond holding (Viceira, 2001; Cocco, Gomes, and Maenhout, 2005). More financial wealth raises risky dollars but lowers the human-capital-to-financial-wealth ratio and therefore the financial risky share. This distinction also underlies the negative risky-share response to financial windfalls documented by Briggs et al. (2023).

A baseline nonholder must pay $\kappa > 0$ to obtain access (Gomes and Michaelides, 2005). Define the optimized utility gain before the fee as $b = p \log(1 + \eta \bar{r}) + (1 - p) \log(1 - \eta \bar{r}) > 0$. When the

corresponding post-fee optimum is feasible, comparing $\log(w_1 + h - \kappa) + b$ with $\log(w_1 + h)$ yields

$$w_1 \geq w^* = \frac{\kappa}{1 - e^{-b}} - h. \quad (2)$$

An earnings gain can consequently move a nonholder across an entry threshold while lowering an incumbent's optimal risky share. The different initial states record whether the cost has already been paid; they need not reflect different preferences or initial wealth.

2.2 A fixed-moment comparison and two predictions

Consider a rotation from symmetric earnings innovations to a distribution with a larger favorable tail. For $w > d > 0$, let

$$Y_0 = \begin{cases} -d, & \text{with probability } 1/2, \\ d, & \text{with probability } 1/2, \end{cases} \quad Y_1 = \begin{cases} -d/2, & \text{with probability } 4/5, \\ 2d, & \text{with probability } 1/5. \end{cases}$$

Both laws have mean zero and variance d^2 . Upper-minus-lower second partial moments increase from zero to $3d^2/5$.

Proposition 1 (Opposite prospective portfolio responses). *Suppose the optimal positions are interior under both earnings laws and, for nonholders, after the fee whenever access is chosen. If $w^* \in (w + d, w + 2d)$, moving from Y_0 to Y_1 raises nonholder entry from zero to $1/5$ and raises their expected future risky share. Among baseline holders it lowers the expected future risky share, while leaving expected risky dollars unchanged.*

The threshold condition holds for an open interval of fees, not one knife-edge value. Appendix A.5 proves the result and supplies feasible numerical parameters. All investors optimize; neither behavioral preferences nor delayed rebalancing is required. Two implications organize the empirical interpretation.

Prediction 1: Initial portfolio state. Within the proposition's threshold region, the same upside-oriented rotation raises expected future risky shares among nonholders and lowers them among holders. The holder-minus-nonholder contrast is therefore negative. The relevant empirical comparison fixes ownership at baseline and retains future exits, rather than selecting households that continue to hold stocks.

Prediction 2: Remaining human capital. At fixed w , dollar-shock scale d , and return parameters, the negative holder-share contrast increases in absolute magnitude with h and converges to zero as h approaches zero, provided positions remain interior. The expression in Appendix A.5 is proportional to $-h$. This is a prediction about incumbent allocation, not a general monotonicity claim for entry: changing h also changes the entry threshold.

2.3 Economic interpretation and scope

Financial inflows and favorable news about future earnings are different changes. In equation (1), higher h raises the risky share at fixed financial wealth. A hypothesis that better future earnings reduce the need for financial risk to reach an income target would therefore require an additional mechanism. The model here has no income target or satiation. Allowing consumption and saving to adjust introduces further channels (Gomes and Michaelides, 2005; Inkmann, Michaelides, and Zhang, 2026; Catherine, Lu, and Paron, 2026).

The proposition concerns earnings realized before allocation. Distributional composition can also matter when access is chosen before earnings are realized. In a participation-cost benchmark with predetermined wealth $w > \kappa$, future income $\bar{y} + \lambda z$, $z \sim F$, $\lambda \geq 0$, increasing strictly concave utility and positive consumption, the entry gain is

$$\Gamma(F; w, \kappa) = \max_{0 \leq a \leq w - \kappa} \mathbb{E} [u(w - \kappa + \bar{y} + \lambda z + ar) - u(w + \bar{y} + \lambda z)]. \quad (3)$$

An interior position satisfies

$$\mathbb{E}[r u'(w - \kappa + \bar{y} + \lambda z + a^* r)] = 0. \quad (4)$$

Holding saving and independent return risk fixed, equal earnings mean and variance need not equate these nonlinear portfolio comparisons. Appendix A.4 derives this result and explains why partial-moment composition is not a stochastic ordering.

The worked rotation preserves moments of dollar innovations; equal variance of adjusted log or arc growth need not preserve dollar-income or lifetime-resource variance. Nor does every increase in upper partial variance imply the proposition's signs. Systematic earnings risk adds a separate dimension through comovement with stock returns (Catherine, 2022; Catherine, Sodini, and Zhang, 2024); the empirical composition measure does not estimate that comovement. The model supplies a state-dependent interpretation and distinguishing predictions, not an identification of the historical distribution with perceived risk or of the regression coefficient with a causal portfolio response.

3 Data and Empirical Design

3.1 Earnings histories and portfolio observations

I combine annual earnings histories from the Current Population Survey (CPS) with subsequent portfolio observations from the Survey of Income and Program Participation (SIPP). The CPS supplies the earnings distributions; SIPP supplies household portfolios. The two-survey design separates the workers who estimate an earnings environment from those whose portfolios are studied.

The main earnings sample contains 83,372 histories with validated longitudinal identifiers and positive wage and total earnings at both endpoints. Workers are aged 25–65 in the origin year. Earnings origins run from 2013 through 2017 and destinations through 2018. Total labor earnings

sum wages, net business earnings, and net farm earnings. The wage comparison uses the identical histories. I use the public Census ASEC amounts consistently at both endpoints and retain only the redesigned 2014 questionnaire subsample. The population therefore remains workers with positive origin wages; it does not cover every entrepreneurial earnings history. Appendix I.9 documents the source linkage and treatment of signed earnings.

Workers are grouped by harmonized industry and education. Within each of the 30 supported industry–education groups, two age bands, 25–44 and 45–65, yield 58 supported earnings cells. The public-data crosswalk and its effective-support rules are reported in Appendix C. The industry–education grouping determines the earnings profile; the finer cells describe variation around that profile.

SIPP reports directly held stocks and mutual funds, financial assets, and retirement balances. Write R_{it} for stocks and mutual funds, F_{it} for financial assets excluding retirement balances, and $s_{it} = R_{it}/F_{it}$ for the nonretirement risky share. I aggregate the available eligible monthly records to annual observations and link consecutive years. The portfolio baseline years are 2019–2023 and outcomes are 2020–2024. Every CPS earnings history therefore precedes both observations in a portfolio transition.

The entry sample contains 27,419 transitions beginning with no stock or mutual-fund holdings. The prospective allocation sample contains 31,569 transitions with usable positive broad and nonretirement financial denominators and both ratios in $[0, 1]$ at both dates: 24,235 begin as nonholders and 7,334 as holders. Genuine future zero holdings remain in this sample, including 2,165 exits by baseline holders. Requiring usable ratios changes the sample relative to the entry regression; missing financial assets are not coded as a zero share. Table 1 describes the prospective sample by initial ownership.

The observed share excludes the unobserved allocation inside retirement accounts, and mutual funds need not consist entirely of equities. The unit is a worker transition involving household-reported assets. A change in a balance can reflect purchases, sales, valuation, or other balance-sheet changes. Appendix C.3 describes denominator support and the reporting unit.

3.2 Age-adjusted earnings risk

Following the earnings-profile adjustment in Catherine, Sodini, and Zhang (2024), I regress annual real log earnings growth on an origin-age cubic separately within industry–education groups:

$$\log y_{i,t+1}^{\text{real}} - \log y_{it}^{\text{real}} = \theta_{0g} + \theta_{1g}a_{it} + \theta_{2g}a_{it}^2 + \theta_{3g}a_{it}^3 + \epsilon_{i,t+1}. \quad (5)$$

Estimation uses CPS person weights. Real growth removes annual CPI-U inflation. The residual ϵ is earnings growth net of the group-specific age profile. This is the standard profile adjustment used for the main results, not the richer earnings-rank prediction exercised later in the appendix.

Table 1. Prospective portfolio sample by initial ownership

	All		Nonholders		Holders	
	Mean	SD	Mean	SD	Mean	SD
Age (years)	43.74	11.27	43.56	11.31	44.34	11.09
Female (%)	47.47	49.94	49.82	50.00	39.71	48.93
College graduate (%)	49.99	50.00	43.12	49.52	72.65	44.57
Nonretirement financial assets (\$1,000)	78.70	317.76	27.67	107.26	247.14	599.47
Retirement balances (\$1,000)	129.46	411.21	83.29	310.00	281.89	616.26
Stocks and mutual funds (\$1,000)	33.96	222.05	0.00	0.00	146.06	442.40
Baseline risky share (%)	10.76	25.12	0.00	0.00	46.27	32.73
Subsequent risky share (%)	11.18	25.77	4.27	16.86	33.99	35.22
Subsequent participation (%)	23.52	42.41	9.60	29.46	69.49	46.05
Person-year pairs	31,569		24,235		7,334	

Notes: Weighted means and standard deviations in the retained prospective sample, using outcome-year SIPP person weights. Ownership is measured at the baseline in 2019–2023; subsequent outcomes are in 2020–2024. Other characteristics and balances refer to the baseline. Dollar balances are nominal. The risky share is stocks and mutual funds divided by financial assets excluding retirement balances. Both dates have positive usable financial denominators and shares in $[0, 1]$; future zero holdings remain. The table describes selected consecutive person-year pairs, not a cross-sectional population of all households.

For each industry–education–age cell c , let μ_c denote mean residual growth. I construct lower and upper centered second partial moments,

$$V_c^- = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon < \mu_c\}], \quad V_c^+ = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon > \mu_c\}], \quad (6)$$

and define total variance $V_c = V_c^- + V_c^+$ and upside composition $C_c^{\text{up}} = V_c^+ - V_c^-$. A larger C^{up} at fixed V reallocates variance from adverse to favorable growth. It does not separately increase the upper component while keeping the lower component unchanged.

The primary measure uses total real log earnings. I repeat the same adjustment for wages and for nominal arc growth, $2(y_{t+1} - y_t)/(y_{t+1} + y_t)$, on the identical positive-endpoint histories. These constructions are not substituted across outcomes according to significance. The main construction follows CSZ’s age-profile method, while addressing a different question: historical earnings composition rather than its comovement with stock returns. CSZ uses after-tax disposable income and a much longer annual history; the present measures are pretax labor earnings in public surveys.

3.3 Portfolio regressions

Entry is positive stock-and-mutual-fund ownership at t conditional on nonholding at $t - 1$. I estimate

$$H_{it} = \beta_E \tilde{C}_{c(i,t-1)}^{\text{up}} + Z'_{i,t-1} \gamma_E + u_{it}, \quad H_{i,t-1} = 0, \quad (7)$$

where a tilde denotes the fixed source-standard-deviation scaling. Z includes total earnings variance, standard deviation, mean growth and its square, household controls, and outcome-year and baseline

industry, education, and age-band effects. Including variance and standard deviation allows curvature in the relation with dispersion. All specifications use outcome-year SIPP person weights.

Household controls are baseline age and its square, sex, log gross assets, transformed first-listed-job earnings, and an indicator for any eligible month with nonpositive decoded first-listed-job earnings. For mean monthly job-1 earnings $\bar{y}^{(1)}$, the income control is $\log(\max\{\bar{y}^{(1)} + 1, 1\})$. Job 1 is the first-listed job, not necessarily the largest job, and the nonpositive-earnings indicator is not separately observed unemployment. Appendix L gives the public field definitions.

For allocation, I let every risk and control coefficient differ by initial ownership:

$$s_{it} = (1 - H_{i,t-1})(\beta_0 \tilde{C}_{c(i,t-1)}^{\text{up}} + Z'_{i,t-1} \gamma_0) + H_{i,t-1}(\beta_1 \tilde{C}_{c(i,t-1)}^{\text{up}} + Z'_{i,t-1} \gamma_1 + \phi s_{i,t-1}) + u_{it}. \quad (8)$$

The common scaling permits a direct test of $\beta_1 - \beta_0$. Baseline holders remain in their initial group if they subsequently exit, so the regression does not select on continued participation. Conditioning on the initial share makes the holder coefficient also the composition coefficient in a regression of share changes with the same controls.

Main standard errors cluster by the 30 industry–education groups, with CR1 correction and a t_{29} reference distribution. This grouping matches the separately estimated earnings profiles and follows CSZ’s group-clustering logic. Finer 58-cell clustering is a sensitivity. Conventional tests condition on the estimated CPS moments; Appendix J separately reconstructs both surveys for the entry comparison. The analysis is exploratory. Assignment before the portfolio window avoids concurrent own-observation mechanics, but does not identify exogenous risk changes or eliminate occupational sorting.

4 Earnings Composition and Household Portfolios

4.1 Entry into stocks and mutual funds

Greater upside composition predicts subsequent portfolio entry even after removing the industry–education age profile from earnings growth. Table 2 reports equation (7) on a common sample of 27,419 baseline nonholder transitions. Column (1) uses total real log earnings growth; the remaining columns compare wages and nominal arc growth on the same validated CPS histories. All four constructions hold the earnings support, portfolio observations, control span, and inference convention fixed.

In the total real-log specification, a source standard deviation of upside composition is associated with 0.875 percentage point more entry (SE 0.279). The wage estimate is 0.782 point (SE 0.307). The corresponding total-earnings and wage arc estimates are 0.695 and 0.740 point. Standard errors cluster at the 30 industry–education groups; the respective p -values are 0.0039, 0.0165, 0.0089, and 0.0086. Relative to the destination-weighted entry rate of 8.82%, the main total-earnings estimate is economically meaningful.

Table 2. Age-adjusted earnings composition and subsequent portfolio entry

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
Upside composition	0.875*** (0.279)	0.782** (0.307)	0.695*** (0.248)	0.740*** (0.263)
Conditional-SD magnitude (pp)	0.573	0.524	0.534	0.582
Group-specific cubic age adjustment	Yes	Yes	Yes	Yes
Risk and baseline controls	Yes	Yes	Yes	Yes
Industry, education, age-band, year FE	Yes	Yes	Yes	Yes
Nonholder transitions	27,419	27,419	27,419	27,419
Observed entry rate (%)	8.82	8.82	8.82	8.82
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30

Notes: Weighted linear probability models of next-year stock-or-mutual-fund entry among baseline nonholders, using outcome-year SIPP person weights. Baselines are 2019–2023; outcomes are 2020–2024. The CPS source comprises 83,372 validated histories, ages 25–65, with positive wage and total-earnings endpoints; origins are 2013–2017 and destinations end in 2018. Total earnings add wages, net business earnings and net farm earnings. Survey-weighted group-specific origin-age cubics are removed before constructing centered moments in 58 industry–education–age cells. Composition is upper minus lower partial variance. Coefficients are percentage points per fixed pre-adjustment source SD for each construction; the conditional-SD row instead uses residual composition after controls. Variance, SD, mean growth and its square are controlled; Section 3.3 specifies household controls and FE. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Inference conditions on CPS moments. Table J.1 reports 58-cell CR1/ t_{57} sensitivity, every raw/adjusted model, and joint CPS–SIPP remeasurement.

The scaling distinguishes a common source-unit comparison from variation left after controls. One standard deviation of composition remaining after the other regressors corresponds to 0.573 point more entry for total real-log earnings, compared with 0.524 for wages and 0.534 and 0.582 for the arc specifications. These are alternative units for the same slopes, not additional results. Because the main support requires positive endpoints for both earnings measures, the entry association survives without a wage-zero mass. It is not a comparison between employment and nonemployment alone.

At fixed $V = V^- + V^+$, raising $C^{\text{up}} = V^+ - V^-$ necessarily combines more upper partial variance with less lower partial variance. Reversing the orientation gives the corresponding downside statement. The positive coefficient is therefore one within-variance contrast, not two independently identified tail effects. The same argument explains why a variance-only summary cannot distinguish the distributions compared here.

4.2 Subsequent allocation by initial ownership

The entry pattern does not imply that earnings opportunities increase every household’s financial risky share. Table 3 estimates equation (8), using the same age-adjusted total-earnings constructions for both initial ownership states. The sample contains 24,235 initial nonholders and 7,334 initial

holders. Every risk coefficient, household control, and fixed effect is interacted with initial ownership, and baseline share enters the holder block.

With real log earnings growth, a common source standard deviation of upside composition predicts a 0.360-percentage-point higher subsequent share among nonholders and a 1.621-point lower share among holders. Both coefficients are significant with industry–education clustering ($p = 0.0172$ and $p = 0.0311$). The directly estimated holder-minus-nonholder difference is -1.980 points ($p = 0.0118$); its two-transformation family-adjusted q -value is 0.0237. This directly tested contrast, rather than a comparison of separate significance levels, supports the relevance of initial portfolio state.

The dependent variable includes genuine future zero holdings. The negative holder association can therefore combine exit with reallocation among continuing holders, as well as valuation changes in reported balances. It is not a direct estimate of net stock sales. The control for the initial share ensures that the same composition slope applies to share changes conditional on the initial portfolio.

The nominal-arc specification gives a positive nonholder slope of 0.264 point ($p = 0.0490$), but its holder slope and direct status contrast are imprecise. Thus, the entry association extends across all four earnings constructions, whereas the prospective holder contrast is established in the real-log construction. This pattern is consistent with Prediction 1’s distinction across initial states, without requiring the model’s sign to hold for every empirical transformation.

4.3 Construction comparisons and uncertainty

The main comparison follows the group-specific age-profile adjustment of Catherine, Sodini, and Zhang (2024). Appendix J reports the complete raw-versus-age-adjusted family, with fixed source scales and matched observations. All four adjusted entry slopes remain significant under both the 30-group and 58-cell conditional-inference conventions and under the complete eight-test conditional family adjustment. This result is not obtained by assigning different earnings measures to different outcomes.

Reconstructing the estimated earnings distributions adds uncertainty. In 499 matched joint remeasurement draws for the entry family, the adjusted coefficients have eight-test q -values of 0.073, and all adjusted 95% percentile intervals include zero. The original-point tests and the resampled intervals answer distinct inferential calculations, so conditional significance should not be read as precision to the same level after first-stage remeasurement. The prospective allocation tests condition on the CPS moments; a joint remeasurement of that comparison is not reported.

Standing allocation provides another useful comparison. On the broader contemporaneous portfolio sample, age-adjusted real-log composition is negatively associated with nonretirement risky shares among current holders for both total earnings and wages. The complete standing-share family, including broad-denominator and Tobit specifications, appears in Appendix J. The arc holder and pooled Tobit marginal effects are imprecise. Tobit links zero holdings and positive allocation through one latent equation; the main analysis instead distinguishes entry from subsequent allocation by initial ownership. The pooled estimate is therefore a restricted benchmark, not a prerequisite for the

Table 3. Earnings-tail composition and subsequent portfolio allocation

	(1) Real log	(2) Nominal arc
Age-adjusted total earnings		
<i>Outcome: next-year nonretirement risky share</i>		
Upside composition: baseline nonholders	0.360** (0.142)	0.264** (0.128)
58-cell sensitivity p	0.0058	0.0178
Upside composition: baseline holders	−1.621** (0.715)	−0.409 (0.642)
58-cell sensitivity p	0.0150	0.4902
Difference: holder minus nonholder	−1.980** (0.737)	−0.673 (0.683)
58-cell sensitivity p	0.0044	0.2887
Difference: two-test BH q , 30 groups	0.0237	0.3325
Difference: two-test BH q , 58 cells	0.0089	0.2887
Baseline-nonholder observations	24,235	24,235
Baseline-holder observations	7,334	7,334
Total consecutive-year observations	31,569	31,569
Assignment cells / inference groups	58 / 30	58 / 30
All risk and other controls $\times$ baseline status	Yes	Yes
Baseline share in holder block	Yes	Yes

Notes: Weighted OLS of future stock-and-mutual-fund balances divided by financial assets net of retirement balances. Baselines are 2019–2023; outcomes are 2020–2024. Both dates require positive financial denominators and valid ratios in $[0, 1]$; genuine future zeros, including exits, remain. Coefficients are percentage points per fixed pre-adjustment cross-cell SD of upside composition in total earnings, common across ownership states within each growth definition. Earnings construction and controls follow Sections 3 and 3.3. All coefficients vary by baseline ownership; lagged share additionally enters the holder block. Weights are destination-year person weights. Parentheses give CR1 SEs clustered by 30 industry–education groups. Stars use unadjusted two-sided t_{29} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sensitivity rows use 58-cell CR1 and t_{57} . BH rows adjust the two growth-specific holder-minus-nonholder tests within each clustering convention. Inference conditions on the estimated CPS moments; no first-stage bootstrap is included.

separate comparisons. The main allocation evidence remains the prospective initial-state comparison, which retains exits and controls the starting portfolio.

A richer, rank-based earnings-prediction exercise asks a separate question from the CSZ age-profile adjustment. It attenuates the entry magnitudes and yields imprecise adjusted slopes, but the paired changes from the raw estimates are also imprecise. Appendix N retains that comparison and the complete own-history and conventional-skewness exercises. These results do not establish that predictable promotions explain the main association, nor do the main estimates identify unanticipated earnings innovations. They bound the interpretation of the historical distribution while leaving the main empirical specification conventional and comparable.

5 Balance-Sheet Channels and Perceived Risk

5.1 Risky balances and the financial-asset denominator

The model makes a distinction between risky dollars and their share of financial assets. A lower share can reflect slower risky-balance growth, a larger denominator, or both. To distinguish these accounting routes, I decompose each observed share change on the same prospective sample. For nonretirement financial assets F and risky balances R , define $\bar{F} = (F_t + F_{t-1})/2$ and $\bar{s} = (s_t + s_{t-1})/2$. Then

$$\Delta s = \frac{\Delta R}{\bar{F}} - \bar{s} \frac{\Delta F}{\bar{F}}. \quad (9)$$

This identity assigns the numerator change to a risky-balance component and the denominator change to an asset component, using midpoint weights. It remains defined for entry and exit when both financial denominators are positive. With the same sample, weights, and full regression design, the two component coefficients sum exactly to the share-change coefficient.

Table 4 reports this decomposition for both earnings constructions and initial ownership states. In the real-log holder comparison, the risky-balance contribution is -3.68 percentage points per source standard deviation of composition ($p = 0.028$). The denominator contribution is $+2.06$ points ($p = 0.060$), giving the -1.62 -point total. The denominator offsets part of the negative association rather than causing it. Thus the holder result is not simply a smaller risky fraction mechanically induced by faster growth in total financial assets.

This is an accounting result, not a decomposition into causal mechanisms. The components inherit changing midpoint weights, and reported risky balances include returns as well as transactions. The holder numerator component has a twelve-test accounting-family q -value of 0.090 . The complete family and alternative clustering are in Appendix K.

Both component regressions retain the baseline controls used for prospective allocation. The decomposition thus locates the lower holder share in the risky-balance component rather than an expanding financial-asset denominator. Together with greater entry, this result connects earnings opportunities to different portfolio adjustments across initial ownership states. The model organizes the sign contrast rather than fitting these accounting contributions: its equal-moment example leaves expected risky dollars unchanged. Appendix K retains the additional normalized-balance diagnostics.

5.2 Elicited earnings beliefs and retirement portfolios

The SCE complements the historical assignment by measuring respondents' earnings beliefs directly. Table 5 relates perceived dispersion and centered tail orientation to standing retirement allocation and reported reallocation. A standard deviation of perceived earnings dispersion is associated with a 4.75 -percentage-point lower defined-contribution equity share when normalized orientation is controlled. Using the sum of the upper and lower semideviations as the scale measure gives a 3.80 -point lower share. The two measures differ in units and support, but both connect perceived earnings dispersion with lower financial risk taking.

Table 4. Earnings-tail composition and the sources of portfolio-share changes

	Real log		Nominal arc	
	(1) Nonholder	(2) Holder	(3) Nonholder	(4) Holder
Baseline ownership				
Risky-share change, Δs	0.360** (0.142)	−1.621** (0.715)	0.264** (0.128)	−0.409 (0.642)
Risky-balance term, $\Delta R/\bar{F}$	0.589** (0.226)	−3.679** (1.595)	0.362 (0.227)	−1.244 (1.339)
Asset term, $-\bar{s}\Delta F/\bar{F}$	−0.229** (0.101)	2.058* (1.050)	−0.099 (0.111)	0.835 (0.847)
Observations in baseline-status block	24,235	7,334	24,235	7,334
Industry–education clusters	30	30	30	30
Fully interacted baseline controls	Yes	Yes	Yes	Yes

Notes: Composition slopes are percentage points per common fixed source SD. The 31,569 consecutive SIPP pairs and fully baseline-status-interacted controls are those in Table 3; future zeros remain. With risky balances R , nonretirement financial assets F , $s = R/F$, and two-date means denoted by bars, $\Delta s = \Delta R/\bar{F} - \bar{s}\Delta F/\bar{F}$. Parentheses report 30-group CR1 standard errors. Stars use unadjusted two-sided t_{29} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. These are conditional associations and an accounting decomposition, not trading flows or causal channels.

A one-standard-deviation greater centered upside-minus-downside tilt is associated with a 0.028-unit higher riskier-minus-safer reallocation score at fixed total semideviation scale. This mainly reflects fewer reported moves toward safer assets. The marginal p -value is 0.036; the original shape-family q -value is 0.054 and the expanded-family q -value is 0.169. The standing-share orientation coefficients are imprecise. Thus the evidence on perceived dispersion and standing allocation is stronger than the tail-orientation evidence on reallocation.

Beliefs precede the Household Finance interview but generally fall within the twelve-month retrospective action window. These estimates do not measure a prospective response to a dated belief revision. Appendix F retains the timing and shape comparisons; Appendix F.5 reports repeated-belief tests, none of whose twelve focal revision and amplitude tests survives family adjustment. Annual SIPP tail-change estimates also have insufficient incremental reproducibility to support the proposed large-risk-change mechanism (Appendix D.2). Historical distributions, perceived levels, and revisions therefore remain distinct empirical objects.

5.3 Implications for household-finance models

The main findings ask an earnings-risk model to match portfolio transitions as well as average risky shares. At a given variance, composition predicts entry; with real-log growth, its allocation association differs by initial market access. Participation costs and nontradable human capital provide a coherent economic route to this contrast, complementing the windfall evidence of Briggs et al. (2021, 2023). A lower financial equity share need not imply greater risk aversion or a dislike of favorable earnings outcomes.

Table 5. Perceived earnings risk, portfolio shares, and reported adjustment

	Semideviation tilt and scale		Normalized orientation and SD	
	(1) Reallocation	(2) DC share	(3) Reallocation	(4) DC share
Upside-minus-downside tilt	0.028** (0.013)	−0.576 (0.664)		
Sum of semideviations	−0.017 (0.018)	−3.805*** (1.011)		
Normalized upside orientation			0.019 (0.014)	0.059 (0.937)
Total standard deviation			−0.016 (0.019)	−4.754*** (0.970)
Shape coefficient: marginal p	0.036	0.390	0.173	0.950
Shape: original-family BH q	0.054	0.390	0.347	0.950
Mean and demographics	Yes	Yes	Yes	Yes
State and module-year FE	Yes	Yes	Yes	Yes
Respondents	2,136	3,332	1,893	2,952
State clusters	50	50	50	50

Reallocation is the reported direction of a DC-account adjustment over the preceding twelve months: +1 for riskier assets, −1 for safer assets, and zero for no change. DC share is the standing equity allocation in percentage points. Beliefs are averaged from reports one to eleven months before the Household Finance module, using monthly sampling weights; these reports generally overlap the retrospective action window. All regressions use official module weights and control for expected earnings growth, age, education, and household income.

Each risk regressor uses its unweighted strict-pre respondent standard deviation. Columns (1)–(2) center lower and upper semideviations on the elicited mean and retain degenerate densities with zero tilt and scale. Columns (3)–(4) normalize the upper-minus-lower semivariance difference by total variance, requiring positive variance. Their samples and risk scales therefore differ. Standard errors are clustered by state. Stars retain unadjusted two-sided t_{49} tests of a zero coefficient: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The displayed q -values retain their original shape-testing families, not a new four-column adjustment; Appendix F reports those families. Expanding the column (1) tilt test to fourteen support/outcome variants gives $q = 0.169$.

The model also provides a way to discipline further mechanism tests. Prediction 2 concerns remaining human capital at fixed financial resources and a fixed dollar-shock scale. Age or financial wealth alone is not that comparison. The exposure-based tests in [Catherine, Sodini, and Zhang \(2024\)](#) motivate the prediction, but the public-data analysis does not estimate a household-specific human-capital value or reproduce their twin design. The existing liquid-asset interactions in Appendix N are inconclusive and are not evidence that the human-capital channel has been identified.

The relevant modeling target is therefore the joint pattern across initial ownership states, with the earnings support and information timing stated explicitly. Richer earnings distributions already matter for quantitative portfolios ([Catherine, 2022](#); [Shen, 2024](#); [Gálvez and Paz-Pardo, 2026](#)). These results supply an empirical comparison for such models rather than a claim that variance is their only state variable. A quantitative application must additionally specify persistence, consumption and saving responses, and the relation between historical earnings variation and beliefs.

6 Conclusion

The composition of labor income risk is informative about household portfolio transitions beyond its variance. Using earnings growth net of industry–education-specific age profiles, greater upside composition predicts stock-and-mutual-fund entry across total earnings, wages, real log growth, and nominal arc growth. In the main total real-log specification, a source standard deviation corresponds to 0.88 percentage point more entry.

Initial ownership changes the allocation comparison. The same real-log composition measure predicts a 0.36-percentage-point higher subsequent risky share among nonholders and a 1.62-point lower share among holders, with a significant direct difference. A model with participation costs and safe nontradable human capital generates this contrast for an earnings rotation that preserves mean and variance. Opportunities can cross an access threshold while lowering existing investors’ financial equity fraction.

The balance-sheet decomposition shows that the negative holder-share association is accounted for by the risky-balance component of share changes; the financial-asset denominator partly offsets it. The holder contrast is imprecise under arc growth, and the historical assignment does not identify exogenous earnings-risk changes. Complete construction and remeasurement comparisons remain in the Online Appendix.

For household finance, the contribution is a state-dependent view of earnings opportunities: the same favorable composition can accompany greater market access and lower incumbent equity shares. Models confronting earnings risk should match these distinct portfolio margins, not only the relation between variance and average risky shares.

Data Availability. The CPS ASEC extract uses licensed IPUMS CPS microdata ([Flood et al., 2025](#)). SIPP ([U.S. Census Bureau, 2025](#)), SCE ([Federal Reserve Bank of New York, 2026](#)), and SCF ([Board of Governors of the Federal Reserve System, 2019](#)) are public. The replication package supplies download links, raw-data build scripts, frozen analysis inputs, and source-to-paper validation; redistribution of the IPUMS extract is subject to its license.

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Online Appendix
for
Beyond Variance:
Asymmetric Labor Income Risk and Portfolio Entry

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A Earnings-Tail Measures and Analytical Results

Let q_p denote the survey-weighted p th quantile of earnings growth g , and let μ and σ denote its weighted mean and standard deviation. The signed quantile-orientation measures in Table A.1 are defined so that positive values indicate a longer or heavier lower tail relative to the upper tail. The unsigned distances L and U measure the lengths of the two tails separately. The variance-composition comparisons instead report the upside-positive contrast $C^{\text{up}} = V^+ - V^-$; conventional third-moment skewness retains its usual sign.

Table A.1. Distributional estimators

Estimator	Definition	Interpretation
Upper tail	$U = q_{.90} - q_{.50}$	Magnitude above the median
Lower tail	$L = q_{.50} - q_{.10}$	Magnitude below the median
90–10 spread	$S = L + U = q_{.90} - q_{.10}$	Interquantile scale
Tail difference	$D = L - U$	Unscaled downside-minus-upside length
Kelley shape	$(L - U)/(L + U)$	Tail difference normalized by the 90–10 spread
Bowley shape	$(2q_{.50} - q_{.25} - q_{.75})/(q_{.75} - q_{.25})$	Central-quantile analogue of Kelley shape
Moment skewness	$-\mathbb{E}[(g - \mu)/\sigma]^3$	Conventional skewness with sign reversed
Tail probability difference	$\Pr(g \leq \mu - \sigma) - \Pr(g \geq \mu + \sigma)$	Downside-minus-upside mass beyond one SD
Conditional tail balance	Tail probability difference divided by its probability sum	Direction conditional on being in either tail
L-skewness	$-\lambda_3/\lambda_2$ from probability-weighted L-moments (Hosking, 1990)	Robust full-distribution orientation
Incremental estimator	Residual from equation (M.5), applied to the named estimator	Cell ranking incremental to zero share, location, and scale

Weighted quantiles sort observations by g , cumulate survey weights, and select the first value whose cumulative weight share meets the requested probability. L-moments are approximated on a probability grid from 0.005 through 0.995. Weighted variance divides by the sum of weights because the target is the weighted empirical distribution in the cell.

For Kelley shape, $K = 1$ is the limiting case in which the 90–10 span lies entirely below the median, while $K = -1$ is the limiting upside case. A point mass at $g = -2$ can move the lower quantile to the support bound. This changes the unconditional distribution’s shape; it is not treated as an outlier correction.

A.1 Destination zeros and the reliability of tail measures

Lemma 1 (A point mass below the positive-earnings support). *Let F have support strictly above a , finite mean μ_F , and variance σ_F^2 . For $p \in [0, 1]$, let $G_p = p\delta_a + (1 - p)F$. Using generalized-inverse quantiles, for $\tau \in (0, 1)$,*

$$q_\tau(G_p) = \begin{cases} a, & \tau \leq p, \\ F^{-1}((\tau - p)/(1 - p)), & \tau > p, \end{cases}$$

while

$$\mu(p) = pa + (1 - p)\mu_F, \quad \text{Var}(G_p) = (1 - p)\sigma_F^2 + p(1 - p)(\mu_F - a)^2.$$

Proof of Lemma 1. Because a lies below the support of F , the mixture distribution is zero below a , jumps to p at a , and equals $p + (1 - p)F(x)$ above a . The generalized inverse is therefore a for $\tau \leq p$. For $\tau > p$, solving $p + (1 - p)F(x) \geq \tau$ gives the stated transformed quantile of F .

The mean follows by iterated expectations. Conditioning on the mixture component and applying the law of total variance gives

$$\text{Var}(G_p) = (1 - p)\sigma_F^2 + p(1 - p)(\mu_F - a)^2.$$

Changing p therefore changes the quantile map and lower moments while leaving the conditional distribution F untouched. The formula imposes no global monotonicity on Kelley shape without additional restrictions on F . $\square$

Lemma 2 (Reliability of a composite statistic). *Suppose the two independent half-sample estimates of a group statistic satisfy*

$$\hat{K}_c^{(h)} = a'Z_c + s_c + e_c^{(h)}, \quad h \in \{1, 2\},$$

where Z_c is a stable vector of lower distributional features, s_c is stable incremental tail variation, and the half-sample errors are mutually independent and independent of (Z_c, s_c) . If their error variances are equal to σ_e^2 , then

$$\text{Corr}(\hat{K}_c^{(1)}, \hat{K}_c^{(2)}) = \frac{\text{Var}(a'Z_c + s_c)}{\text{Var}(a'Z_c + s_c) + \sigma_e^2}.$$

Reliability can therefore converge to one as sampling error vanishes even when $\text{Var}(s_c) = 0$.

Proof of Lemma 2. Independence implies that the covariance between the two half-sample estimates is $\text{Var}(a'Z_c + s_c)$. Each half-sample variance is $\text{Var}(a'Z_c + s_c) + \sigma_e^2$. Dividing covariance by the common variance gives the displayed correlation. Setting $\text{Var}(s_c) = 0$ and taking σ_e^2 to zero establishes the final statement. $\square$

A.2 Properties of arc growth

For nonnegative earnings y_0 and y_1 , with $y_0 > 0$, arc growth is defined at $y_1 = 0$ and equals -2 . Writing $y_1 = y_0(1 + x)$,

$$g^{\text{arc}} = \frac{2x}{2+x} = x + O(x^2), \quad \log(y_1/y_0) = x + O(x^2).$$

Arc growth is invariant to the reporting unit, unlike $\log(y+1)$ growth, and retains transitions from positive to zero earnings, unlike positive-to-positive log growth. Its bounded support and mass at the lower bound are therefore part of the economic distribution being measured.

A.3 A qualitative orientation benchmark

This benchmark shows how distributional asymmetry can matter for portfolio choice while clarifying why it does not sign the empirical quantile-tail coefficients. Let x denote terminal financial wealth before labor income and define

$$y = 1 + \sigma_y \eta, \quad \mathbb{E}\eta = 0, \quad \mathbb{E}\eta^2 = 1, \quad m_3 = \mathbb{E}\eta^3,$$

with $x + y > 0$ almost surely. For CRRA utility, let

$$V(x; m_3) = \mathbb{E}u(x + y), \quad A_V(x; m_3) = -\frac{V''(x; m_3)}{V'(x; m_3)}.$$

Within a smooth local family of shock laws that holds the first two moments fixed while varying m_3 , suppose consumption remains bounded away from zero locally and derivatives of the higher-moment remainder are regular. A third-order expansion around $\sigma_y = 0$ gives

$$\frac{\partial A_V(x; m_3)}{\partial m_3} = -\frac{1}{2}\gamma(\gamma+1)(\gamma+2)\sigma_y^3(x+1)^{-4} + O(\sigma_y^4).$$

The leading term is negative; under the stated regularity conditions, the derivative is therefore negative for sufficiently small σ_y . Thus, locally and within this approximation, a higher third central moment lowers derived absolute risk aversion.

For an interior unconstrained risky position in the local mean–variance approximation, write terminal financial wealth as $x = wR_f + w\alpha(R - R_f)$, with the risky return independent of labor income. A second-order expansion around wR_f gives the approximate terminal-wealth certainty-equivalent gain

$$G(w; m_3) \simeq \frac{(\mathbb{E}R - R_f)^2}{2A_V(wR_f; m_3) \text{Var}(R)}.$$

The preceding comparative static raises this approximate gain locally. If a fixed cost creates a regular participation boundary, households sufficiently close to that boundary can change participation. This

is not a global threshold theorem: portfolio constraints, a discrete wealth grid, and the cost itself can create flat regions and jumps.

The calculation does not sign the effect of the empirical lower or upper quantile distance. Holding mean and variance fixed, widening $q_{0.50} - q_{0.10}$ need not lower m_3 , and widening $q_{0.90} - q_{0.50}$ need not raise it; either transformation can move the opposite tail and higher moments. Nor does it sign a general change in upside semivariance composition, which weights signed squared deviations rather than cubes. The calculation is a qualitative third-moment mechanism, not a prediction for the regression coefficients or their magnitudes.

An exact example separates the two objects. Consider the ordered support

$$x = (-1/2, 1/2, -1, 1, -3/2, 3/2).$$

The first law assigns probability $100/600$ to each point. The second assigns the probability vector $(124, 76, 79, 121, 106, 94)/600$. Both have mean zero and variance $7/6$; both assign total probability $1/3$ to each absolute magnitude, so every even moment agrees. Yet moving from the first law to the second increases $\mathbb{E}[x|x|]$ from zero to $1/200$ and reduces $\mathbb{E}[x^3]$ from zero to $-3/400$. The standardized third moment also falls because variance is unchanged. Thus even fixing the complete absolute-shock distribution does not make semivariance composition and third-moment skewness interchangeable.

A separate mirror comparison isolates a stronger distributional contrast. Let F be a centered, unit-variance law formed from CPS arc-growth residuals and let F^- be the law of the negative of every realization. The two laws have the same distribution of absolute shocks and therefore identical even moments, while their odd moments have opposite signs. In the reported numerical one-period CRR grid, replacing F with F^- changes the optimal risky position in some states. This establishes that matching every even moment need not make portfolio choice invariant. The exercise is not fitted to the SIPP entry rate and is not used to validate the empirical signs, magnitudes, or operative mechanism.

A.4 Entry value at fixed earnings moments

This section derives the comparison used in Section 2. It is an organizing benchmark, not an estimated model or a new participation-cost mechanism. Let $w > \kappa \geq 0$, let the distribution of risky excess returns r be fixed and independent of the earnings state z , and define

$$c_0(z) = w + \bar{y} + \lambda z, \quad c_1(z, r; a) = c_0(z) - \kappa + ar.$$

Assume $\lambda \geq 0$, a fixed compact support, $u' > 0$, $u'' < 0$, and consumption bounded away from zero for every feasible $a \in [0, w - \kappa]$ and every distribution considered locally. Oriented earnings comparisons take $\lambda > 0$. The derivatives below require the corresponding smoothness and domination conditions. The income mapping is affine by assumption; it does not identify observed arc growth with an innovation in consumption resources.

The value of access. For a given position, define the earnings-state utility difference

$$k_a(z) = \mathbb{E}_r[u(c_1(z, r; a)) - u(c_0(z))] = \mathbb{E}_r \left[\int_0^{ar-\kappa} u'(c_0(z) + t) dt \right]. \quad (\text{A.1})$$

The inner integral is signed when $ar - \kappa < 0$. Thus favorable and adverse financial payoffs receive the marginal-utility weights of the earnings states in which they occur. The participation gain is $\Gamma(F; w, \kappa) = \max_a \int k_a(z) dF(z)$.

Consider a valid local probability path $F_\theta = F_0 + \theta\nu$, where F_0 is centered and the signed measure ν satisfies

$$\int d\nu(z) = \int z d\nu(z) = \int z^2 d\nu(z) = 0. \quad (\text{A.2})$$

Probability mass, mean, and variance are then fixed along the path. At a unique maximizing position a^* , the envelope theorem and equation (A.2) imply

$$\begin{aligned} \partial_\theta \Gamma(F_\theta; w, \kappa)|_{\theta=0} &= \int k_{a^*}(z) d\nu(z) \\ &= \int [k_{a^*}(z) - k_{a^*}(0) - k'_{a^*}(0)z - \frac{1}{2}k''_{a^*}(0)z^2] d\nu(z). \end{aligned} \quad (\text{A.3})$$

The cancellation is exact, not a truncation of expected utility. If k_a is three times continuously differentiable on the convex hull of the support and zero, the bracket equals

$$\frac{z^3}{2} \int_0^1 (1-t)^2 k'''_{a^*}(tz) dt, \quad k'''_a(z) = \lambda^3 \{ \mathbb{E}_r[u'''(c_1(z, r; a))] - u'''(c_0(z)) \}.$$

Matching the first two moments therefore removes only the quadratic part of the state-value comparison. It need not remove the remainder. When $\lambda = 0$, the earnings law drops out entirely, an exposure restriction of this benchmark.

The derivative of upside composition along the same centered path is $\int z|z| d\nu(z)$. This is a different functional from equation (A.3). Its sign alone does not sign the gain from entry. The local third-moment example and the exact partial-moment counterexample in Appendix A.3 illustrate why a fully specified distributional comparison is necessary.

The positive position and the participation boundary. If the optimal position is interior, differentiating its first-order condition gives

$$\partial_\theta a^*|_0 = - \frac{\int \mathbb{E}_r[r u'(c_1(z, r; a^*))] d\nu(z)}{\mathbb{E}_{F_0, r}[r^2 u''(c_1(z, r; a^*))]}. \quad (\text{A.4})$$

The denominator is strictly negative when risky returns are nondegenerate. The numerator uses a different earnings-state function from the entry-value derivative. At fixed w and κ , the share $a^*/(w - \kappa)$ scales this dollar derivative by $1/(w - \kappa)$. Binding position constraints require one-sided

comparisons instead. An observed balance-sheet share additionally reflects its denominator, valuation changes, and selection into holding.

Suppose a regular monetary-cost boundary $\kappa^*(\theta)$ satisfies $\Gamma(F_\theta; w, \kappa^*(\theta)) = 0$, with an interior maximizing position and positive expected marginal utility. The envelope theorem yields $\partial_\kappa \Gamma = -\mathbb{E}[u'(c_1)]$, so

$$\left. \frac{d\kappa^*}{d\theta} \right|_0 = \frac{\partial_\theta \Gamma|_0}{\mathbb{E}[u'(c_1)]}. \quad (\text{A.5})$$

For nonholders with otherwise identical states and a smooth cost distribution held fixed along the path, the entry-probability derivative multiplies this expression by the cost density at κ^* . A low density near the boundary can produce a small entry response even if the gain changes. Conversely, observing entry does not identify the cost density or the underlying change in the value of financial risk. The empirical assignment may also sort households with different costs and preferences into different earnings groups.

Allowing accumulated financial resources to change adds $\partial_w \Gamma dw/d\theta$ to the fixed-resource derivative, even before other state variables adjust. This distinction is also important in the dynamic portfolio-saving comparisons of [Catherine, Lu, and Paron \(2026\)](#); it prevents a conditional portfolio estimate from becoming a mechanical wealth-accumulation counterfactual.

The joint distribution with returns. Drop independence while holding both marginal distributions fixed. Conditional on having paid the entry cost, the marginal value of a risky dollar at zero is

$$\begin{aligned} M &= \mathbb{E}[r u'(w - \kappa + \bar{y} + \lambda z)] \\ &= \mathbb{E}[r] \mathbb{E}[u'(w - \kappa + \bar{y} + \lambda z)] + \text{Cov}(r, u'(w - \kappa + \bar{y} + \lambda z)). \end{aligned} \quad (\text{A.6})$$

Changing the coupling of earnings and returns changes the covariance term without changing earnings variance, either partial moment, or unconditional skewness. This is distinct from changing the earnings marginal in equation (A.3). Moreover, $M > 0$ is an incentive for a positive position after paying the cost, not a sufficient condition for paying it. The cyclical-tail exposure in [Catherine, Sodini, and Zhang \(2024\)](#) concerns the joint distribution, whereas the historical composition regressors in this paper describe the earnings marginal.

The replication code checks the fixed-moment identities exactly and verifies the envelope, position, and regular cost-boundary derivatives against reoptimized finite-distribution comparisons. A separate same-marginal exercise checks equation (A.6). These are reproducible mathematical checks, not fitted examples or additional empirical findings.

A.5 Entry and allocation under a fixed-moment tail rotation

This subsection derives the portfolio rules and proves Proposition 1 in Section 2.1. The household observes a transitory earnings innovation before choosing its next portfolio. Financial resources are $w_1 = w + Y$, remaining nontradable income is the safe payoff $h > 0$, and utility is logarithmic.

Independent risky excess returns equal $\bar{r}$ with probability $p \in (1/2, 1)$ and $-\bar{r}$ otherwise. Investors cannot borrow or short stocks. A baseline holder has already paid the one-time access cost; a baseline nonholder has not. All quantities are in arbitrary monetary units.

Optimal allocation and access. For a holder, expected utility from risky dollars a is

$$p \log(w_1 + h + \bar{r}a) + (1 - p) \log(w_1 + h - \bar{r}a).$$

Strict concavity and the first-order condition give

$$a^*(w_1) = \eta(w_1 + h), \quad \eta = \frac{2p - 1}{\bar{r}}.$$

Assume the positions considered satisfy $0 < a^*(w_1) < w_1$. The optimal risky share is then $\alpha^*(w_1) = \eta(1 + h/w_1)$: risky dollars increase with resources, while their financial-wealth share decreases. The value of access before a fee is $\log(w_1 + h) + b$, where

$$b = p \log(1 + \eta\bar{r}) + (1 - p) \log(1 - \eta\bar{r}) > 0.$$

Paying κ replaces w_1 by $w_1 - \kappa$. When the corresponding optimum is feasible, access is worthwhile if

$$\log(w_1 + h - \kappa) + b \geq \log(w_1 + h),$$

which yields equation (2). This is a sunk entry cost, not a recurring cost incurred by incumbent investors.

Proof of Proposition 1. The moment identities follow directly from the two probability laws. The symmetric law has upper and lower partial variances $d^2/2$ each; the upside-oriented law has $4d^2/5$ above zero and $d^2/5$ below. A threshold in $(w + d, w + 2d)$ is reached only in the $Y_1 = 2d$ state, where the chosen share is strictly positive.

For incumbent investors,

$$\mathbb{E}\left[\frac{1}{w + Y_0}\right] = \frac{w}{w^2 - d^2}, \quad \mathbb{E}\left[\frac{1}{w + Y_1}\right] = \frac{w + 3d/2}{w^2 + 3wd/2 - d^2}.$$

Consequently,

$$\mathbb{E}[\alpha^*(w + Y_1)] - \mathbb{E}[\alpha^*(w + Y_0)] = -\frac{3\eta h d^3/2}{(w^2 - d^2)(w^2 + 3wd/2 - d^2)} < 0. \quad (\text{A.7})$$

Both denominators are positive for $w > d$. Expected risky dollars are $\eta(w + \mathbb{E}Y_j + h) = \eta(w + h)$ under either law. Finally, the required threshold follows from the open fee interval

$$(w + d + h)(1 - e^{-b}) < \kappa < (w + 2d + h)(1 - e^{-b}).$$

The feasible numerical example below shows that this interval and the interior allocation conditions can hold simultaneously. $\square$

A feasible illustration. Set $w = 3$, $d = 1$, $h = 1$, $p = 0.55$, and $\bar{r} = 0.20$. Then $\eta = 0.5$ and $b = 0.00500837$. The fee $\kappa = 5.5(1 - e^{-b}) = 0.02747715$ places the entry threshold at 4.5. All optimal positions satisfy the borrowing and short-sale constraints, including the nonholder position after paying the fee.

Table A.2. Illustrative portfolio outcomes under equal earnings mean and variance

	Symmetric law	Upside-oriented law
Mean earnings innovation	0	0
Variance of earnings innovation	1	1
Upper-minus-lower partial variance	0	0.6
Nonholders: probability of entry (%)	0	20
Nonholders: expected future risky share (%)	0	12.01
Holders: expected future risky share (%)	68.75	68.00
Holders: expected risky dollars	2	2

Notes: Exact outcomes from the illustrative model, not estimates or a calibration to SIPP. Both groups begin with financial wealth $w = 3$ and safe remaining human income $h = 1$; initial holders have already paid the entry cost. Earnings innovations are $(-1, 1)$ with equal probabilities under the symmetric law and $(-0.5, 2)$ with probabilities $(0.8, 0.2)$ under the upside-oriented law. Returns and earnings are independent. Nonholder shares include nonentry zeros and use financial resources net of the access fee when entry occurs.

The proposition concerns this particular tail rotation and an open set of fees and feasible portfolio parameters. It does not order arbitrary distributions by their upper-minus-lower partial variance. Nor does the negative expected-share comparison imply that each investor sells stocks: expected risky dollars are identical across laws. Income is realized before allocation, and h remains fixed. An increase in persistent future earnings instead raises the share in equation (1) at given financial resources. Finally, the illustration preserves moments of dollar innovations, whereas the empirical allocation comparison conditions on moments of adjusted real-log or arc earnings growth. The model supplies an economic interpretation and differentiating predictions, not an estimated conversion between those objects.

B Reliability Design and Additional Results

B.1 Person-level splitting

Each repetition draws one half indicator for every unique person, attaches it to all records of that person, rebuilds weighted cell distributions independently in each half, re-estimates any projection separately within each half, and correlates the common eligible cells. Eligibility is frozen from the full sample; each half must also clear its fixed support floor. The same split assignment is used across paired construction comparisons.

The sequential reliability estimates use 30 deterministic person splits; comparisons across alternative estimators use 50. Splitting SIPP records rather than people would place some workers in both halves because a worker can contribute several monthly year-apart transitions. The matched construction comparison instead draws one eligible transition per person before imposing destination-positive restrictions.

Table B.1 compares positive-to-positive and zero-inclusive growth measures. Each comparison uses independent-person splits.

Table B.1. Positive-to-positive and zero-inclusive Kelley reliability

Data design	Growth construction	Cells	Raw reliability
CPS age-refined cells	Arc, zeros retained	60	0.735
CPS age-refined cells	Arc, positive-to-positive	60	0.018
CPS age-refined cells	Log, positive-to-positive	60	0.051
SIPP one transition/person	Arc, zeros retained	58	0.602
SIPP one transition/person	Log($y + 1$), zeros retained	58	0.630
SIPP one transition/person	Arc, positive-to-positive	58	0.147
SIPP one transition/person	Log, positive-to-positive	58	0.120

Entries are Spearman–Brown transforms of mean independent-person half-sample correlations. The CPS construction uses the education mapping in Appendix C. The SIPP comparison selects one year-apart transition per person before construction-specific destination restrictions and uses identical split seeds in every row.

B.2 Support frontier

Table B.2. CPS reliability by effective sample support

Effective support floor	Common cells	90–10 spread	Raw Kelley	Mean/SD residual Kelley
100	63	0.922	0.728	0.379
200	60	0.929	0.746	0.334
400	58	0.921	0.741	0.368
800	51	0.937	0.791	0.297
1,200	34	0.951	0.839	0.344

Estimates use 50 global person splits in the broader CPS cell construction. The final column removes the mean and standard deviation but not the destination-zero share. Table O.2 uses the portfolio crosswalk and additionally removes the zero share; its residual measure is therefore a different conditional contrast.

At an effective support floor of 400, early-versus-late Pearson correlations are 0.731 for the upper tail, 0.794 for the lower tail, 0.840 for the spread, 0.603 for raw Kelley, and 0.109 for mean/SD-residual Kelley. The corresponding rank correlations are 0.648, 0.785, 0.790, 0.622, and 0.225. These time-stability comparisons change both people and calendar periods. Split-sample reliability, by contrast, changes people while holding the period fixed.

Table B.3. Alternative measures of earnings-distribution asymmetry

Estimator	Raw rel.	Incr. rel.	Raw early/late	Incr. early/late	Min. 2018–24 BH q
Kelley	0.740	0.370	0.631	0.230	0.058
Bowley	0.768	−0.051	0.625	0.249	0.824
Moment skewness	0.794	0.233	0.733	0.290	0.244
L-skewness	0.685	0.178	0.597	0.050	0.381

Reliability is the Spearman–Brown transform of the mean global-person split correlation. Incremental signals remove the linear zero share, quadratic mean and standard deviation, and education and age components, with the projection re-estimated in every split. Early/late entries are Pearson correlations. The last column reports the smallest conditional false-discovery-rate q across four outcomes in the broader 2018–2024 sample: participation, the positive-financial-assets share, the all-worker share, and the holder share. No measure passes the 5% family-adjusted threshold. The Kelley minimum corresponds to the positive-assets share, whose plug-in first-stage-uncertainty p -value is 0.115. The main text reports the strict-post Kelley family; the alternative-estimator comparison was not rerun in that window. The complete estimator-by-outcome results are retained in the replication files.

B.3 Alternative centered estimators

The alternative measures do not strengthen the portfolio-level evidence. None of the twelve Bowley, moment-skewness, and L-skewness coefficients survives within-estimator family adjustment. The controlled-raw and residualized parameterizations yield the same p -values, as required by Frisch–Waugh–Lovell. The negative estimated incremental reliability for Bowley shape indicates that this design does not detect a stable component; it is not a negative population reliability.

B.4 Assignment, transition-selection, and cross-survey sensitivities

Table B.4 separates sensitivity to analyst-generated person splits and transition selections from formal sampling uncertainty. It also replaces the overlapping-month SIPP object with complete calendar-year job-1 earnings and directly compares CPS and SIPP cells over common origin years.

Table B.4. Reliability and cross-survey sensitivity ranges

Design	Raw Kelley	Incremental Kelley	Zero share
<i>Panel A. Spearman–Brown reliability: center [P10, P90]</i>			
CPS exact crosswalk, 30 person splits	0.740 [0.679, 0.802]	0.370 [0.109, 0.549]	0.940 [0.916, 0.956]
SIPP, 30 one-transition selections	0.566 [0.501, 0.630]	0.092 [−0.072, 0.239]	0.829 [0.814, 0.851]
SIPP, complete-year job-1 earnings	0.442 [0.213, 0.607]	0.128 [−0.203, 0.369]	0.774 [0.703, 0.859]
<i>Panel B. Direct common-year CPS–SIPP Pearson correlation: mean [P10, P90]</i>			
Pooled common-year cells	0.482 [0.440, 0.522]	0.053 [−0.025, 0.137]	0.619 [0.598, 0.633]

Notes: Brackets are analyst-generated sensitivity ranges, not confidence intervals. The CPS row varies the 30 frozen person-half assignments; its center applies the Spearman–Brown transform to the mean half-sample correlation. The second row reports the median and P10–P90 range across 30 deterministic choices of one monthly 12-month-apart transition per person, with 30 person splits per choice. The complete-year row sums twelve finite nonnegative decoded monthly job-1 earnings observations, uses December origin characteristics, selects one adjacent-year transition per person, and reports the range across 30 person splits; it contains 40,635 people in 57 supported cells. Panel B compares 58 common cells over the eight origin years shared by both surveys and varies the SIPP transition selection 30 times. CPS measures total annual wage-and-salary earnings, whereas SIPP measures job-1 earnings; the direct correlation is therefore a comparability sensitivity, not validation against an identical construct. Incremental Kelley is residualized on the zero share, quadratic mean and standard deviation, education, and age within each design.

Table B.5. Strict out-of-time prediction across SIPP transition selections

Later SIPP destination outcome	Early CPS signal	Median [P10, P90]	Positive / $q < 0.05$
Any monthly job-1 zero	Raw Kelley	0.297 [0.068, 0.450]	29 / 0
Any monthly job-1 zero	Incremental Kelley	0.154 [−0.013, 0.338]	26 / 0
Any monthly job-1 zero	Destination-zero share	0.965 [0.104, 1.662]	28 / 0
Employer-arrangement zero	Raw Kelley	0.200 [0.081, 0.328]	30 / 1
Self-employment-arrangement zero	Raw Kelley	0.059 [−0.045, 0.165]	22 / 0
Zero with any no-job spell	Raw Kelley	−0.006 [−0.043, 0.057]	14 / 0

Entries are percentage-point coefficients per one early-CPS signal standard deviation across 30 deterministic choices of one eligible 2018–2023 SIPP origin transition per person, selected without conditioning on destination state. Destination outcomes occur in 2019–2024, after the final destination year entering the CPS signal. Each cell-year regression uses 360 cell-years in 60 signal cells, official person weights, baseline earnings and demographics, destination-year and additive industry, education, and age-band fixed effects, and standard errors clustered by signal cell. The incremental specification also includes the CPS zero share and quadratic mean and dispersion. The final column counts positive coefficients and selections whose four-outcome Benjamini–Hochberg q -value is below 0.05. Brackets are transition-selection sensitivity ranges, not sampling confidence intervals. SIPP arrangement classes are mutually exclusive; the no-job-spell flag can coexist with an arrangement within a month.

C CPS-to-SIPP Crosswalk and Portfolio Construction

C.1 Exact public industry mapping

Table C.1. CPS IND1990 to SIPP broad-industry crosswalk

SIPP code	Sector	CPS IND1990 support
1	Agriculture	10–32
2	Mining	40–50
3	Construction	60
4	Manufacturing	100–392
5	Trade	500–691
6	Transportation and utilities	400–432
7	Information	440–472
8	Finance, insurance, and real estate	700–712
9	Professional and business services	721–760 and 861–893
10	Education and health	800–860
11	Leisure and hospitality	761–791
12	Other services	No clean IND1990 analogue; unmatched
13	Public administration	900–932
14	Military	940–960; excluded ex ante

Education is harmonized to high school or less, some college, and college; IPUMS CPS code 73 is a high-school diploma and belongs to the first group. Age is divided into 25–44 and 45–65. The 12 matched industries, three education groups, and two age bands give 72 candidate cells. Base support requires pooled CPS Kish support of at least 200 and at least 100 in each early and late period; 65 cells pass. The early-signal rule requires early CPS Kish support of 200; 60 cells pass.

The SIPP source contains 7,026,846 person-months for 263,280 people. Age, weight, and industry restrictions leave 2,495,083 worker-months for 109,401 people. Annualization produces 224,910 worker-years for 108,606 people; matching to supported pooled CPS cells leaves 212,635 worker-years for 104,093 people. The primary strict-post sample contains 90,110 worker-years for 52,115 people during 2019–2024 in 60 signal cells. All outcomes therefore follow the CPS destination years, which end in 2018. The broader 2018–2024 timing comparison contains 107,555 worker-years for 61,162 people in the same 60 cells.

C.2 Portfolio outcomes

Let M_{it} denote the observed stock-and-mutual-fund balance and F_{it} reported financial assets. The level outcomes are

$$\text{PART}_{it} = \mathbf{1}\{M_{it} > 0\}, \quad (\text{C.1})$$

$$\text{SMF}_{it}^+ = 100M_{it}/F_{it} \quad \text{for } F_{it} > 0, \quad (\text{C.2})$$

$$\text{SMF}_{it}^{\text{all}} = 100M_{it}/F_{it} \quad \text{when } F_{it} > 0, \text{ and } 0 \text{ otherwise}, \quad (\text{C.3})$$

$$\text{SMF}_{it}^{\text{holder}} = 100M_{it}/F_{it} \quad \text{for } M_{it} > 0, F_{it} > 0. \quad (\text{C.4})$$

The ratio is not total household equity exposure. Mutual funds need not be pure equity, retirement balances enter reported financial assets without their internal allocation being observed, and employer-stock compensation cannot be cleanly removed.

C.3 Portfolio sample and measurement

The person-year measure averages available monthly balance-sheet records within the calendar year. Table C.2 describes the strict-post sample before outcome-specific exclusions and reports the broader timing window for comparison. Asset-imputation flags are unavailable in the cleaned source, so the analysis cannot distinguish imputed from directly reported balances. The observed records do permit checks of missingness, denominator support, ratio bounds, and duplication.

Table C.2. SIPP portfolio sample and measurement diagnostics

Sample characteristic	Value	Definition
Strict-post primary frame	90,110	Worker-years on 52,115 people during 2019–2024 in 60 supported cells
Broader timing-sensitivity frame	107,555	Worker-years on 61,162 people during 2018–2024 in the same cells
Strict-post positive financial assets	93.40%	Observations entering the positive-assets share
Strict-post weighted participation	19.49%	Positive observed stock-and-mutual-fund balance
Strict-post all-worker share equal to zero	80.49%	Nonholders, including nonpositive financial-asset years
Strict-post fewer than 12 eligible monthly records	15.77%	Annual mean uses the available within-year records
Strict-post positive-assets ratio outside [0, 100]	0.00%	No mechanical ratio-tail deletion is required
Broader-frame person-years in multiworker household-years	61.09%	More than one eligible worker shares the household-year
Broader-frame multiworker household-years with same asset tuple	0.87%	Exact cross-worker duplication is rare

Values precede outcome-specific missing-value exclusions. The primary sample covers 2019–2024; the two household-record rows explicitly use the broader 2018–2024 window. The asset tuple comprises the annual stock-and-mutual-fund balance, financial assets, total assets, and net worth. Monthly asset values are constant within each retained person-year in the cleaned source. The participation rate uses official person-year weights; all other shares are unweighted sample diagnostics. The replication files retain the underlying definitions, weighted ratio quantiles, and sample counts.

The strict-post sample contains 90,110 worker-years in 61,658 household-years and 36,333 households, identified by the composite SIPP key SSUID plus SHHADID. Using SSUID alone would combine distinct current households in 811 sample-unit-years, affecting 1,651 household-years. Exactly 36,049 household-years contain one worker satisfying all sample and support restrictions. Table C.3 first clusters the full sample by household and signal cell, then restricts it to these single-supported-worker household-years. The comparison tests sensitivity to observing several eligible workers in the same household; it does not identify the household’s financial decision maker.

Table C.3. Household decision-unit sensitivity, strict-post sample

	All supported workers		One supported worker	
	(1) Raw	(2) Incremental	(3) Raw	(4) Incremental
<i>Kelley measure</i>				
<i>Panel A. Participation</i>				
Kelley downside shape	0.0038 (0.0073)	−0.0092* (0.0048)	0.0053 (0.0086)	−0.0108* (0.0058)
BH <i>q</i> -value	0.879	0.080	0.538	0.091
Worker-years	90,110	90,110	36,049	36,049
<i>Panel B. Share among workers with positive financial assets</i>				
Kelley downside shape	0.0608 (0.1836)	−0.2610*** (0.0976)	0.2704 (0.2272)	−0.3123** (0.1501)
BH <i>q</i> -value	0.879	0.039	0.344	0.091
Worker-years	84,161	84,161	33,228	33,228
<i>Panel C. Share among all workers</i>				
Kelley downside shape	0.0800 (0.1833)	−0.2341** (0.0985)	0.2648 (0.2189)	−0.2742* (0.1382)
BH <i>q</i> -value	0.879	0.042	0.344	0.091
Worker-years	90,110	90,110	36,049	36,049
<i>Panel D. Share among current holders</i>				
Kelley downside shape	−0.0967 (0.6303)	−0.1525 (0.4292)	0.9947 (0.8708)	−0.2065 (0.8321)
BH <i>q</i> -value	0.879	0.724	0.344	0.805
Worker-years	17,578	17,578	6,497	6,497

Coefficients are per one early-CPS standard deviation of the raw or incremental Kelley measure specified in the column header. Participation is a probability; share outcomes are percentage points. All regressions use the strict-post 2019–2024 sample. Standard errors in parentheses cluster by the composite SIPP household key (SSUID plus SHHADID) and signal cell. Superscripts use unadjusted two-sided conditional tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Benjamini–Hochberg *q*-values adjust the four outcomes separately within each column. Columns (3)–(4) retain household-years with exactly one worker in the supported analytic sample, not necessarily the household’s sole earner or financial decision maker. The selected person’s official weight is retained because no dedicated household weight is available in the cleaned source. These estimates do not propagate CPS first-stage uncertainty.

The all-worker share coefficient is -0.234 with household rather than person clustering (SE 0.099; $q = 0.042$), and -0.274 in the single-supported-worker sample (SE 0.138; $q = 0.091$). The negative point estimate therefore does not require multiple supported workers per household-year, although the smaller sample is less precise. Both comparisons condition on the estimated CPS measures and retain person weights; they are not household-representative estimates.

C.4 Quantile orientation across margins and earnings supports

Table C.4 reports the complete standing-outcome and transition families for raw and incremental Kelley orientation. Table C.5 varies the zero-endpoint construction for the entry comparison. The main text discusses both comparisons in Section 4; they use the original quantile-based assignment, not the age-adjusted prospective allocation specification in Table 3.

Table C.4. Assigned tail orientation, portfolio levels, and transitions

<i>Panel A. Standing outcomes, 2019–2024</i>				
	(1) Participation	(2) Share: positive financial assets	(3) Share: all workers	(4) Share: current holders
Raw Kelley	0.380 (0.727)	0.061 (0.185)	0.080 (0.185)	−0.097 (0.638)
Conditional BH q	0.880	0.880	0.880	0.880
Incremental Kelley	−0.915* (0.478)	−0.261*** (0.096)	−0.234** (0.097)	−0.153 (0.424)
Conditional BH q	0.081	0.035	0.039	0.721
Observations	90,110	84,161	90,110	17,578
People	52,115	49,175	52,115	11,867
CPS signal cells	60	60	60	60

<i>Panel B. Consecutive-year transitions; both observations in 2019–2024</i>			
	(5) Signed participation change	(6) Nonholder to positive holdings	(7) Holder to zero holdings
Raw Kelley	−0.335 (0.428)	−1.093** (0.526)	−2.236 (1.538)
Conditional BH q	0.437	0.126	0.227
Incremental Kelley	−0.731*** (0.260)	−1.047*** (0.272)	−0.813 (1.152)
Conditional BH q	0.010	< 0.001	0.483
Observations	35,112	27,752	7,360
People	23,170	19,300	5,308
CPS signal cells	60	60	60

Notes: Coefficients and standard errors are percentage points per one cross-cell standard deviation. Raw and incremental estimates in each outcome column come from separate regressions. Both include household controls and year, industry, education, and age-band fixed effects; incremental specifications also include the CPS zero share and quadratic mean and dispersion. Panel B assigns lagged signals and controls, uses lagged-state risk sets, and additionally controls for lagged participation in column (5). Regressions use current-year SIPP weights. Parentheses report conditional standard errors clustered by person and CPS signal cell. Stars use unadjusted conditional two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. BH q -values adjust exactly the four standing outcomes or three transitions within each specification (Benjamini and Hochberg, 1995). All outcomes strictly postdate CPS destination earnings through 2018. Shares among all workers or workers with positive financial assets include nonholders and are not holder-only intensive-margin outcomes.

Table C.5. Tail coefficients under alternative zero-endpoint constructions

	(1)	(2)	(3)	(4)
Early-CPS construction	A. Baseline	B. Exclude self-employed	C. Zero: no weeks worked	D. Interior $q_{0.10}$
<i>Panel A. Lower and upper tails entered jointly</i>				
Lower distance, L	−2.712*** (0.646)	−0.907 (0.887)	−0.758 (0.895)	−1.215 (1.352)
Upper distance, U	1.917** (0.846)	1.392 (0.902)	1.660** (0.817)	2.306** (1.114)
<i>Panel B. Total span and composition entered jointly</i>				
Total span, $L + U$	2.721 (1.850)	1.557 (1.640)	1.845 (1.372)	4.466 (2.793)
Downside composition, $L - U$	−4.456*** (1.530)	−1.730** (0.857)	−1.767** (0.757)	−4.458** (1.845)
<i>Panel C. Incremental orientation</i>				
Incremental Kelley	−1.047*** (0.272)	−0.325 (0.230)	−0.432* (0.243)	−0.683** (0.337)
Cross-cell S.D. of L	0.398	0.249	0.231	—
Lagged controls and fixed effects	Yes	Yes	Yes	Yes
Observations	27,752	27,752	27,752	23,858
CPS signal cells	60	60	60	49

Notes: The dependent variable is one-year entry into positive reported stock-and-mutual-fund holdings among lagged nonholders. Each panel–column combination is a separate regression. Coefficients are percentage points per one cross-cell standard deviation; parentheses report conditional standard errors clustered by SIPP person and lagged CPS cell. Columns (1)–(3) use the same 27,752 transitions. Column (2) excludes CPS transitions whose destination class of worker is self-employed (CLASSWKR 13 or 14); column (3) excludes destination wage zeros with positive weeks worked. Their early-CPS quantiles, zero share, mean, dispersion, and incremental Kelley are rebuilt and restandardized. Column (4) retains baseline measures and scaling but restricts the portfolio sample to the 49 cells with a 10th percentile above the arc-growth bound of -2 . Thus its lower-distance S.D. is not reported as a newly standardized construction. All regressions use current SIPP weights and the same lagged controls and fixed effects as Table N.3. Incremental Kelley is the residual in equation (M.5); Panel C does not add total span. The S.D. of L is Kish-weighted and in arc-growth units. Stars use unadjusted conditional two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. These checks do not propagate CPS first-stage uncertainty.

C.5 Transition durability and positive-balance thresholds

Table C.6 tests whether the entry association persists beyond a single-year classification or a small positive balance. Sustained entry requires zero holdings at $t - 1$ and positive holdings at both t and $t + 1$. Threshold entry requires a nominal stock-and-mutual-fund balance to move from no more than \$1,000 or \$5,000 to above that amount. Each outcome is estimated in the primary sample and among workers whose assigned signal cell is unchanged.

Table C.6. Durability and positive-balance threshold checks

Sample	Sustained entry		Cross \$1,000		Cross \$5,000	
	(1) Primary	(2) Same cell	(3) Primary	(4) Same cell	(5) Primary	(6) Same cell
Incremental Kelley	-0.697** (0.293)	-0.578** (0.268)	-1.104*** (0.222)	-1.131*** (0.273)	-0.453** (0.221)	-0.537** (0.260)
Conditional p -value	0.0205	0.0349	< 0.0001	0.0001	0.0445	0.0436
BH q -value	0.041	0.045	< 0.001	< 0.001	0.045	0.045
Baseline controls and fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Worker transitions	9,331	7,328	28,686	22,242	29,823	23,132

Coefficients and standard errors in parentheses are percentage points per one early-CPS incremental-signal standard deviation. Specifications retain the lagged signal, lagged controls, current-year weight, fixed effects, and two-way person and lag-cell clustering of the primary transition design. Superscripts use unadjusted conditional two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. BH q -values adjust the complete six-coefficient family. These checks condition on the CPS first stage. Sustained entry requires positive holdings at both t and $t + 1$ after zero holdings at $t - 1$ and is observed through 2023. Threshold entry moves from a balance at or below the stated nominal amount to one above it; thresholds are not inflation-adjusted.

The entry association is not concentrated in one outcome year. Omitting each year in turn produces estimates from -1.203 to -0.881 percentage point, with a largest conditional p -value of 0.0031 . Restricting to stable households gives -1.057 point (SE 0.275 ; three-outcome $q = 0.001$); selecting the eligible worker with the highest mean lagged job-1 earnings in each household transition gives -0.869 point (SE 0.360 ; $q = 0.056$). These estimates assess sensitivity to the observational unit, rather than testing a household risk-pooling mechanism.

Table C.7. Finite-cell inference for the entry tail decomposition

Component	Coef.	Two-way p	Wild p	Wild BH q
Lower distance, $q_{0.50} - q_{0.10}$	-2.712***	< 0.001	0.0097	0.0194
Upper distance, $q_{0.90} - q_{0.50}$	1.917**	0.0272	0.0865	0.0865
Downside composition, $L - U$	-4.456***	0.0051	0.0327	—

Coefficients are percentage points per one early-CPS cross-cell standard deviation in the locked 27,752-transition entry risk set. The conventional column uses two-way person and signal-cell clustering. The wild column uses 99,999 null-imposed Rademacher wild-bootstrap draws with weights at the 60 CPS signal cells and retains the two-way covariance structure (Roodman et al., 2019). The BH family contains the lower and upper coefficients; signed composition is an exact rotation of the same two-dimensional column space and is reported separately. Superscripts use the conventional two-way-clustered tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. This check conditions on the frozen CPS tail estimates; Table N.3 separately reports CPS first-stage uncertainty.

Table C.8. Formal contrasts with other asset-entry margins

Comparison entry	Lower contrast	Lower BH q	Upper contrast	Upper BH q	Joint p / BH q	N
Bank deposits	-3.038	0.689	4.717	0.689	0.585 / 0.750	2,712
Bonds	-2.512***	0.086	2.013*	0.279	0.0035 / 0.017	26,913
Retirement assets	-3.033	0.324	3.566	0.324	0.110 / 0.274	12,312
Home equity	-0.485	0.821	0.310	0.838	0.920 / 0.920	13,137
Vehicle equity	-2.265	0.689	2.356	0.689	0.600 / 0.750	7,864
Bonds, one worker per lag household	-2.253**	0.234	2.492**	0.234	0.020 / 0.102	20,337

Each row restricts the locked worker-transition frame to records whose household reports neither stock-and-mutual-fund holdings nor the comparison asset in the lag year; N counts worker transitions. The dependent variable is stock-and-mutual-fund entry minus comparison-asset entry, so a coefficient is a formal cross-outcome slope difference on the same observations. Coefficients are percentage points per one early-CPS cross-cell standard deviation. Plug-in covariance adds the covariance of coefficients across CPS person-within-cell resamples to the locked two-way-clustered SIPP covariance. Stars use the unadjusted plug-in two-sided tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Lower and upper BH q -values adjust one family of ten scalar contrasts; joint q -values adjust the five two-dimensional tests. The first five rows retain all eligible workers, cluster by person and signal cell, and use 200 CPS draws. Because the asset reports are household-level, the final row re-estimates the complete five-comparison family with 1,000 CPS draws after selecting the eligible worker with the highest mean lagged job-1 earnings in each lag-household transition and clusters by lag household and signal cell; only its bond row is displayed. The all-worker bond vector passes its joint family, but the one-worker version narrowly misses the 10% joint-family threshold and neither of its scalar contrasts survives the ten-coefficient family. These are exploratory diagnostics, not literal placebos: labor-income tails can affect other assets for economic reasons, and neither a rejection nor a null contrast establishes a stock-and-mutual-fund mechanism.

Table C.9. Entry estimates under alternative weights, coverage, and controls

	(1) Cell-year	(2) Lag-year weights	(3) Unweighted	(4) Full-year records	(5) Quadratic zero share	(6) Cubic zero share
Incremental Kelley	−0.985*** (0.297)	−0.975*** (0.272)	−0.806*** (0.242)	−0.862** (0.345)	−0.982*** (0.273)	−0.901*** (0.298)
Conditional <i>p</i> -value	0.0016	0.0007	0.0015	0.0152	0.0007	0.0037
Observations	300	27,752	27,752	20,522	27,752	27,752

The dependent variable is entry into stock-and-mutual-fund holdings. Coefficients and standard errors in parentheses are percentage points per one early-CPS incremental-signal standard deviation. Column (1) uses 300 lag-cell-by-outcome-year observations and signal-cell-clustered inference with 59 reference degrees of freedom; all other columns use worker transitions and two-way person and lag-cell clustering. Column (4) requires twelve eligible monthly records in both years. Columns (5)–(6) add powers of the zero-incidence control to the outcome equation without redefining the first-stage signal. Superscripts use unadjusted two-sided conditional tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. These checks were conducted post hoc; their p -values are not multiplicity-adjusted and omit CPS first-stage uncertainty.

Nor does any single signal cell or represented industry account for the negative coefficient. Omitting each of the 60 cells produces estimates from -1.223 to -0.890 percentage point; the largest unadjusted conditional p -value is 0.0020. Omitting each of the 11 represented broad industries gives estimates from -1.316 to -0.700 point, with a largest p -value of 0.0155. These are influence summaries, not 60- or 11-test confirmatory families.

D Additional Evidence on the Entry Association

This section examines within-person variation, sample heterogeneity, the relation between assigned and perceived tails, and alternative assignment cells. Unless stated otherwise, regressions use the strict-post entry risk set, baseline controls and fixed effects, current SIPP weights, and two-way clustering by person and lagged CPS cell. These estimates condition on the CPS first-stage measures. Section N.7 discusses their implications for the main result.

D.1 Out-of-cell prediction

The in-sample tail decomposition does not establish improved forecasting for workers in other earnings-risk cells. I assess that question by leaving out each of the 60 signal cells in turn, estimating the SIPP coefficients on the other 59, and predicting entry in the omitted cell. The three comparisons use the same 27,752 transitions and contrast the separate-tail model with controls alone, total tail span, and incremental Kelley shape. The model family was fixed in the analysis script before its prediction results were read; this was not a preregistered design.

The separate-tail model reduces the Brier score by only 0.039 percent relative to total span ($p = 0.571$, $q = 0.695$). Its improvement over controls alone is 0.179 percent, but does not pass the 5% three-comparison adjustment ($q = 0.091$). Clipping predicted probabilities to $[0, 1]$ leaves the span comparison small and imprecise: the improvement is 0.025 percent ($p = 0.698$). Thus the

Table D.1. Out-of-cell predictive value of separate earnings tails

Comparison model	Brier score	Improvement (percent)	<i>p</i> -value	BH <i>q</i>
Controls only	0.077509	0.179	0.030	0.091
Total 90–10 span	0.077400	0.039	0.571	0.695
Incremental Kelley	0.077380	0.013	0.695	0.695
Separate lower and upper tails	0.077370	—	—	—

Each cell is held out once. Scores use current SIPP weights and the original linear-probability predictions in the 27,752-transition sample, which contains 19,300 people, 2,566 entries, and 60 signal cells. A lower Brier score is better. Improvement is the percentage reduction in the comparison model's score obtained by the separate-tail model. Two-sided tests use cell-clustered uncertainty for the difference in squared prediction errors; BH *q*-values adjust the complete three-comparison family. These comparisons condition on the CPS measures and do not address sorting, first-stage measurement error, perceptions, or causality. The total-span comparison is not a test against variance or third-moment skewness.

coefficient decomposition reveals conditional associations, but this exercise does not demonstrate an out-of-cell forecasting gain.

D.2 Annual changes in the earnings distribution

A change in the earnings environment could be associated with portfolio adjustment even when a historical level is most informative about entry. Assessing this distinction requires measuring changes in the same distributional statistic, rather than subtracting quantities built with different estimators or averaging windows. I examine that requirement in the public SIPP before linking annual tail revisions to portfolio changes.

The construction uses first-listed-job earnings in the 2018–2025 public files. Each annual transition matches the same person and calendar month one year apart, with positive origin earnings and nonnegative destination earnings; missing earnings are not coded as zero. One eligible month is selected per person and origin year before restricting the data to the fixed 58 industry–education–age cells. Origin characteristics determine the cell. The resulting 53,082 transitions yield distributions ending in 2018–2024 and adjacent distributional changes ending in 2019–2024. Weighted means, centered partial moments, and inverse-CDF quantiles are recomputed identically at both dates. These are changing cohorts of earnings histories, not observations of individual belief revisions; the first-listed slot need not identify the same employer across waves.

Table D.2 reports support and split-sample reproducibility. Of 348 possible cell-year changes, 78 have person-level Kish support of at least 100 at both dates. The 2021 and 2022 changes each have only six such cells. One hundred global household splits keep each full sample-unit lineage together across all years and cells. Requiring Kish support of at least 50 at all four half-sample endpoints leaves 61–75 paired cell-year changes in pooled comparisons and only 4–6 in each of 2021 and 2022. The latter cannot meet the declared eight-cell minimum for a within-year correlation.

The pooled median Pearson half-sample correlation is 0.270 for upper-distance changes and –0.038 for lower-distance changes. For the upside-minus-downside semivariance change it is 0.105. Conditioning on initial composition and on levels and changes in mean, variance, and zero incidence,

with update-year fixed effects, reduces the median to 0.085. The incremental signed-quadratic composition change has a median correlation of 0.062 after also conditioning on linear composition and parallel nonlinear variance changes. Each of these five measures has both positive and negative partition correlations; this refers to the full range, not the table’s 10th–90th percentiles. These partition summaries are not confidence intervals. Limited support in 2021 and 2022 and weak incremental reproducibility prevent this construction from resolving the proposed share-adjustment mechanism. No portfolio-response estimate follows from this source screen; the result is a measurement limitation, not evidence that shares do not respond to earnings-tail changes.

Table D.2. Support and split-sample stability of annual SIPP earnings-tail changes

<i>Panel A: Cell-update support</i>				
Update year	Full cells	Median paired	Paired range	All splits
2019	22	19	16–22	14
2020	22	19	16–22	14
2021	6	5	4–6	4
2022	6	5	4–6	4
2023	13	12	10–13	7
2024	9	7.5	6–9	5
<i>Panel B: Pooled half-sample Pearson correlations</i>				
Measure	Median r	10th pct.	90th pct.	Negative/defined
Composition, ΔC	0.105	-0.043	0.241	18/100
Variance, ΔV	0.207	0.060	0.346	6/100
Mean growth, $\Delta \mu$	0.167	-0.010	0.293	11/100
Zero incidence, Δp_0	-0.020	-0.163	0.158	55/100
Upper span, ΔU	0.270	0.091	0.422	4/100
Lower span, ΔL	-0.038	-0.168	0.134	62/100
Quantile orientation, $\Delta(U - L)$	0.054	-0.108	0.178	32/100
Total span, $\Delta(U + L)$	0.108	-0.061	0.266	20/100
Nonlinear composition, $h(x)$	0.139	-0.042	0.280	19/100
Incremental composition, ΔC	0.085	-0.037	0.202	22/100
Incremental nonlinear composition, $h(x)$	0.062	-0.099	0.244	32/100

The source is public SIPP first-listed-job pay, with positive origin pay, nonnegative destination pay, origin age 25–65, and positive weights at both endpoints. Earnings distributions use origin person weights. One eligible same-person, same-month twelve-month transition is selected per person and origin year before applying the fixed 58-cell map. Seven distributions end in 2018–2024; each update compares adjacent distributions constructed with the same estimator. Missing pay is not zero. Changing source cohorts and job slots are not individual belief updates.

Full cells require person Kish effective size at least 100 at both distribution dates. Each of 100 globally balanced full-SSUID splits keeps entire lineage histories together. Paired cells additionally require Kish size at least 50 at all four half-by-date endpoints. All splits is the intersection across the 100 partitions. Pooled samples have 61–75 cell-updates (median 69); 48 pass in every partition. These partition percentiles are descriptive, not confidence intervals or portfolio-response tests.

$C = V^+ - V^-$, $V = V^+ + V^-$, μ is mean bounded earnings growth, and p_0 is the fraction with recorded zero destination pay. $U = q_{90} - q_{50}$ and $L = q_{50} - q_{10}$. Moments are re-centered and quantiles re-estimated in each half and date. Nonlinearity is $h(x) = x|x|$, where $x = \Delta C/\tau$ and τ is the fixed cross-cell SD of C in 2018. Incremental ΔC removes the complete span of initial C, V, μ, p_0 , changes in V, μ, p_0 , an intercept, and update-year indicators separately in each half. Incremental $h(x)$ additionally removes x and $h(\Delta V/\tau)$. Correlations use equal cell-update weights and retain negative values. No portfolio outcomes enter.

D.3 Within-person variation and sample heterogeneity

Table D.3. Within-person changes in the assigned tails

Design	Lower L	Upper U	ΔL	ΔU	Obs.	Persons
Table N.3 baseline (lagged cell)	−2.712 (0.646)	1.917 (0.846)			27,752	19,300
Lagged levels plus changes, full frame	−2.623 (0.664)	1.988 (0.847)	0.058 (0.408)	0.315 (0.397)	27,752	19,300
Movers only, changes given lagged levels	−1.669 (1.242)	5.347 (1.163)	−0.057 (0.567)	0.563 (0.546)	6,231	5,338
Current-cell assignment, full frame	−0.416 (0.440)	0.609 (0.413)			27,752	19,300
Current-cell assignment, movers only	−0.042 (0.567)	0.647 (0.547)			6,231	5,338
Person fixed effects, lagged cell	−0.578 (1.303)	4.266 (1.987)			14,241	5,789
Person fixed effects, current cell	−0.046 (0.964)	−1.171 (0.752)			14,241	5,789

Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS component; standard errors in parentheses. A mover is a transition whose signal cell differs between the lagged and current observation (6,231 transitions, 22.5 percent); ΔL and ΔU are the current minus lagged standardized distances and equal zero for stayers. The full-frame change specification also includes a mover indicator, whose coefficient is 2.057 (SE 0.439). Person-fixed-effect rows use the 5,789 workers with at least two entry-risk transitions, 2,226 of whom change cell, and replace the year and lagged-cell fixed effects by person and year effects. Among movers the standard deviations of ΔL and ΔU are 1.21 and 1.27 and their correlation is 0.56.

Table D.4. Hold-out year and split-sample heterogeneity

Sample	Lower L	Upper U	Obs.	Entry rate
<i>Panel A. Outcome years</i>				
Locked frame, outcomes 2020–2024	–2.712 (0.646)	1.917 (0.846)	27,752	0.088
Hold-out, 2018 to 2019 transitions	–0.343 (1.079)	–0.462 (0.987)	6,033	0.066
Pooled, outcomes 2019–2024	–2.419 (0.593)	1.573 (0.738)	33,785	0.085
<i>Panel B. Lagged bank-account balances, locked frame</i>				
Below \$1,000	–0.526 (0.754)	1.981 (0.877)	9,759	0.045
\$1,000 to \$7,500	–1.012 (0.948)	1.565 (0.983)	8,903	0.077
Above \$7,500	–6.204 (1.681)	0.643 (1.782)	9,090	0.143
<i>Panel C. Retirement account at $t - 1$, locked frame</i>				
None	–1.892 (0.610)	0.031 (0.699)	14,036	0.061
Owns	–3.108 (1.232)	3.566 (1.121)	13,716	0.115
<i>Panel D. Age band, locked frame</i>				
25–44	0.950 (1.249)	3.523 (1.785)	13,765	0.091
45–65	–2.885 (0.698)	3.593 (2.095)	13,987	0.084

Coefficients are percentage points per one cross-cell standard deviation; standard errors in parentheses. The hold-out row uses consecutive 2018 to 2019 worker-years in the same 60 supported cells with the same lagged assignment, controls, and clustering; the strict design excludes them because CPS destination earnings extend through 2018. Bank-balance terciles use weighted cutpoints from the locked frame and lagged annual means of the public SIPP monthly balance. Joint interaction tests that hold controls and fixed effects common across bank-balance terciles do not reject equal lower-tail or upper-tail coefficients within the locked frame ($p = 0.385$ and 0.890), within the 2018–2019 hold-out frame ($p = 0.661$ and 0.665), or within the pooled 2019–2024 outcome frame ($p = 0.373$ and 0.862), respectively. A continuous interaction with the inverse hyperbolic sine of the balance gives -0.154 (SE 0.080) for the lower and -0.089 (SE 0.097) for the upper tail. Adding the worker's own realized arc growth over the transition year does not move the upper coefficient in the bottom tercile (1.981 to 1.978).

Table D.5. Assigned historical CPS tails and individual SCE earnings expectations

	(1) Lower distance	(2) Upper distance	(3) Fall > 4%	(4) Rise > 4%	(5) S.D.	(6) Job-loss probability
Perceived outcome						
Assigned lower tail, L	0.122 (0.097)	0.094 (0.088)	0.0070 (0.0051)	–0.0093 (0.0146)	0.149 (0.118)	0.070 (0.853)
Assigned upper tail, U	–0.137 (0.062)	–0.154 (0.064)	–0.0080 (0.0031)	–0.0065 (0.0079)	–0.176 (0.074)	–0.918 (0.516)
Joint p -value	0.018	0.016	0.016	0.635	0.015	0.115
Controls and fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Respondent-months	75,669	75,669	75,863	75,863	75,863	68,813

Assigned tails use CPS origins through 2017. Of 306 candidate state-by-education-by-age cells, 144 meet the Kish support rule of 200; standardized cell measures match to 75,863 SCE responses from 10,861 respondents aged under 60 in 143 cells. All regressions include the assigned zero share, quadratic mean and dispersion, household-income and numeracy categories, and state, education, age-category, and year fixed effects, with SCE sampling weights and cell-clustered standard errors in parentheses. Perceived lower and upper distances are median-minus-25th and 75th-minus-median quantiles, respectively. Distances and standard deviation are percentage points of expected earnings growth; the fall and rise probabilities are shares. Job loss concerns the next twelve months. In outcome-standard-deviation units, the largest coefficient is -0.07 for the assigned upper tail in column (2).

Table D.6. Finer assignment cells: occupation groups

Cell scheme	Cells	Raw rel.	Incr. rel.	Lower L	Upper U	Incr. Kelley	Obs.
Industry, education, age (paper)	60	0.710	0.390	−2.712 (0.646)	1.917 (0.846)	−1.047 (0.272)	27,752
Industry, occupation, education	69	0.671	0.315	−0.666 (0.725)	0.316 (0.860)	−0.369 (0.227)	26,290
Industry, occupation, education, age	96	0.529	0.180	−1.218 (0.976)	0.552 (0.906)	−0.438 (0.286)	23,735
Occupation, education, age	18	0.817	0.344	−1.259 (0.531)	−1.738 (0.976)	−0.269 (0.134)	27,751

Occupation groups are professional and managerial (CPS OCC1990 3–199; SIPP 2018 codes 10–3655), sales, office, and administrative support (203–389; 4700–5940), and service, production, construction, and transportation (403–889; 3700–4655 and 6005–9760). SIPP workers receive their modal annual occupation group. Cells are early-CPS cells clearing Kish support of 200. Reliability is the Spearman–Brown transform of the mean correlation across 30 random global person splits generated inside the script, which is why the paper’s scheme shows 0.710 and 0.390 rather than the 0.687 and 0.397 of the frozen deterministic splits in Table O.2. Coefficients are percentage points per one cross-cell standard deviation under the scheme’s own standardization, with lagged assignment, the scheme’s zero-share, mean, and dispersion controls, the paper’s household controls and fixed effects, and clustering by person and lagged cell. Observations fall when a worker-year lacks a supported cell under the finer scheme.

E Portfolio Levels in the Broader Timing Window

Table E.1. Portfolio levels and alternative Kelley parameterizations, 2018–2024

Outcome	Raw	Raw + controls	Incremental	BH q
Participation	0.004 (0.007)	−0.022* (0.012)	−0.008* (0.005)	0.106
Share, positive financial assets	0.097 (0.162)	−0.552** (0.219)	−0.209** (0.083)	0.058
Share, all workers	0.115 (0.163)	−0.489** (0.227)	−0.185** (0.086)	0.071
Share, current holders	0.200 (0.590)	0.070 (0.993)	0.027 (0.376)	0.944

This table is the broader 2018–2024 timing-sensitivity window, not the strict-post primary sample. Conditional two-way-clustered standard errors are in parentheses. Superscripts denote unadjusted two-sided conditional significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$; they omit BH adjustment and CPS first-stage uncertainty. Raw Kelley is standardized using early supported cells. “Raw plus controls” adds the linear CPS zero share and quadratic mean and dispersion; Incremental Kelley is the standardized residual from that projection. Those two columns are FWL-equivalent—their t statistics, p -values, and four-outcome q -values agree—and are not independent tests. Plug-in first-stage-uncertainty p -values are 0.182 for participation, 0.115 for the positive-assets share, and 0.168 for the all-worker share.

Raw Kelley coefficients are positive and imprecise in the broader 2018–2024 sample. Conditioning on the zero share, location, and dispersion reverses the sign for participation and the two broad share measures, but no person-year coefficient passes the 5% four-outcome adjustment. Adding CPS first-stage uncertainty further weakens inference. Aggregating to 420 signal-cell-years in 60 cells yields an incremental positive-assets-share coefficient of −0.224 (SE 0.085; four-outcome BH $q = 0.044$). The participation, all-worker-share, and holder-share coefficients are −0.007, −0.170, and 0.092, with q -values of 0.198, 0.125, and 0.816. The positive-assets result is thus specific to a conditional aggregation comparison in the broader timing window. It does not establish a holder-allocation response or overcome sorting, imperfect portfolio measurement, or the distinction between historical and perceived risk.

F Perceived Earnings Risk and Portfolio Adjustment

The SCE provides complementary evidence on perceived risk, which need not coincide with risk assigned from earnings histories. Guiso, Jappelli, and Terlizzese (1996) relates directly elicited income uncertainty to household portfolios; Wang (2023), Caplin et al. (2026), and Arellano et al. (2026) document the distinction between perceived and realized earnings risk. Related evidence connects elicited asset-return skewness to allocations conditional on other expected moments (Drerup, Wibrat, and Zimpelmann, 2023) and investor beliefs to the direction of adjustment conditional on trading (Giglio et al., 2021). Dew-Becker (2024) separates forward-looking from realized skewness and uses relative upper and lower semivariance to distinguish orientation from scale. Closely related concurrent work by van den Akker et al. (2026) finds that a longer perceived earnings right tail predicts administrative stock and bond investment. These studies motivate the comparisons below, but their results do not establish the interpretation of the SCE fixed-threshold imbalance: its centering, timing, and associated portfolio outcomes must be assessed separately.

F.1 Fixed thresholds are not centered shape

Lemma 3 (Translation of a fixed-threshold imbalance). *Let X have distribution function F , with density f positive at the relevant thresholds, and let $c > 0$. For the translated variable $X + m$, define*

$$A_c(m) = \Pr(X + m < -c) - \Pr(X + m > c) = F(-c - m) + F(c - m) - 1.$$

Then $A'_c(m) = -f(-c - m) - f(c - m) < 0$, although every translation-invariant centered shape measure is unchanged.

Proof. Continuity gives $\Pr(X + m < -c) = F(-c - m)$ and $\Pr(X + m > c) = 1 - F(c - m)$. Subtraction yields the expression; differentiation yields the derivative. A translation-invariant centered-shape functional takes the same value for X and $X + m$. $\square$

The public SCE density bins use fixed percentage-growth thresholds. The imbalance subtracts probability above +4 percent from probability below −4 percent. In the state-education-year reconstruction, this threshold imbalance is correlated −0.941 with the forecast mean. Additive cubics in the mean and standard deviation explain 89.4 percent of its variation, leaving 32.6 percent of its raw standard deviation. The residual correlates 0.018 with centered Bowley shape and 0.029 with centered moment skewness. Residualizing a fixed-threshold statistic therefore does not recover the same ranking as a centered shape estimator.

F.2 Lagged group beliefs and subsequent SIPP portfolio changes

I first aggregate SCE beliefs by state, education, and year and relate these lagged beliefs to subsequent SIPP portfolio changes. The matched sample contains 42,375 consecutive transitions for 26,005 people in 31 states and 366 state-education-outcome-year cells. Regressions include state-by-year,

state-by-education, and education-by-year fixed effects, perceived mean and dispersion, and baseline household controls; standard errors are clustered by state. The comparison establishes temporal ordering across surveys, but not a causal effect of beliefs.

Table F.1. Lagged SCE beliefs and subsequent SIPP active share changes

	Subsequent active share change		
	(1)	(2)	(3)
Fixed-threshold residual	0.1543 (0.2136)		
Centered Bowley downside shape		−0.3471* (0.1997)	
Centered moment downside skewness			−0.0121 (0.1745)
Focal coefficient <i>p</i> -value	0.476	0.093	0.945
Mean, dispersion, and household controls	Yes	Yes	Yes
State-year, state-education, education-year FE	Yes	Yes	Yes
Worker transitions	42,375	42,375	42,375
State clusters	31	31	31

Beliefs in year $b - 1$ are assigned to SIPP portfolio changes from b to $b + 1$ among state-education stayers. Coefficients are percentage points per one standard deviation of the assigned belief measure. Regressions use baseline SIPP weights and control for baseline income, wealth, age, sex, risky share, and participation. Standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Split reliability is 0.198 for the raw threshold, 0.147 for its leave-year-out residual, 0.068 for Bowley shape, and 0.121 for moment skewness. The fixed-threshold coefficient is imprecise, and the two centered measures do not provide consistent evidence of a subsequent share response.

F.3 Belief timing and reported retirement reallocation

The Household Finance module records the direction of defined-contribution retirement-account reallocation over the preceding twelve months: +1 for a move into riskier assets, −1 for a move into safer assets, and 0 for no change. Because the measures in this subsection are downside-positive, a positive coefficient associates more perceived downside with a riskier reported adjustment. Belief timing matters. Among respondents with a reallocation answer and a same-calendar-year belief report, 87.1 percent have at least one belief report after the module month. Moreover, no same-calendar-year report predates the start of the twelve-month action window. The preferred construction averages only reports one through eleven months before the module and uses the official module weight with forecast mean and standard deviation controls. These reports precede the interview, but usually overlap the period in which the reported action occurred.

Table F.2. Perceived earnings risk, belief timing, and retirement portfolios

	(1) Pre-module	(2) No realized shock	(3) Realized income fall	(4) Post-module	(5) Joint
<i>Panel A. Fixed-threshold imbalance; outcome: reported reallocation</i>					
Pre-module imbalance	0.067** (0.031)	0.013 (0.051)	0.240*** (0.083)		0.109** (0.049)
Post-module imbalance				−0.052 (0.041)	−0.091* (0.047)
Observations	2,136	1,270	190	1,805	1,749
<i>Panel B. Centered Bowley downside shape; outcome: reported reallocation</i>					
Pre-module shape	−0.024* (0.013)	−0.037** (0.017)	0.015 (0.035)		−0.022 (0.015)
Post-module shape				−0.010 (0.015)	0.001 (0.017)
Observations	2,136	1,270	190	1,804	1,748
Forecast mean, dispersion, and demographics	Yes	Yes	Yes	Yes	Yes
Module-year and state fixed effects	Yes	Yes	Yes	Yes	Yes
<i>Panel C. Perceived dispersion; outcome: standing DC equity share</i>					
Shape control		(1) Fixed threshold	(2) Bowley	(3) Moment skewness	
Perceived standard deviation		−3.633*** (1.002)	−3.766*** (1.006)	−4.734*** (0.961)	
Forecast mean and demographics		Yes	Yes		Yes
Module-year and state fixed effects		Yes	Yes		Yes
Observations		3,332	3,332		2,952

Standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. All specifications use the official Household Finance quarterly weight; demographics comprise age, education, and household income. The comparisons are exploratory and are not family-adjusted. In Panels A and B, the outcome is +1 for a riskier reallocation, −1 for a safer reallocation, and zero for no change during the preceding twelve months. Pre-module beliefs are recorded in months −11 to −1; post-module beliefs, in months +1 to +11, cannot have caused the reported action. Column (5) includes both terms. Centered moment skewness on the same sample grid produces coefficients from −0.083 to +0.026; only the realized-fall subsample and joint post-module term have $p < 0.05$. Panel C uses pre-module beliefs and reports percentage points of standing DC equity share per one-standard-deviation increase in perceived earnings dispersion. These are descriptive associations with an accumulated portfolio stock.

Table F.2, Panel A, shows that the fixed-threshold association depends on both the sample and the belief window. The pre-module coefficient is 0.013 after excluding intervening realized shocks and 0.240 in the 190-person large-income-fall subsample. This exploratory contrast does not decompose the full-sample coefficient or establish a subgroup difference. In the joint specification, pre-module and post-module beliefs enter with opposite signs (+0.109 and −0.091). Averaging reports from both sides of the interview can therefore combine different associations. Nor can an income fall after a belief report explain that earlier belief mechanically: selection, common shocks, or overlap between the action and shock windows may account for the pattern. Together with Lemma 3, these results show why a fixed-threshold coefficient cannot be read as a prospective response to centered tail orientation.

Panel B gives a more conventional background-risk sign for pre-module Bowley shape, but moment skewness does not corroborate it. The dispersion comparison in Panel C is more stable: a one-standard-deviation increase in perceived earnings dispersion is associated with a 3.633–4.734 percentage-point lower standing DC equity share across the three shape controls. Thus the SCE contains a clear descriptive scale association, whereas the evidence on centered shape remains sensitive to its definition and timing.

F.4 Centered upper and lower semimoments

To measure centered tails directly, I reconstruct each respondent-month’s ten-bin public density. Open tails are assigned ± 16 percentage points, the three reports whose integer bins sum to 99 or 101 are renormalized, and lower and upper semideviations are centered on the respondent’s elicited mean. Denote these semideviations by d_i^- and d_i^+ . The regression includes their sum and their difference, or tilt, $d_i^+ - d_i^-$. Holding the sum fixed, a positive tilt coefficient associates a shift in perceived tail magnitude from downside toward upside with a riskier reported reallocation. This gap is measured in semideviation units; it is not a scale-free orientation measure.

Table F.3. Perceived earnings-tail tilt and reported retirement reallocation

	Reported retirement reallocation					DC share
	(1) Primary	(2) Equal months	(3) No large change	(4) Post- module	(5) Normalized	(6) Primary
Upside-minus-downside semideviation tilt	0.0276** (0.0128)	0.0284** (0.0127)	0.0435*** (0.0137)	0.0047 (0.0178)		−0.576 (0.664)
Normalized semivariance orientation					0.0189 (0.0137)	
Focal coefficient p -value	0.036	0.030	0.003	0.793	0.173	0.390
BH q -value	0.054	—	0.008	—	0.347	0.390
Scale, mean, and demographics	Yes	Yes	Yes	Yes	Yes	Yes
Module-year and state fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,136	2,136	1,270	1,805	1,893	3,332

The reallocation outcome equals +1 for a reported move into riskier assets, −1 for a move into safer assets, and zero for no change over the preceding twelve months. Column (6) measures standing DC equity share in percentage points. Except in column (2), repeated belief reports are averaged within respondent using the public monthly SCE sampling weight. The standardized tilt enters jointly with total semideviation scale, expected earnings growth, age, education, household income, and the displayed fixed effects. Column (5) instead divides the upper-minus-lower semivariance difference by total variance and controls for total standard deviation. Regressions use official Household Finance quarterly weights; standard errors in parentheses are clustered by state. Superscripts denote unadjusted two-sided significance: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

The primary $q = 0.054$ adjusts the three displayed focal shape tests in its outcome family; expanding to fourteen support and outcome variants gives $q = 0.169$. The no-large-change restriction excludes a reported large income, health, or wealth change and a respondent or spouse retirement-contribution cut; it is a post hoc subgroup and its stronger coefficient is scale dependent. The post-module report cannot have caused the retrospective action, but the formal pre-minus-post contrast is also imprecise ($p = 0.191$). Beliefs recorded before the module usually fall inside the retrospective action window. The separate lower and upper semideviations are correlated 0.95 and are therefore not presented as independent discoveries. The result is exploratory and descriptive: it does not survive the full support family, normalized orientation is null, standing share is null, and adding the tilt changes five-fold cross-validated RMSE by only −0.065%.

The semideviation results suggest an upside association, but do not establish a robust perceived-shape effect. Entered jointly, lower semideviation has a coefficient of -0.107 (SE 0.047 ; $p = 0.026$) and upper semideviation a coefficient of $+0.087$ (SE 0.045 ; $p = 0.060$). Their correlation of 0.95 motivates reporting the gap and sum rather than treating the two estimates as independent findings. In the full sample, the tilt mainly predicts fewer reported moves into safer assets; the increase in moves into riskier assets is imprecise, and the association among active movers alone is also imprecise. The full testing family, normalized-orientation comparison, and standing-share result in Table F.3 limit the interpretation. These estimates neither establish that perceived risk dominates assigned histories nor supply an independent confirmation of the CPS–SIPP entry result.

F.5 Belief revisions and the size of updates

The level comparisons leave open whether larger changes in the perceived earnings distribution are more closely associated with portfolio adjustment. I examine this possibility using repeated SCE earnings densities. For each respondent, I select the earliest and latest valid reports one to eleven months before the Household Finance interview, requiring at least six months between reports. Both endpoint distributions must have positive variance. Together with valid baseline demographics and positive weights, these restrictions define 1,132 respondents before portfolio outcomes are selected. Valid outcome reports leave 539 respondents for reported reallocation and 827 for the standing DC equity share.

For each of three shape measures, let x_i be the change between endpoints divided by the measure’s baseline standard deviation in this common endpoint cohort. The linear specification includes x_i and the baseline shape level, together with baseline and changes in scale and expected earnings growth. An amplitude specification adds $x_i|x_i|$ and the analogous signed quadratic change in scale. The conditional slope is then $\beta_x + 2\beta_a|x_i|$: the amplitude coefficient allows it to vary with the absolute size of the revision, without choosing a large-update cutoff. A positive amplitude coefficient need not increase the absolute response when the linear coefficient is negative. All specifications include baseline age, education, and household-income categories, state and module-year fixed effects, and official module weights. The endpoint and specification rules were fixed for this exploratory exercise, after earlier analyses had been examined; they are not a preregistered design.

Table F.4 reports the complete comparison. Linear revisions in centered tilt and normalized orientation are imprecisely associated with both outcomes. Their amplitude coefficients for reported reallocation are positive, with marginal p -values of 0.093 and 0.058 , respectively. The downside-positive fixed-threshold revision has a positive standing-share coefficient with an unrounded p -value just below 0.100 . None of the twelve focal tests survives the family adjustment: the smallest BH q -value is 0.399 . The table also retains the supplementary joint tests of the revision and amplitude coefficients.

These results do not establish a stronger response to large changes in perceived tail risk. The outcomes are reported adjustment and standing shares, not changes in shares. Although both belief reports precede the interview, they overlap the retrospective action window, and the date of any

adjustment is unknown. Conditioning on the initial belief does not supply a person fixed effect or eliminate common shocks. This exercise therefore complements the level comparisons without establishing either a causal revision effect or an attention mechanism.

Table F.4. Earnings-belief revisions and retirement portfolios

Shape measure	Reported reallocation			Standing DC equity share		
	(1) Tilt	(2) Orient.	(3) Threshold	(4) Tilt	(5) Orient.	(6) Threshold
<i>Panel A. Linear revisions</i>						
Revision, x	-0.024 (0.031)	0.012 (0.024)	0.088 (0.126)	0.717 (1.202)	1.613 (1.398)	5.673* (3.380)
Focal marginal p	0.435	0.633	0.487	0.554	0.255	0.100
Focal BH-twelve q	0.664	0.690	0.664	0.664	0.611	0.399
<i>Panel B. Revisions and their amplitude</i>						
Revision, x	-0.093* (0.052)	-0.046 (0.041)	0.132 (0.114)	0.250 (2.413)	0.064 (2.454)	3.523 (4.074)
Amplitude, $x x $	0.032* (0.019)	0.025* (0.013)	-0.039 (0.028)	0.294 (0.881)	0.654 (0.978)	0.897 (1.120)
Focal marginal p	0.093	0.058	0.174	0.740	0.507	0.427
Focal BH-twelve q	0.399	0.399	0.523	0.740	0.664	0.664
Joint p : $x, x x $	0.193	0.092	0.288	0.699	0.459	0.219
Respondents	539	539	539	827	827	827
State clusters	46	46	46	48	48	48

Reallocation is -1 for safer assets, zero for no change, and $+1$ for riskier assets over the preceding twelve months. Equity share is the standing DC allocation in percentage points, not its change. Columns (1)–(3) use the same respondents, as do columns (4)–(6). Each panel reports separate regressions.

The earliest and latest valid earnings densities lie one to eleven months before the module and six to ten months apart. Tilt is the upper-minus-lower semideviation gap; orientation is the upper-minus-lower semivariance difference divided by total variance. Both are upside-positive. The threshold measure is $\Pr(g < -4) - \Pr(g > 4)$ and is downside-positive. Revisions use each measure's baseline standard deviation in the common 1,132-person endpoint cohort. Positive variance at both endpoints is required for this common comparison.

All models control for baseline shape, baseline and changes in scale and expected growth, baseline age, education, and household income, and state and module-year fixed effects, using official module weights. Scale is the semideviation sum in tilt models and standard deviation otherwise. Panel B also includes the signed quadratic change in scale. State-clustered CR1 standard errors appear in parentheses. Stars use unrounded marginal two-sided t_{G-1} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The twelve-test BH family comprises six linear-revision tests in Panel A and six amplitude tests in Panel B. Joint tests in Panel B are supplementary $F_{2,G-1}$ tests. This exploratory comparison does not identify responses to belief revisions or test attention directly.

G Uncertainty in the Estimated CPS Risk Measures

Suppose the portfolio regression assigns $\hat{K}_c = T(\hat{G}_c)$. Conventional regression inference conditions on the observed $\hat{K}_c$. Clustering by cell allows second-stage residuals to covary within a cell, but does not account for the sampling variation of the nonlinear first-stage statistic.

The CPS bootstrap resamples unique people within each frozen signal cell. If a draw repeats a person m times, weighted first moments use mw and Kish denominators use mw^2 , not $(mw)^2$.

Every draw rebuilds weighted mean, standard deviation, zero share, quantiles, support selection, equation (M.5), standardization, SIPP assignment, and the outcome regression. Across 200 draws, all requested estimates succeed; early support ranges from 59 to 61 cells and pooled support from 63 to 66. The strict-post level coefficient draws are not perfectly centered on the corrected point estimates. For participation, their mean is -0.0062 versus a point estimate of -0.0092 (mean-minus-point 0.0029 ; 2.5th–97.5th percentiles $[-0.0145, 0.0013]$). For the all-worker share, the corresponding values are -0.150 versus -0.234 (mean-minus-point 0.084 ; percentiles $[-0.363, 0.045]$).

The transition bootstrap additionally reassigns both annual observations and rebuilds the risk sets in each draw. For nonholder-to-positive-holdings entry, the draw mean is -0.650 point versus a point estimate of -1.047 points, a mean-minus-point displacement of 0.397 point. The across-draw standard deviation is 0.258 point, 198 of 200 coefficients are negative, and the 2.5th–97.5th percentile range is $[-1.125, -0.138]$ points. For signed change the corresponding standard deviation is 0.250 point; for exit it is 0.845 point. These ranges are descriptive summaries of rebuilt-CPS coefficients with SIPP held fixed, not confidence intervals, probability statements, or bias corrections; their displacement further motivates treating the variance combination as a sensitivity.

Table G.1 reports the corresponding resampling distributions for all six coefficients in the entry parameterization comparison. Some distributions are materially displaced from the original estimates, especially those involving the upper tail or total span. The added variance should therefore be interpreted as a first-stage sensitivity calculation, not as a recentered estimate or a bootstrap confidence interval.

Table G.1. CPS resampling distributions for the entry tail specifications

Regressor	Original estimate	CPS-resampled coefficients			
		Mean	S.D.	P2.5	P97.5
<i>Panel A. Lower and upper distances</i>					
Lower distance	−2.712	−2.138	0.562	−3.269	−1.137
Upper distance	1.917	0.707	0.619	−0.530	2.134
<i>Panel B. Total span and signed composition</i>					
Total span	2.721	0.209	1.331	−2.286	3.187
Lower-minus-upper composition	−4.456	−2.076	0.986	−4.486	0.089
<i>Panel C. Total span and incremental Kelley</i>					
Total span	−0.755	−1.410	0.770	−2.981	−0.110
Incremental Kelley	−0.870	−0.430	0.293	−1.035	0.128

Entries are percentage-point coefficients per one draw-specific cross-cell standard deviation. Each of 200 CPS person-within-cell resamples rebuilds the risk measures, projections, and standardization while holding the 27,752 SIPP entry transitions fixed. All 200 draws succeed. P2.5 and P97.5 are percentiles of the rebuilt-CPS coefficients, not confidence limits for joint CPS–SIPP inference. The table reports the complete three-parameterization comparison; no coefficient is recentered or bias-corrected.

Given the independence of CPS and SIPP, I combine the two sources of variance as follows:

$$\widehat{\text{Var}}_{\text{plug-in}}(\hat{\beta}) = \widehat{\text{Var}}_{\text{SIPP}}(\hat{\beta} \mid \hat{G}) + \widehat{\text{Var}}_{\text{CPS}}(\hat{\beta}).$$

The first term is the conditional two-way-clustered SIPP variance evaluated at the observed signal, and the second is the across-draw variance induced by resampling CPS people and rerunning the stated first-stage pipeline. This is not a joint nonparametric bootstrap of CPS and SIPP. It propagates CPS signal-estimation uncertainty within the frozen cell schema, crosswalk, and observed SIPP sample; it does not cover crosswalk or functional-form choice, SIPP sample-reconstruction uncertainty, or a latent-shape correction. The resampler also does not model cross-cell covariance from a person appearing in multiple cells, but no CPS person does so in the frozen annual-link sample. Reported plug-in p -values use a two-sided standard-normal reference. The primary strict-post first-stage results use this calculation; Table G.2 reports the corresponding design sensitivity.

Table G.2. Strict-post power for the early incremental signal

Outcome	80% minimum detectable effect		Power at observed magnitude	
	Conditional	CPS plug-in	Conditional	CPS plug-in
Participation	1.34	1.79	0.481	0.298
All-worker asset share	0.273	0.409	0.672	0.361

Entries use the strict-post 2019–2024 sample and are per one standard deviation of the early CPS incremental signal. Both MDE columns are in percentage points. The first pair gives two-sided 5%-size, 80%-power minimum detectable effects under a normal approximation; the second gives power at the absolute magnitude of the observed coefficient. “Conditional” treats the estimated CPS signal as fixed. “CPS plug-in” uses the conditional variance plus the CPS first-stage bootstrap variance and remains a sensitivity rather than joint CPS–SIPP inference. The table deliberately reports no latent-signal correction: weak cross-survey transport and possible nonclassical error preclude an equivalence mapping from the assigned signal to latent orientation.

Table G.3. Reporting group-estimated earnings-risk measures

Layer	Required disclosure	Diagnostic reported here
Population and support	Define origin population, destination zeros, growth transformation, cell schema, weights, and effective support.	Positive-origin workers; zero-inclusive arc growth; exact crosswalk; Kish support.
Reproducibility	Split globally by person and report rank and time stability.	Person-split Spearman–Brown reliability and early-versus-late correlations.
Incremental content	State the conditional interpretation and report the sequential relation with zero share, mean, and dispersion.	Raw; after zero share; after zero share, mean, and dispersion.
Downstream inference	Preserve support, nonlinear estimation, projection, standardization, and assignment inside first-stage resampling.	Independent CPS person bootstrap combined with conditional SIPP variance.

H Related Applications of Estimated Risk Measures

Table [H.1](#) places the measurement issues in a broader set of finance applications. Each study estimates a risk measure from histories or groups and uses it in an analysis of portfolios, consumption, or performance. Their data and identification strategies differ, and rich administrative histories can substantially reduce the constraints that arise in public survey cells. The common consideration is that construction and estimation of the risk measure affect the interpretation of the subsequent analysis.

Table H.1. Finance applications using estimated risk measures

Study	Estimated risk measure	Application	Relation to the present analysis
Ranish (2013)	Permanent variance, disaster exposure, and cyclical labor-income risk estimated in PSID histories and mapped to SCF households	Household stockholding and leverage	Direct two-sample group assignment; bootstraps estimated risk, multiple imputation, and generated-regressor uncertainty
Angerer and Lam (2009)	Occupation-by-education permanent and transitory income risk	Household risky-asset participation and shares	Early public group-risk assignment; does not isolate a nonlinear tail-orientation contrast
Fagereng, Guiso, and Pistaferri (2018)	Firm-level shocks to individual wage income	Household risky shares	Administrative and instrumental-variables benchmark for the effect of wage risk rather than a public-cell measurement design
Chang et al. (2022)	Individual income-volatility breaks and firm-volatility instruments	Household portfolio choice	Causal second-moment benchmark; distinct from higher-moment reliability and transport
d'Astous and Shore (2024)	Permanent income volatility shifted by university-admission discontinuities	Household participation and risky shares	Quasi-experimental sorting benchmark; stronger identification but a different risk object
Shen (2024)	Regional earnings-risk skewness	PSID consumption and risky portfolio shares	Nonlinear group risk assigned to households; cell support and incremental shape are relevant interpretation gates
Catherine, Sodini, and Zhang (2024)	Cyclical skewness and unconditional third moments in Swedish registers	Household equity participation and shares	Closest empirical comparison; the unconditional skewness–share relation attenuates within twins
Catherine (2022)	Estimated non-Gaussian, state-contingent income process	Life-cycle portfolio model	Shows the economic object that a reduced-form group statistic seeks to proxy
Gálvez and Paz-Pardo (2026)	Nonlinear earnings process estimated from PSID histories	Structurally estimated consumption, housing, participation, and risky shares	Quantitative nonlinear-risk benchmark, not a test of public group-assigned composition
Akey, Robertson, and Simutin (2026)	Methodologically revised Fama–French factor vintages	Mutual-fund performance, stock alphas and betas, and corporate-finance conclusions	Finance analogue showing that construction revisions can change downstream inference

Notes: The table contrasts risk constructs, portfolio margins, and identification strategies with the present historical composition–entry comparison. It distinguishes substantive portfolio benchmarks from measurement and inference precedents.

I Variance Composition and Joint Inference

I.1 The original wage benchmark

This appendix preserves the original-linkage wage benchmark and the matched wage–total extension. These comparisons use different earnings links and inference from the validated comparison in Section N.1. Table I.1 reports the wage comparison using upside variance composition, $C_c^{\text{up}} = V_c^+ - V_c^-$. In the original 60 cells, its coefficient is 3.390 percentage points per cross-cell standard deviation. Almost 94% of the SIPP-weighted variation in that regressor is absorbed by the other regressors. A one-standard-deviation difference in the remaining composition is therefore associated with 0.799 point more entry, a more representative magnitude for the conditional comparison than the full cross-cell shift.

Restricting to the common 59 cells changes the coefficient little. Reconstructing the earnings distribution from positive destinations within those cells also leaves it positive: 0.972 point per construction-specific standard deviation, or 0.640 point per standard deviation of residual composition. The estimates are not measured in a common cross-cell unit, so their ratio is not an attenuation estimate. The positive point relation is therefore not confined to a construction retaining wage zeros.

The upside and downside descriptions are economically symmetric. At fixed variance, more upside composition necessarily means less downside composition. Reversing the regressor’s orientation gives equal-magnitude negative coefficients, identical standard errors, and identical two-sided tests; it is not a second empirical confirmation. Nor do the estimates establish that upper semivariance independently increases entry with lower semivariance unchanged. Both separate partial-moment coefficients are negative in the original sample, and the upper one is imprecise. Their difference at fixed total variance is the interpretable composition contrast.

The uncertainty about this contrast is greater than its conditional significance suggests. Independent-person reliability of incremental variance composition is only 0.22 in the original cells and 0.26 in the positive-destination construction. In 2,000 draws, rebuilding the CPS moments moves the respective coefficient means from 3.390 to 1.798 and from 0.972 to 0.544 points. Joint resampling of both surveys produces 95% percentile intervals of $[-0.346, 3.932]$ and $[-0.232, 1.343]$, although basic intervals exclude zero. Section N.6 examines this disagreement. The upside framing preserves every aspect of the uncertainty calculation: it does not turn the conditional comparison into an inference-insensitive rejection of equal portfolio associations at fixed variance.

Explicit treatment of the 2014 survey design preserves the point pattern. Retaining only its redesigned origin subsample gives 0.835 point more entry per residual standard deviation of upside composition in the zero-inclusive comparison and 0.646 point more entry in the positive-destination comparison. Refitting the original moments on the same SIPP observations shows that these results are not generated by the small change in supported cells. The zero-inclusive comparisons retain local CPS-plus-two-way-SIPP evidence; the positive-destination comparison is borderline, and broader crossed-reweighting inference is inconclusive. Omitting 2014 entirely attenuates all three magnitudes while preserving their signs. Appendix Table I.6 reports every declared convention and testing family.

Table I.1. Variance composition and portfolio entry

	(1) Zero-inclusive Baseline	(2) Zero-inclusive Common	(3) Positive-to-positive Common
Earnings distribution CPS cell frame			
Upside composition, C^{up}	3.390*** (1.243)	3.265** (1.229)	0.972** (0.483)
Per residual S.D. (percentage points)	0.799	0.767	0.640
Conditional two-way SIPP: S.E.	0.692	0.662	0.328
Local IF + two-way SIPP: BH q	0.015	0.015	0.049
Local IF + product SIPP: S.E.	1.589	1.585	0.591
Local IF + product SIPP: BH q	0.066	0.066	0.105
CPS-only remeasurement mean	1.798	1.776	0.544
Joint remeasurement S.D.	1.060	1.084	0.398
Joint percentile 95% interval	[-0.346, 3.932]	[-0.351, 3.952]	[-0.232, 1.343]
Total variance V and $\sqrt{V}$	Yes	Yes	Yes
Other controls and fixed effects	Yes	Yes	Yes
Destination-zero probability	Yes	Yes	Not applicable
Observations	27,752	27,545	27,545
People	19,300	19,157	19,157
CPS signal cells	60	59	59

Notes: Entry is a transition from zero to positive reported stock-and-mutual-fund holdings in the strictly later SIPP window. For arc growth g centered at its cell mean μ , $V^- = \mathbb{E}[\min(g - \mu, 0)^2]$, $V^+ = \mathbb{E}[\max(g - \mu, 0)^2]$, $V = V^- + V^+$, and $C^{\text{up}} = V^+ - V^-$. At fixed V , higher C^{up} means more upside and less downside variance: one composition contrast, not two independent effects. Each column regresses entry on C^{up} and the scale block V , $\sqrt{V}$, quadratic mean growth, lagged individual controls, and year, industry, education, and age-band fixed effects, using SIPP weights. Coefficients and uncertainty are percentage points per original construction-specific cross-cell S.D.; the residual-S.D. row uses the variation remaining after the regression controls. Parentheses combine local CPS IF and person-and-cell-clustered SIPP uncertainty. Stars use the corresponding two-sided t tests with cells minus one degrees of freedom: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The conditional S.E. row omits CPS uncertainty. Local IF propagates CPS moment-estimation noise through the first-order influence function, including estimation of the centering mean, and combines it separately with two-way clustered or crossed-product-bootstrap SIPP covariance. Each combined BH q row adjusts the same three composition tests, using a t reference with cells minus one degrees of freedom. The remeasurement means and joint intervals use 2,000 draws with fixed original scales; the joint engine also applies positive crossed person-and-cell SIPP weights. All three joint percentile intervals cross zero. These are exploratory assignment associations; displaced redraw means and nonlinear attenuation prevent treating the joint redraw S.D. as stronger inferential evidence.

I.2 Wages and total labor earnings: original links

The wage-based comparison leaves open whether its interpretation depends on excluding other labor earnings. To address this question, I recover wages, net business earnings, and net farm earnings from the original public Census ASEC files. Total labor earnings sum these three components. Wages and totals come from the same release at both endpoints; only the redesigned 2014 origin subsample is retained. I keep the original positive-wage-origin worker population and compare the two measures on identical CPS histories. Positive total origins and nonnegative total destinations are required in both constructions; a second comparison additionally requires positive wage and total destinations. The exercise therefore broadens measured earnings without adding a different initial worker population or treating business losses as zero.

Table I.2 reports the original-linkage benchmark: four models on a common 58-cell sample of 27,419 subsequent SIPP entry transitions. With zero destinations retained, upside variance composition remains positively associated with entry for wages and total earnings. The total-earnings coefficient is 2.203 percentage points per cross-cell standard deviation, or 0.684 point per residual standard deviation, compared with 0.789 point for wages. Both survive local CPS-plus-two-way-SIPP inference within the complete four-test family ($q = 0.037$). The association is thus not confined to a wage-only measure that omits business and farm income. Section N.1 separately rebuilds this matched comparison using validated earnings links; these local-inference tests are not reassigned to the rebuilt estimates.

The positive-destination comparisons also have positive point estimates: 0.647 point per residual standard deviation for wages and 0.655 for total earnings. Both have four-test $q = 0.052$, making them suggestive rather than 5% family rejections. Under the broader SIPP product covariance, all four comparisons have $q = 0.111$. Formal wage–total slope differences, expressed in common wage-composition units and retaining cross-model covariance from both surveys, are imprecise. These tests neither establish equality nor show that total earnings generate a stronger portfolio response. The extension addresses the scope of measured labor earnings; it does not resolve weak incremental reliability, endogenous assignment, or sensitivity to the covariance approximation. Appendix I.9 records the source linkage, signed-earnings restrictions, and complete inference design.

Allowing dependence among household members leaves the comparison virtually unchanged. Clustering the CPS moment influences by household and the SIPP regression scores by sample-unit lineage and assignment cell gives four-test $q = 0.038$ for both zero-inclusive constructions and $q = 0.052$ for both positive-destination constructions. This permits common earnings disturbances and portfolio reporting across different workers assigned to different risk cells. It is a useful check on the effective unit of observation, not a resolution of endogenous risk assignment or the nonlinear-resampling discrepancy. Appendix Table I.7 reports the comparison.

I.3 Second-moment construction and conditional support

The variance components in equation (M.1) use the same early-CPS linked earnings histories and original assignment cells as the quantile comparison. Each early CPS person contributes one linked annual pair. The 60-cell sample contains 106,404 pairs. I use survey-weighted cell means and partial moments, with $V = V^- + V^+$ and $C^{\text{up}} = V^+ - V^-$. Both identities hold before any standardization. Cell standard deviations used to scale regressors are weighted by Kish effective support capped at 1,000. Original scales remain fixed in all remeasurement draws. The underlying downside-oriented estimates are reflected exactly for upside reporting: means and coefficients change sign, interval endpoints change sign and order, and standard errors and two-sided tests remain unchanged.

Positive-to-positive histories retain positive origin earnings and require positive destination earnings. One of the original 60 cells has positive-destination Kish support below 200, leaving 59 cells. Both the common-cell unconditional and positive-destination regressions use exactly 27,545 SIPP entry-risk observations on 19,157 people. Only the CPS distribution and its associated controls

Table I.2. Total labor earnings and variance composition

Earnings Destinations	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
Upside composition, C^{up}	3.272** (1.235)	2.203** (0.908)	0.982* (0.495)	1.036** (0.508)
Per residual S.D. (percentage points)	0.789	0.684	0.647	0.655
Local IF + two-way SIPP: BH q	0.037	0.037	0.052	0.052
Local IF + product SIPP: S.E.	1.611	1.183	0.606	0.622
Local IF + product SIPP: BH q	0.111	0.111	0.111	0.111
Incremental composition reliability	0.197	0.276	0.239	0.249
Total minus wage slope	Zero-inclusive		Both positive	
Common-unit coefficient	−0.610 (0.702)		0.028 (0.122)	
Local IF + two-way SIPP: BH q	0.777		0.816	
Local IF + product SIPP: BH q	0.844		0.844	
Observations	27,419	27,419	27,419	27,419
People	19,081	19,081	19,081	19,081
CPS signal cells	58	58	58	58

Notes: The outcome is subsequent entry from no reported stock-and-mutual-fund holdings. Wages and total labor earnings (wages plus net business and farm earnings) come from the same Census ASEC release at both endpoints. The population retains the original positive-wage-origin workers. Both constructions require positive total origins and nonnegative total destinations; columns (3)–(4) additionally require both wage and total destinations positive. Negative total earnings are excluded and recorded, not recoded to zero. The 2014 origins retain only the redesigned questionnaire subsample; later years are unchanged. Exact record linkage passes, but 232 high-income 2014 wage values differ from the retained IPUMS version, so these are not hybrid or silently harmonized earnings. All four models use identical later SIPP observations in the four-way effective-support intersection. Each includes C^{up} , total variance and its square root, quadratic mean growth, its own zero probability when applicable, and the same household controls and fixed effects. Coefficients use construction-specific cross-cell standard deviations. Parentheses combine local CPS influence-function and two-way SIPP covariance; stars use two-sided t_{57} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The broader covariance substitutes 1,000 paired person-by-cell SIPP reweightings. Four composition tests and two wage–total differences have separate complete BH families within each covariance convention. Differences rescale the total-earnings slope to the corresponding wage-composition unit and retain cross-model CPS and SIPP covariance; different conditional risk controls prevent a causal decomposition. Reliability uses 200 paired person partitions, not a sampling confidence interval. All results are exploratory.

change. In the positive-destination specification the zero-share control is identically zero and omitted. The full- and positive-support coefficients use different original standard deviations and cannot be compared as a common-unit attenuation ratio.

The weighted SIPP projection removes all regressors other than standardized variance composition, including the variance and standard-deviation scale block. Its residual standard deviation is 0.236 original cross-cell units in the baseline and 0.658 in the positive-destination construction. Multiplying by the corresponding coefficient gives the residual-standard-deviation comparisons in Table I.1. These are normalizations of the original estimates, not additional regressions or new tests. Independently splitting CPS people into halves 200 times gives incremental-composition Spearman–Brown reliability of 0.220, 0.253, and 0.261 across the three constructions. The cell-level reliability projection includes quadratic mean growth, variance, standard deviation, zero incidence when applicable, and industry,

education, and age-band components. Split-to-split dispersion is a conditional partition sensitivity, not a sampling confidence interval.

I.4 Uncertainty in the assigned earnings distributions

The historical distributions are estimated from a separate sample, not known without error. The CPS resampling exercise reconstructs weighted quantiles, support, zero incidence, mean and dispersion, residualization, standardization, and assignment before re-estimating the SIPP regressions. Following the two-sample household-finance logic of [Ranish \(2013\)](#), this carries uncertainty in the measured earnings environment into the portfolio calculation. The reported plug-in variance adds the coefficient variance across CPS resamples to the conditional two-way-clustered variance. [Table I.3](#) shows the full three-outcome transition family.

Entry retains the clearest association. For incremental Kelley, the CPS resampling standard deviation is 0.258 point and the plug-in standard error is 0.375 point, giving $p = 0.0052$ and three-outcome $q = 0.0157$. A t_{59} reference instead gives $p = 0.0070$ and $q = 0.0211$. Signed participation change has plug-in $q = 0.064$, while exit remains imprecise. The corresponding standing all-worker share and participation have plug-in p -values of 0.109 and 0.153. At 5% size and 80% power, the measured-signal minimum detectable differences are 0.409 point for the all-worker share and 1.79 points for participation ([Appendix Table G.2](#)). These are limits on detection, not evidence of zero portfolio responses.

The separate two-tail calculation keeps the 27,752 entry transitions and 60 assignment cells fixed and reconstructs the tail measures within each of 200 CPS draws. The pair is jointly distinguishable from zero under the plug-in calculation ($p = 0.0022$), although the upper coefficient alone has $p = 0.068$. In a distinct 99,999-draw null-imposed wild bootstrap at the 60 cells, the lower and upper two-component BH q -values are 0.019 and 0.087; signed composition has $p = 0.033$. The two exercises address different sources of uncertainty. Neither combines a CPS remeasurement draw with a resampled SIPP outcome process; the joint exercise below does so.

The behavior of the CPS draws is important for interpreting these calculations. Their mean coefficients are -2.138 , $+0.707$, and -2.076 points for the lower tail, upper tail, and signed composition, compared with the original estimates of -2.712 , $+1.917$, and -4.456 . The mean incremental-Kelley entry coefficient is likewise -0.650 rather than -1.047 point. The remeasurement distributions are not centered on the original estimates, and SIPP is held fixed. Their means, sign frequencies, and percentiles are therefore diagnostics, not bias-corrected estimates or confidence intervals.

Allowing both surveys to vary makes the inference sensitive to the treatment of this displacement. In 2,000 joint draws, the lower, upper, and signed-composition means are -1.906 , 0.698 , and -2.028 points. Their 95% percentile intervals all contain zero; for composition the interval is $[-5.755, 1.481]$. Basic intervals, which reflect the draw distribution around the original estimate, instead exclude zero for all three. The same disagreement appears for the smooth variance components, so a quantile at the zero-earnings boundary cannot be its only source. [Appendix Table I.4](#) reports both constructions, rather than selecting one after seeing its significance.

Table I.3. Portfolio transitions after rebuilding the CPS first stage

	(1) Signed participation change	(2) Zero to positive holdings	(3) Positive to zero holdings
Incremental Kelley orientation	−0.731** (0.360)	−1.047*** (0.375)	−0.813 (1.429)
Conditional S.E.	0.260	0.272	1.152
Conditional BH q	0.010	< 0.001	0.483
CPS coefficient S.D.	0.250	0.258	0.845
Plug-in p	0.043	0.005	0.569
Plug-in BH q	0.064	0.016	0.569
Lagged controls and fixed effects	Yes	Yes	Yes
Lagged participation control	Yes	No	No
Observations	35,112	27,752	7,360
People	23,170	19,300	5,308
CPS signal cells	60	60	60
CPS resamples	200	200	200

Notes: Each column is a separate regression using consecutive SIPP observations in 2019–2024, lagged signals and controls, and current-year SIPP weights. Entries are percentage points per one early-CPS cross-cell standard deviation. All columns include lagged zero share, quadratic mean and dispersion, household controls, and year, lagged industry, education, and age-band fixed effects. Parentheses report plug-in standard errors. The 200 deterministic CPS person-within-cell resamples rebuild weighted quantiles, Kish support selection, residualization, standardization, assignment to both SIPP observations, and the transition risk sets. Every draw produces all three estimates; supported cells range from 59 to 61. SIPP is held fixed; two-way clustering by person and lagged CPS cell supplies the conditional variance. Plug-in S.E.² = conditional S.E.² + CPS coefficient S.D.², with a two-sided standard-normal reference. This is a sensitivity, not a joint nonparametric bootstrap or inference for an error-free latent signal. Conditional and plug-in BH q -values each adjust exactly the three displayed transitions. Stars use unadjusted plug-in tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Adding CPS remeasurement can also reduce the variance of the refitted coefficients: for the upper quantile distance, the joint resampling standard deviation is 1.088 points, compared with 1.500 when only SIPP varies. Noisy regressors alter the conditional comparison and can contract both the center and spread of the refitted coefficients. A smaller original-point p -value from the joint draws is consequently not evidence of greater precision.

For the smooth variance components, a local influence-function calculation provides a separate check without refitting at a noisier set of cell moments. It includes uncertainty in the estimated centering mean and propagates all moment covariances through the same regression. In the original 60 cells, the resulting CPS standard error is 1.033 points, almost twice the 0.527-point standard deviation of nonlinear CPS redraws. Adding conditional two-way-clustered SIPP variance gives a total standard error of 1.243 points and three-comparison $q = 0.015$. The positive-destination comparison has $q = 0.049$. Replacing the SIPP component with the person-by-cell product-reweighting covariance gives $q = 0.066$ and $q = 0.105$, respectively. These are distinct first-order covariance approximations, not alternative estimates of the coefficient. Their disagreement with the nonlinear draws is a stability diagnostic, not a bias correction. The distinction between a conditional composition association and a stable population comparison is material to the paper’s conclusion.

I.5 Joint resampling and interval construction

The joint exercise fixes the SIPP observations, original cell support, and coefficient units. Within each original CPS cell, it draws multinomial counts over the observed people and rebuilds the weighted distribution. Independently, each SIPP person and lagged assignment cell receives a mean-one exponential weight. Their product multiplies the original SIPP survey weight, and the weighted regression is re-estimated. Positive multipliers preserve the original design support. The CPS-only, SIPP-only, and joint versions use matched underlying random draws, with 2,000 draws per version. For the quantile model, span and composition are exact fixed linear rotations of the jointly estimated lower and upper coefficients; each draw preserves fitted values and the joint Wald comparison.

Crossed reweighting allows dependence along the person and assignment-cell dimensions (Owen and Eckles, 2012). Its general conservative rationale does not imply exact finite-sample coverage for these nonlinear generated-regressor estimates. CPS resampling is conditional on the matched weighted sample, not a reconstruction of Census sampling strata and primary sampling units. Draw-specific support is recorded but does not redefine the original comparison.

For an original coefficient $\hat{b}$ and draw quantiles $q_{.025}^*, q_{.975}^*$, Table I.4 reports the percentile interval $[q_{.025}^*, q_{.975}^*]$ and the basic interval $[2\hat{b} - q_{.975}^*, 2\hat{b} - q_{.025}^*]$. It also reports the original-point normal-reference test using the draw standard deviation. The percentile construction inherits the displaced coefficient distribution; the basic construction reflects its displacement around the original estimate. Neither is designated as definitive when they disagree. The full replication output retains all engines, draw means, covariances, and 90% as well as 95% intervals.

I.6 Local propagation of moment uncertainty

The smooth partial moments admit a first-order check that does not refit the regression at a noisier distribution. Write $d = g - \mu$, $m_- = \mathbb{E}[\min(d, 0)]$, and $m_+ = \mathbb{E}[\max(d, 0)]$. The moment influence functions are

$$\psi_\mu(g) = d, \tag{I.1}$$

$$\psi_-(g) = \min(d, 0)^2 - V^- - 2m_-d, \tag{I.2}$$

$$\psi_+(g) = \max(d, 0)^2 - V^+ - 2m_+d, \tag{I.3}$$

$$\psi_0(g) = \mathbf{1}\{g = -2\} - p_0. \tag{I.4}$$

The last terms in the partial-moment expressions account for estimation of the centering mean. In a cell with weights w_i , let $h_i = w_i\psi(g_i)/\sum_j w_j$. The local covariance under the within-cell multinomial draw is $\sum_i h_i h_i' - (\sum_i h_i)(\sum_i h_i)'/n$. For the positive-destination distribution, excluded zero observations contribute zero and the denominator sums positive-destination weights. The regression Jacobian propagates the full covariance of these primitive moments, retaining the identities $V = V^- + V^+$ and $\sigma = \sqrt{V}$ and the original scales.

Table I.4. Joint remeasurement and the choice of interval

	(1)	(2)	(3)	(4)
Coefficient row	Original	Draw mean	Draw S.D.	Normal p
Interval row	Percentile	95%	Basic	95%
<i>Panel A. Quantile distances: baseline 60 cells</i>				
Lower distance	-2.712 [−4.462, 0.872]	-1.906	1.323 [−6.297, −0.962]	0.040
Upper distance	1.917 [−1.377, 2.870]	0.698	1.088 [0.964, 5.211]	0.078
Total tail span	2.721 [−4.667, 5.882]	0.456	2.645 [−0.440, 10.109]	0.304
Tail composition	-4.456 [−5.755, 1.481]	-2.028	1.872 [−10.394, −3.157]	0.017
<i>Panel B. Centered second partial moments: upside composition</i>				
Zero-inclusive, 60 cells	3.390 [−0.346, 3.932]	1.831	1.060 [2.847, 7.126]	0.001
Zero-inclusive, 59 cells	3.265 [−0.351, 3.952]	1.816	1.084 [2.577, 6.881]	0.003
Positive-to-positive, 59 cells	0.972 [−0.232, 1.343]	0.569	0.398 [0.601, 2.177]	0.014

Notes: Each contrast uses 2,000 paired CPS and SIPP draws with the original cell assignment, outcome rows, and coefficient units fixed. CPS persons are sampled within cells; independent mean-one exponential person and lagged-cell multipliers are multiplied into SIPP weights. Values are percentage points per original cross-cell S.D. Panel A retains quantile composition $D = L - U$; Panel B reports upside variance composition $C^{up} = V^+ - V^-$. Reversing either contrast would reverse estimates and interval endpoints without changing its two-sided tests; this is not independent evidence. The upper row gives the original estimate, redraw mean, redraw S.D., and the two-sided standard-normal reference $p = 2\Phi(-|\hat{\beta}/SD_b|)$. The lower row gives the percentile interval in columns (1)–(2) and the basic interval $[2\hat{\beta} - Q_{0.975}, 2\hat{\beta} - Q_{0.025}]$ in columns (3)–(4). The normal reference ignores mean displacement; basic intervals extrapolate it as a bias correction; percentile intervals retain it. All seven percentile intervals contain zero. These interval conventions are sensitivity diagnostics, not interchangeable confirmatory tests. The crossed-weight procedure has no finite-sample exact guarantee, and weighted quantiles have atoms. Panel B avoids quantile jumps but remains sensitive to noisy composition and nonlinear coefficient attenuation; adding CPS noise can reduce redraw variance. No interval convention is selected because it excludes zero. Panel A uses 27,752 transitions; Panel B uses 27,752, 27,545, and 27,545 transitions, respectively.

The resulting local CPS covariance is combined separately with the locked two-way SIPP covariance and with the SIPP-only crossed-reweighting covariance. Independence of the two surveys justifies adding their first-order contributions; it does not make either covariance approximation exact. The reported reference uses a t distribution with the number of original cells minus one degrees of freedom, followed by BH adjustment over the three composition comparisons within each covariance combination. Numerical perturbations of the actual weighted moment estimator and every cell-by-moment regression derivative verify the analytic calculation.

Local CPS standard deviations exceed the nonlinear CPS redraw standard deviations by factors of 1.96, 1.94, and 1.83 in the three constructions. The discrepancy persists for smooth moments and therefore cannot be attributed solely to quantile jumps. It indicates curvature in the downstream

regression when little composition variation remains after conditioning. These calculations do not correct measurement error, identify latent household risk, or supply a bias-corrected coefficient.

I.7 Separating earnings-zero states

The earnings-state exercise uses the same 59-cell common support for three comparisons. The first retains unconditional quantiles and the total zero share. The second replaces that share with separate probabilities of zero wage earnings with no weeks worked and zero wage earnings despite some weeks worked. The third uses positive-to-positive quantiles, positive-distribution mean and dispersion, and the same two unconditional zero-state probabilities. CPS annual weeks worked classify the states; neither is labeled an exogenous job-loss or business-loss shock.

All four displayed coefficients in each model—lower distance, upper distance, span, and signed composition—form one twelve-scalar family. The lower–upper vectors form a separate three-model joint family. The span–composition model is an exact reparameterization, not additional identifying evidence. Each of 500 CPS draws reconstructs the distribution and controls on the fixed common SIPP sample. Plug-in covariance adds the draw covariance to the conditional two-way SIPP covariance; scalar reference tests use t_{58} and joint tests use $F_{2,58}$.

The two unconditional vectors retain joint plug-in evidence, while no scalar coefficient survives its complete family adjustment. For positive-to-positive quantiles, incremental signed-composition reliability is -0.113 across 200 independent-person splits. Its near-zero portfolio coefficient is therefore not a persuasive test against economically relevant positive-earnings orientation. The positive upside-variance-composition coefficient in Table I.1 concerns a different, smoother feature of that distribution, whose reliability is higher but still limited. Neither construction resolves sorting into the assigned earnings environments.

I.8 The 2014 survey design

The 2014 ASEC randomly assigned households to traditional and redesigned income questionnaires. Both subsamples have independently calibrated population weights. Combining them without adjustment gives earnings origin year 2013 a 28.10% share of the original early-pair weight. Retaining only the redesigned subsample reduces that share to 14.81%, consistent with the official recommendation for income comparisons with 2015 onward.¹ I therefore examine this restriction using the original IPUMS wage amounts, without introducing the Census wage-version differences discussed below. Two separately declared sensitivities omit 2014 or halve all its weights. The latter produces a population-normalized average of the two questionnaire regimes; it does not harmonize their instruments or correct selection into longitudinal linkage.

Each convention reconstructs the three original variance-composition comparisons, retaining their original units and requiring effective support of 200 before estimating portfolio coefficients. The

¹See IPUMS CPS, “2014 3/8 ASEC File Information.” The redesign did not extend to the job-income questions, but its independently population-weighted samples still require an explicit pooling convention.

Table I.5. Earnings tails and the composition of destination zeros

Earnings distribution Zero-state controls	(1) Zero-inclusive Total zero rate	(2) Zero-inclusive Separate states	(3) Positive-to-positive Separate states
<i>Panel A. Lower and upper distances</i>			
Lower distance, L	-2.129 (1.097)	-2.331 (1.173)	1.536 (0.898)
BH q (12-test family)	0.112	0.112	0.139
Upper distance, U	2.312 (1.230)	2.367 (1.249)	1.891 (1.280)
BH q (12-test family)	0.112	0.112	0.193
<i>Panel B. Exact span and composition rotation</i>			
Tail span, $S = L + U$	3.996 (2.936)	4.002 (3.000)	3.193 (1.571)
BH q (12-test family)	0.204	0.204	0.112
Downside composition, $D = L - U$	-4.941 (2.040)	-5.124 (2.070)	0.007 (0.540)
BH q (12-test family)	0.112	0.112	0.990
Joint L, U BH q (three models)	0.019	0.019	0.140
CPS mean: upper	0.786	0.791	1.018
CPS mean: composition	-2.101	-2.122	-0.031
Lagged controls and fixed effects	Yes	Yes	Yes
Observations	27,545	27,545	27,545
CPS signal cells	59	59	59

Notes: All columns use the same 59 CPS cells and SIPP lagged-nonholder sample. Column (1) includes total destination-zero probability; columns (2)–(3) replace it with no-work/wage-zero and worked-without-wage probabilities, measured on the full positive-origin CPS frame using annual weeks worked. The latter is not an identified business-loss state. Column (3) estimates tails and mean/SD controls conditional on positive destination wages; no person is excluded from the SIPP risk set by that CPS condition. All columns retain quadratic growth mean/SD, lagged individual controls, and year, industry, education, and age-band fixed effects. Coefficients are percentage points per construction-specific cross-cell S.D.; scaling weights use full-frame Kish support in every column. Parentheses report plug-in S.E.s from conditional person-and-cell variance plus coefficient variance from 500 CPS redraws, with SIPP fixed. Stars, if any, use BH q over all 12 displayed scalar tests: *** $q < 0.01$, ** $q < 0.05$, * $q < 0.10$. Marginal tests use a t_{58} reference; joint tests use $F_{2,58}$ and a separate three-model BH family. The rotations are equivalent regressions, not independent evidence. Positive-only composition reliability is 0.038 before controls and -0.113 after residualization across 200 person splits, failing the declared measurement gate. Its imprecise coefficient does not establish economic irrelevance. The zero controls do not address nonlinear boundary mechanics or endogenous assignment.

redesigned-only construction leaves 59 cells in the zero-inclusive comparison and 58 cells on common positive-destination support. An additional accounting refit uses the original moments on each identical SIPP sample, separating sample changes from changes in the assigned distribution. Table I.6 reports the complete nine-model exercise. The redesigned-only comparisons form a three-test family; the other two conventions form a six-test sensitivity family. All-nine adjustments are retained as well. Local CPS covariance includes estimation of the centering mean; the broader SIPP covariance uses 1,000 matched positive person-by-cell reweightings.

All nine upside-composition point estimates remain positive. The redesigned-only zero-inclusive comparisons retain local CPS-plus-two-way evidence, including within the full nine-test family,

Table I.6. The 2014 ASEC design and variance composition

Earnings frame	Original moments	C^{up}	Per residual S.D.	q : two-way	q : product	Cells
<i>A. Redesigned 2014 subsample only</i>						
With zeros, full $N = 27,626$	3.410	3.338*** (1.194)	0.835	0.015	0.064	59
With zeros, common $N = 27,419$	3.276	3.192*** (1.196)	0.791	0.015	0.064	58
Positive, common $N = 27,419$	0.967	0.923* (0.467)	0.646	0.053	0.114	58
<i>B. Omit interview year 2014</i>						
With zeros, full $N = 27,752$	3.390	2.293* (1.342)	0.545	0.101	0.193	60
With zeros, common $N = 27,545$	3.265	2.426* (1.359)	0.575	0.101	0.193	59
Positive, common $N = 27,545$	0.972	0.699 (0.420)	0.501	0.101	0.193	59
<i>C. Half of every 2014 weight</i>						
With zeros, full $N = 27,752$	3.390	3.493** (1.380)	0.773	0.042	0.145	60
With zeros, common $N = 27,545$	3.265	3.425** (1.353)	0.759	0.042	0.145	59
Positive, common $N = 27,545$	0.972	1.013** (0.494)	0.640	0.089	0.193	59

Notes: The outcome is stock-and-mutual-fund entry from nonholding. $C^{\text{up}} = V^+ - V^-$ is upper minus lower centered semivariance. Models retain original IPUMS wages, V , $\sqrt{V}$, quadratic mean growth, lagged household controls, fixed effects, and SIPP weights. Positive/common uses positive CPS destinations and the same SIPP sample as zero-inclusive/common. Cells retain original support and Kish counts of at least 200. Coefficients are percentage points per original cross-cell S.D.; residual S.D.s remove all other regressors. Original-moment estimates refit the old assignment on identical rows and units; they are not additional tests. Parentheses combine local CPS influence-function uncertainty, including estimation of the centering mean, with person-and-cell-clustered SIPP covariance. Stars use unadjusted two-sided t references with cells minus one degrees of freedom: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The product column instead combines local CPS covariance with 1,000 crossed person-by-cell SIPP reweightings. BH q -values adjust the three Panel A tests and, separately, the six Panel B–C tests, fixed before these exploratory estimates. With a single nine-test adjustment, two-way q -values are 0.032 for zero-inclusive A/C, 0.079 for positive A/C, and 0.101 for B; corresponding product q -values are 0.109, 0.171, and 0.193. Neither covariance is full Census survey-design inference or a causal test.

whereas the positive-destination comparison is borderline. None of the redesigned-only results survives a 5% family threshold under the broader product covariance. Omitting 2014 attenuates the associations and does not preserve 5% family significance. Thus the original pooling convention does not explain the direction of the result, but these checks do not establish an inference-insensitive positive-earnings finding.

I.9 The same-source total-earnings comparison

The public Census files for interview years 2014–2019 preserve wages, net business income, net farm income, and their signed earnings total. I link these records to the retained IPUMS extract using the documented ASEC ordering method, checking record counts, person order, household

boundaries, identifiers, demographics, work measures, and weights.² All wage comparisons in 2015–2019 agree exactly. In 2014, all non-wage checks pass, but 232 high-income wage values differ from the retained IPUMS version. Each lies at a documented earnings-disclosure threshold; the particular source-version revisions remain unresolved. I preserve the failed exact-wage check, flag the record-linked release separately, and use Census wages and Census total earnings consistently. No IPUMS-wage/Census-business hybrid is constructed, and the discrepant records are not selectively discarded.

Before portfolio estimation, the design was amended to use the redesigned 2014 origin subsample and the same Census earnings source. These choices followed the source and sampling-design audit, not inspection of new portfolio coefficients. The restriction leaves 90,657 original wage-worker histories. Eleven have negative total origins and 61 have negative total destinations, with one overlap, leaving 90,586 nonnegative-destination histories with positive total origins. The matched positive-destination comparison contains 84,902 histories. Negative totals are retained in an exclusion ledger and never recoded to zero. Effective support of 200 in all four constructions leaves 58 common cells and fixes the SIPP sample before estimation. These populations exclude histories with negative total labor earnings and do not represent workers without positive origin wages. Business or farm losses offset by other earnings can remain in the sample.

The four regressions use identical SIPP observations and household controls but their own earnings-distribution mean, variance, standard deviation, composition, and zero rate when applicable. Original construction-specific scales remain fixed throughout inference. Across 200 paired person partitions, incremental-composition reliabilities are 0.197 and 0.276 for zero-inclusive wages and totals, and 0.239 and 0.249 for the corresponding positive-destination distributions. Partition ranges cross zero; the total-earnings measure is not a high-precision risk benchmark.

For the wage–total contrasts, local CPS covariance retains all four estimating-function contributions from the same person. The SIPP covariance similarly stacks the four person-level regression scores before clustering by person and assignment cell; its diagonal is independently verified against each conditional regression. A separate 1,000-draw product-reweighting covariance uses the same person and cell multipliers in all four models. Each total-earnings coefficient is rescaled into the corresponding wage-composition unit before differencing, and both covariance stages retain the cross-model terms. Four composition tests and two differences form separate complete families under each covariance convention. None of these comparisons is a causal decomposition or an equivalence test.

I.10 Dependence within households and sample units

Members of the same household can contribute earnings histories in different assignment cells, while several SIPP workers can share portfolio-reporting circumstances. I therefore aggregate the existing CPS coefficient influence functions by the validated longitudinal household identifier rather than

²IPUMS describes the procedure in “[IPUMS FAQs: How can I merge my IPUMS CPS file with an NBER CPS file?](#)” The replication retains each original Census file, its dictionary, source URL, checksums, and the complete row-level comparison.

treating all people as independent. The 89,951 retained CPS case-influence records map to 62,085 households. In SIPP, I replace person clustering with clustering by the validated original sample-unit lineage, retaining assignment-cell clustering and the corresponding lineage–cell intersection subtraction. The 27,419 records on 19,081 people belong to 13,849 lineages. A lineage can include different current addresses or spawned households; it is not labeled a current household. These changes retain all point estimates and allow covariance across members assigned to different cells.

Table I.7 compares the expanded covariance with the original two-way and product-reweighting results. The zero-inclusive and positive-destination four-test q -values change negligibly, and the wage–total differences remain imprecise. The complete outputs also expand each survey’s dependence group separately. Neither identifier reconstructs the Census strata and primary sampling units, and the exercise does not supply a full survey-design correction. It remains a local influence-function sensitivity conditional on the retained sample, not a correction for noisy latent regressors or endogenous sorting.

Table I.7. Household dependence and the composition–entry comparison

	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
Earnings Destinations				
Upside composition, C^{up}	3.272** (1.244)	2.203** (0.911)	0.982* (0.495)	1.036** (0.508)
Household/lineage covariance: BH q	0.038	0.038	0.052	0.052
Original person covariance: S.E.	1.235	0.908	0.495	0.508
Original person covariance: BH q	0.037	0.037	0.052	0.052
Original product covariance: S.E.	1.611	1.183	0.606	0.622
Original product covariance: BH q	0.111	0.111	0.111	0.111
Observations	27,419	27,419	27,419	27,419
SIPP people	19,081	19,081	19,081	19,081
SIPP sample-unit lineages	13,849	13,849	13,849	13,849
CPS assignment cells	58	58	58	58

Notes: The dependent variable is entry into reported stock-and-mutual-fund holdings. Coefficients, construction-specific units, samples, survey weights, risk controls (variance, standard deviation, quadratic mean growth, and zero incidence when applicable), household controls, and fixed effects are unchanged from Table I.2. Parentheses add CPS local influence-function covariance clustered by validated household to SIPP covariance clustered by validated sample-unit lineage and assignment cell. The CPS mapping contains 62,085 households; a SIPP lineage can contain multiple current or spawned households. Stars use unadjusted two-sided t_{57} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each displayed q row adjusts all four models. Original person-based and product-reweighting covariances remain separate benchmarks. These are local dependence sensitivities, not complete Census strata/PSU survey-design inference. The companion output retains both intermediate covariance combinations and both wage–total contrasts.

I.11 Uncertainty in the economic increment

The residual-standard-deviation magnitude already reported in the paper is estimated from both surveys. Write r_C for composition after projecting out every other regressor, $B = \sum_i w_i r_{Ci}^2$, $W_0 = \sum_i w_i$, and $A = \sum_i w_i r_{Ci} H_i$. Then the original slope is $\hat{\beta} = A/B$, whereas the reported magnitude is

$M = \hat{\beta}\sqrt{B/W_0} = A/\sqrt{BW_0}$. The normalization can be uncertain and correlated with the estimated slope.

I differentiate this entire expression, including all generated CPS risk controls, centering means, SIPP weights, and nuisance projections. CPS case influences are aggregated by household; SIPP influences are clustered by sample-unit lineage and assignment cell. For a multiplicative perturbation of SIPP weight i , with full-regression residual e_i and $s = \sqrt{B/W_0}$, the influence is

$$h_{Mi} = s \frac{w_i r_{Ci} e_i}{B} + \frac{\hat{\beta} s}{2} \left(\frac{w_i r_{Ci}^2}{B} - \frac{w_i}{W_0} \right).$$

The second term captures variation in the reported increment. The CPS derivative similarly includes the change in B induced by every generated regressor. Direct perturbations of primitive moments and individual weights verify both derivatives.

Table I.8 reports the resulting local intervals. The total-earnings magnitude of 0.684 point has a 95% t_{57} interval of [0.134, 1.235]. These supplementary tests must not be substituted for the original coefficient tests. Since $s > 0$, $M = 0$ and $\beta = 0$ are the same population null, but finite-sample delta-Wald tests change under nonlinear normalization. The original positive-destination coefficient family remains borderline, and the broader product-covariance comparison remains inconclusive. This audit supplies uncertainty for an existing economic magnitude, not an independent finding or a resolution of nonlinear remeasurement.

Table I.8. Uncertainty in the residual-composition magnitude

	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
Earnings Destinations				
Magnitude per residual S.D. (pp)	0.789*** (0.292)	0.684** (0.275)	0.647** (0.302)	0.655** (0.293)
Full-normalization 95% interval	[0.205, 1.372]	[0.134, 1.235]	[0.043, 1.251]	[0.069, 1.241]
Supplementary magnitude-test BH q	0.031	0.031	0.036	0.036
Holding residual S.D. fixed: S.E.	0.300	0.283	0.326	0.321
Original slope-test BH q	0.038	0.038	0.052	0.052
Broader original slope-test BH q	0.111	0.111	0.111	0.111

Notes: The displayed object is the existing entry coefficient times the weighted standard deviation of composition after removing all other regressors, not a newly estimated portfolio model. All four constructions retain the same 27,419 observations and 58 cells. Parentheses propagate both the coefficient and this normalization, including CPS moment means and nuisance projections and SIPP weights. Covariance permits CPS household and SIPP sample-unit-lineage/assignment-cell dependence. Stars and supplementary q -values use unadjusted two-sided t_{57} magnitude tests and their complete four-test BH family: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The coefficient and magnitude have the same population zero restriction, but finite-sample delta-Wald tests are not invariant to nonlinear normalization. These tests do not replace the original slope tests, whose household-expanded and broader product-covariance families are retained below. In particular, the positive-destination slope family remains borderline. Intervals are local delta-method approximations, not nonlinear bootstrap intervals or a correction for measurement error.

I.12 Observable progression and conventional skewness

Two outcome-free checks examine the economic content of the composition statistic on the original-linkage benchmark. First, predictable earnings progression can contribute to within-cell dispersion even after removing the cell mean. I predict arc growth out of fold using a fixed basis of origin-year wage-rank splines, exact age, sex, origin-year, industry, and education effects. Five folds keep all members of each CPS household together. Each of 200 separate household-disjoint reliability splits reconstructs ranks and refits all predictors within each half. The residuals are centered again within cells. Their zero control remains the probability of zero *raw earnings*, not the probability of a zero regression residual.

Panel A of Table I.9 shows positive held-out earnings-prediction gains in all four constructions. Adjusted composition remains incrementally reproducible, but only weakly so. The adjustment changes the positive-earnings rankings substantially. These residuals remove progression associated with observed covariates; they are not elicited forecasts or identified unanticipated or permanent shocks. Conditioning on a noisy origin wage can also absorb measurement-error mean reversion, and year effects remove common realized growth.

Second, a comparison beyond variance need not be a comparison beyond conventional skewness. Panel B holds the original mean, scale, zero, and assignment controls fixed and adds $m_3/V^{3/2}$. Conditional correlations with upside composition are high. Only zero-inclusive total earnings passes the fixed incremental-information gates under both the CPS support metric and the exact SIPP household-control-profiled metric. The wage-only and positive-destination constructions have negative incremental-composition reliability. Their failed gates are retained, not treated as evidence that conventional skewness is economically sufficient. The full outputs report the unnormalized third moment as well, without selecting a definition on portfolio significance.

Table I.9. Observable progression and incremental earnings shape

Earnings Destinations	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
<i>Panel A. Removing observable progression</i>				
Held-out growth MSE reduction (%)	8.1	10.6	23.0	22.5
Raw-adjusted composition correlation	0.945	0.923	0.595	0.598
Original incremental reliability	0.184	0.268	0.212	0.215
Adjusted incremental reliability	0.188	0.221	0.270	0.267
<i>Panel B. Composition given conventional skewness</i>				
Conditional correlation with skewness	0.901	0.904	0.970	0.967
Incremental reliability: CPS metric	-0.032	0.223	-0.115	-0.045
Incremental reliability: SIPP metric	-0.109	0.182	-0.136	-0.078
Passes both information gates	No	Yes	No	No

Notes: These are measurement diagnostics, not portfolio coefficients or significance tests. All constructions retain the same 58 cells. Panel A predicts arc growth in five household-disjoint folds using origin-year wage-rank splines, exact-age polynomials, sex, origin-year, industry, and education effects. Ranks and predictions are re-estimated independently in each of 200 household-disjoint reliability halves. Residual distributions are re-centered within cells; zero incidence remains the raw earnings-zero probability. Incremental reliability removes mean and squared mean, variance, standard deviation, zero incidence when applicable, and industry/education/age components using the declared CPS projection. Panel B additionally removes standardized third-moment skewness, $m_3/V^{3/2}$, using either fixed CPS support weights or the exact SIPP household-control-profiled metric without fitting portfolio outcomes. Its correlation row uses the latter metric and the lower-moment controls. Gates require at least ten residual degrees of freedom in the full sample and every half, one percent of metric-specific target variation remaining, finite results in all 200 splits, and positive Spearman-Brown reliability under both metrics. Negative reliability means no detected incremental signal, not negative population reliability. Partition ranges, retained in the outputs, are not sampling confidence intervals. Passing a gate does not establish a precise signal or a valid instrument.

I.13 Validated earnings links and observable progression

The progression comparison requires equal positive CPSIDV values at the two earnings endpoints. This identifier validates longitudinal demographic consistency using the underlying CPS rotation history.³ Among the eligible original histories in the matched 58 cells, the restriction retains 88,862 of 89,881 zero-inclusive histories and 83,372 of 84,287 positive-destination histories, in 61,480 source households. All 58 cells retain the required support. The original quantile and variance comparisons, local covariance estimates, and skewness comparison are not rebuilt by this restriction and remain explicitly separate benchmarks.

I refit the same fixed earnings-prediction basis on the retained histories before estimating the eight portfolio models. All four constructions retain positive held-out prediction gains, but incremental-composition reliability remains weak: 0.212, 0.215, 0.259, and 0.263 for zero-inclusive wages and totals and positive-destination wages and totals. All 200-split sensitivity ranges cross zero; these partition ranges are not confidence intervals. The four progression-adjusted models and their four raw-growth references use identical 58-cell support and 27,419 SIPP transitions. Each pair

³See IPUMS CPS, “[Validated Longitudinal Identifier](#).” Equality at two annual endpoints is the operational restriction here. An unequal identifier is not proof that this particular annual pair links different people.

retains the original-linkage raw-composition standard deviation for coefficient units, fixed throughout inference. The adjusted model conditions on its own residual-growth mean, squared mean, variance, and standard deviation, plus the probability of zero raw earnings when applicable. Its residual-standard-deviation magnitude is recomputed after removing every other regressor. A paired slope difference compares descriptive specifications with different conditioning distributions; it is not a causal mediation decomposition.

Exactly 1,000 matched draws rebuild both stages. One mean-one exponential multiplier per CPS household is shared across people, cells, and constructions. Multipliers are generated on the original ordered frame of 62,062 households and applied to the 61,480 retained households; the original random streams therefore also pair the two linkage versions exactly. Each draw reconstructs the weighted origin-year wage ranks, the fixed predictor basis, all five held-out fits, and the residual distributions. Original household-fold membership and spline knots remain fixed. Origin-covariate ranks use all folds, so this is not an untouched new-sample forecast. Independently, a SIPP sample-unit-lineage multiplier and an assignment-cell multiplier multiply the original survey weight; household controls are reprofiled under these weights. The original portfolio observations and supported cells are never selected again within a draw. Both earnings measures use the same multipliers within each support comparison.

The outputs retain CPS-only, SIPP-only, and joint versions of all eight models and all four adjusted-minus-row slope contrasts. Within each engine, adjusted coefficients, raw references, and contrasts form separate complete four-test BH families. Original-point t_{57} approximations use the draw standard deviation. Percentile and basic intervals are shown together, alongside draw means and displacement; the full outputs also retain 90% and 99% intervals and magnitude uncertainty. Neither the reference tests nor one favorable interval construction is treated as definitive when nonlinear remeasurement shifts the distribution. These are dependence sensitivities, not a full complex-survey bootstrap or a correction for endogenous assignment.

Table [I.10](#) reports the joint intervals for all eight slopes and four paired contrasts. Adjusted slopes remain positive, but none survives its four-test family adjustment. Every adjusted percentile interval includes zero; basic slope intervals exclude zero for both zero-inclusive constructions, barely so for total earnings. For the raw references, every percentile interval includes zero and every basic interval excludes it, with draw-mean attenuation of roughly 42–46%. Joint raw-coefficient standard deviations are smaller than their SIPP-only counterparts even though both stages vary. Rebuilding noisy regressors can contract the coefficient distribution rather than supply additional precision.

All paired adjusted-minus-row slope differences are negative at the original estimates but imprecise under the t , percentile, and basic 95% intervals. Their four-test q -values are 0.281. The adjusted residual-standard-deviation magnitudes are similarly inconclusive, with four-test q -values ranging from 0.445 to 0.498 and all 95% point-centered and percentile intervals containing zero. Only the zero-inclusive wage magnitude has a basic interval excluding zero; the corresponding total-earnings magnitude does not, despite its barely positive basic slope interval. Neither a statistically established attenuation nor a progression-adjusted population association follows from this exercise.

Table I.11 reports validated-minus-original-linkage slope differences using the same 1,000 draws. These paired differences retain covariance between the two source versions, rather than treating them as independent estimates. Every joint 95% point-centered, percentile, and basic interval includes zero. The saved results retain the corresponding own-residual-standard-deviation magnitude comparisons, with the same qualitative conclusion. Source validation therefore leaves the raw point pattern close to its predecessor but does not establish equivalence or resolve the uncertainty. The original-linkage progression results and all source-specific inference remain in the replication output.

Table I.10. Observable-progression adjustment: intervals and paired comparisons

	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
<i>Earnings Destinations</i>				
<i>Panel A. Adjusted composition slope</i>				
Slope	1.793* (1.041)	1.259 (0.981)	0.380 (0.499)	0.394 (0.520)
Separate four-test BH q	0.362	0.409	0.451	0.451
95% t_{57} interval	[-0.292, 3.877]	[-0.705, 3.222]	[-0.620, 1.379]	[-0.646, 1.435]
95% percentile interval	[-1.109, 3.025]	[-1.296, 2.508]	[-0.727, 1.256]	[-0.768, 1.321]
95% basic interval	[0.560, 4.694]	[0.009, 3.814]	[-0.497, 1.486]	[-0.532, 1.557]
Draw mean minus original point	-0.866	-0.659	-0.145	-0.148
<i>Panel B. Raw composition slope</i>				
Slope	3.242*** (1.067)	2.260** (0.858)	0.974** (0.403)	1.029** (0.412)
Separate four-test BH q	0.014	0.019	0.019	0.019
95% t_{57} interval	[1.105, 5.380]	[0.543, 3.978]	[0.166, 1.781]	[0.203, 1.855]
95% percentile interval	[-0.329, 3.851]	[-0.562, 2.972]	[-0.262, 1.357]	[-0.265, 1.420]
95% basic interval	[2.634, 6.814]	[1.549, 5.083]	[0.590, 2.210]	[0.638, 2.323]
Draw mean minus original point	-1.483	-1.013	-0.411	-0.444
<i>Panel C. Adjusted-minus-raw slope contrast</i>				
Paired slope difference	-1.450 (1.093)	-1.002 (0.920)	-0.594 (0.491)	-0.635 (0.504)
Separate four-test BH q	0.281	0.281	0.281	0.281
95% t_{57} interval	[-3.639, 0.739]	[-2.844, 0.841]	[-1.577, 0.389]	[-1.643, 0.374]
95% percentile interval	[-3.132, 1.395]	[-2.443, 1.156]	[-1.298, 0.578]	[-1.353, 0.649]
95% basic interval	[-4.294, 0.232]	[-3.159, 0.439]	[-1.765, 0.111]	[-1.918, 0.084]
Draw mean minus original point	0.617	0.354	0.266	0.296

Notes: Entries use Table N.1's fixed original raw-composition slope units (percentage points). All panels use 27,419 SIPP observations, 19,081 people, 13,849 lineages, and 58 cells. Parentheses are S.D.s from 1,000 paired joint full-pipeline draws. CPS weights are generated on the original 62,062-household random-number frame, then restricted to 61,480 validated source households; SIPP weights are lineage-by-cell products. Original-point-centered t_{57} tests are approximate and not null-imposed bootstrap tests; stars use strict unrounded thresholds: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each panel has a separate BH-four family. Percentile bounds are empirical 2.5th and 97.5th percentiles; basic bounds subtract their reversed order from twice the original point. All eight slope percentile intervals contain zero, whereas both zero-inclusive adjusted basic intervals exclude zero; no method is selected for its significance. Panel C retains paired cross-model covariance; these descriptive contrasts are not mediation effects. Appendix I.13 gives the procedure. Saved outputs retain all three source engines, 90/95/99% intervals, covariances, and own-S.D. magnitude inference. This 58-cell source repair does not rebuild original 60/59-cell estimates or original-link local inference.

Table I.11. Validated-link versus original-source portfolio associations

Earnings Destinations	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
<i>Panel A. Adjusted composition: validated minus original</i>				
Paired slope difference	−0.112 (0.174)	0.002 (0.163)	−0.116 (0.088)	−0.116 (0.093)
Separate four-test BH q	0.697	0.993	0.435	0.435
95% t_{57} interval	[−0.460, 0.236]	[−0.325, 0.328]	[−0.292, 0.060]	[−0.302, 0.070]
95% percentile interval	[−0.419, 0.279]	[−0.377, 0.261]	[−0.265, 0.101]	[−0.283, 0.110]
95% basic interval	[−0.503, 0.195]	[−0.257, 0.380]	[−0.334, 0.033]	[−0.341, 0.051]
Draw mean minus original point	0.061	−0.030	0.064	0.063
<i>Panel B. Raw composition: validated minus original</i>				
Paired slope difference	−0.029 (0.180)	0.057 (0.135)	−0.009 (0.065)	−0.007 (0.068)
Separate four-test BH q	0.916	0.916	0.916	0.916
95% t_{57} interval	[−0.389, 0.331]	[−0.214, 0.328]	[−0.139, 0.122]	[−0.143, 0.129]
95% percentile interval	[−0.332, 0.409]	[−0.218, 0.305]	[−0.124, 0.142]	[−0.127, 0.155]
95% basic interval	[−0.468, 0.273]	[−0.191, 0.332]	[−0.159, 0.107]	[−0.169, 0.112]
Draw mean minus original point	0.048	−0.024	0.009	0.010

Notes: The outcome, 27,419 SIPP rows, household controls, control specification, 58 assignment cells, and construction-specific original raw-composition units are fixed across source definitions. Generated risk controls are rebuilt for each source definition. Entries are validated-minus-original slopes in percentage points. Validated histories require equal positive endpoint CPSIDV; rejection does not prove that an original link is false. Both definitions use identical 1,000 CPS and SIPP multiplier streams, including the original 62,062-household CPS frame before restriction. Parentheses retain the full paired covariance, not the sum of marginal variances. Stars use unadjusted, original-point-centered approximate two-sided t_{57} tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, with separate raw/adjusted BH-four families. All three 95% interval methods are displayed; every joint source-difference interval includes zero. These are not equivalence tests or causal effects of linkage errors. Each version's own conditional-S.D. magnitude differences, all source engines, and complete 90/95/99% interval families are retained in the replication outputs. The adjusted positive-destination wage and total magnitude differences are −0.062 and −0.059 percentage points (joint BH $q = 0.507$ and 0.507). This repair concerns only the matched partial-moment models; it does not validate or replace the paper's original 60/59-cell quantile/variance models or original-link local-inference calculations.

I.14 Portfolio composition conditional on standardized skewness

The incremental-information results authorize one exploratory portfolio comparison: zero-inclusive total earnings. I add standardized skewness, $S_c = m_{3,c}/V_c^{3/2}$, to the existing composition model on the unchanged 58 cells and 27,419 entry transitions. Composition and every original risk control retain their original scales; skewness uses its full-sample cross-cell standard deviation, fixed throughout inference. Composition conditional on skewness and skewness conditional on composition form a symmetric two-test family. Their joint zero restriction is a separate $F_{2,57}$ test. No alternative third-moment regression is selected after inspecting these results.

Local CPS uncertainty propagates the centering mean through both shape statistics. For $x = g - \mu$, the third-central-moment influence is $x^3 - m_3 - 3Vx$; the chain rule for S_c also includes the variance influence. Aggregating all primitive influences by CPS household preserves their cross-cell and cross-moment covariance. SIPP covariance clusters by sample-unit lineage and assignment cell, including the intersection subtraction. A separate sensitivity combines the same local CPS covariance

with 1,000 matched SIPP lineage-by-cell positive reweightings. Both conventions retain the full two-coefficient covariance and testing family.

Table I.12. Total-earnings composition conditional on conventional skewness

Dependent variable	(1)	(2)
	Primary	Broader product
Two-survey covariance		
Upside composition, C^{up}	3.782* (2.261)	3.782 (3.045)
Symmetric marginal-family BH q	0.200	0.439
Standardized skewness, $m_3/V^{3/2}$	-1.234 (1.705)	-1.234 (2.315)
Symmetric marginal-family BH q	0.472	0.596
Joint $F(2, 57)$ statistic	3.471	1.996
Joint zero-restriction p -value	0.038	0.145
Risk-scale and zero-incidence controls	Yes	Yes
Household controls and fixed effects	Yes	Yes
Observations	27,419	27,419
SIPP people	19,081	19,081
SIPP sample-unit lineages	13,849	13,849
CPS households in influence ledger	62,085	62,085
CPS assignment cells	58	58

Notes: Both columns report the same single regression on the zero-inclusive total-labor-earnings frame of Table I.2. Composition and standardized skewness enter jointly, alongside variance, standard deviation, quadratic mean growth, zero probability, lagged household controls, and year, industry, education, and age-band fixed effects. Coefficients are percentage points per fixed full-CPS cross-cell standard deviation; skewness uses weights capped at a person Kish count of 1,000. Parentheses in column (1) combine CPS local influence-function covariance clustered globally by household with SIPP sample-unit-lineage/assignment-cell covariance, subtracting lineage-cell intersections. The CPS calculation propagates estimation of the centering mean and all moment covariances. Column (2) replaces the SIPP component with covariance from 1,000 lineage-by-cell product reweightings, retaining the same CPS component. Stars use each column's unadjusted two-sided t_{57} test of a zero coefficient and strict unrounded thresholds: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The two marginal coefficients form one symmetric BH family within each column; the joint F test is separate. Only this earnings construction passed the prior outcome-free incremental-skewness information gates; all three other construction failures remain visible in Table I.9. This is an exploratory gate-selected comparison, not a correction for the project's complete search history. Conditional-only inference and all product draws remain in the companion output.

Neither individual coefficient survives its two-test adjustment. The primary joint test rejects at 5%, but the broader joint test does not. Thus this exercise does not establish an incremental composition association beyond conventional skewness. The failed information gates in the other three constructions remain part of the result; they are not replaced by their conditional portfolio significance.

I.15 An early-SIPP total-earnings validation attempt

An additional measurement check uses annual total earnings from the four waves of the 2014 SIPP panel. It retains complete annual earnings records and adjacent within-panel transitions, with positive origins and nonnegative destinations, and compares total earnings with job-1 earnings on matched records. Negative monthly amounts enter the annual sums; negative annual destinations are recorded separately and excluded, not recoded to zero. The resulting sample contains 29,307 transitions on 15,785 people in the original 60 cells. Repeated transitions remain together in person splits, and effective support uses each person's summed weight. Only 18 cells meet the original effective-support threshold of 200. On this overlap, CPS wage variance correlates with SIPP total-earnings variance at 0.577, but upside variance composition correlates at only 0.162. Its raw split reliability is 0.336 across 200 person partitions. The declared conditional projection leaves too few residual degrees of freedom to assess incremental transport. I therefore do not use this exercise as independent validation or proceed to portfolio regressions with these measures. Historical SIPP total earnings also contain a documented business-profit collection and processing problem, preventing their treatment as an error-free benchmark.⁴

⁴The Census Bureau's March 24, 2022 user note, "[Business Profits Data Collection and Processing Issue](#)," describes the possible repeated reporting of annual profits across employment spells and its effects on total earnings. No assumed correction is applied here.

I.16 Own prior earnings: construction and complete comparisons

The own-history comparison conditions on a worker's earnings before the baseline portfolio observation. If destination year is d , nonholding is measured in $d - 1$ and the earnings years are $d - 3$ and $d - 2$. The supported panel paths produce destinations in 2020, 2022, 2023, and 2024; no 2021 destination is supplied by a substitute path. The primary sample requires self or proxy interview status in all 24 prior monthly records. These are not 24 independent interviews or a guarantee of unimputed earnings. Complete histories retain 6,639 of the original 27,419 nonholder transitions, on 6,639 workers and 5,291 sample-unit lineages in 58 cells. Original destination survey weights are retained; they are not history-longitudinal weights.

I sum the twelve monthly total-person earnings recodes, TPEARN, within each prior calendar year. They include all jobs and signed business profits or losses, not just first-listed-job wages or household disposable income. Exact nominal sums determine zero and negative indicators. Native missing values do not become zeros; a structural zero is used only when the source record establishes a valid no-job state. Incomplete years are excluded by the fixed support rule, not annualized. Amounts can contain allocation, suppressed-component, disclosure, and business-profit processing error. I preserve the source restrictions and warnings rather than treating an observed recode as clean earnings.

Annual CPI-U converts the two totals to 2022 dollars. For levels Y_0, Y_1 , the primary block contains $\text{asinh}(Y_0/10,000)$, $\text{asinh}(Y_1/10,000)$, signed arc change $2(Y_1 - Y_0)/(|Y_1| + |Y_0|)$, and separate zero and negative indicators at both dates. The arc is zero when both totals are zero and otherwise uses the stated denominator. It is bounded signed growth, not percentage growth or an identified surprise. The decline-kink sensitivity replaces the arc by $\max\{\text{asinh}(Y_0/10,000) - \text{asinh}(Y_1/10,000), 0\}$, retaining the other controls. Neither block estimates a personal tail distribution, earnings persistence, or a loss followed by recovery.

Two fixed source sensitivities accompany the primary sample. The panel-2022 processing sensitivity retains 1,275 primary histories, with earnings in 2021–2022, baseline holdings in 2023, and entry in 2024. Its post-fix processing label does not certify error-free income. The Type-Z-inclusive sample contains 7,406 domain-valid histories, including legal Type-Z months, and nests the primary sample. It changes interview coverage, not the accuracy of reported amounts. Both sensitivities retain 58 assignment cells. Sample membership and both history blocks were fixed before the new amounts and portfolio estimates were inspected; this is an exploratory follow-up, not an original preregistration.

For each of the three samples, baseline, signed-change, and decline-kink specifications retain all eight raw/adjusted wage/total and zero-inclusive/positive CPS constructions. The separate eight-model full-sample reference makes 80 models in total. Within every history comparison, observations, destination weights and all original noncomposition controls remain the same; the history block is added without replacing variance, standard deviation, mean, mean squared, applicable zero incidence, personal controls or fixed effects. Positive CPS destinations restrict the source earnings distribution, not the recipient history sample. The original composition units are held fixed, and each model's residual-standard-deviation magnitude is recomputed after profiling all its controls. The 48

augmented-minus-matched-baseline slope comparisons retain their joint cross-model covariance; their differences are not mediation shares.

Exactly 1,000 matched draws rebuild CPS ranks, prediction fits and moments while allowing dependence within CPS households and within SIPP lineages and assignment cells. CPS-only, SIPP-only and joint calculations retain the original random-weight frames before recipient subsetting. The joint calculation is the primary uncertainty summary. Stars use the declared approximate original-point two-sided t_{57} tests; BH8 adjusts all eight raw/adjusted constructions within each sample, control block and target. Paired differences have their own eight-test families. The original four-construction adjustments remain in the saved output for continuity. Point-centered, percentile and basic intervals are retained at 90%, 95% and 99%; no interval method is selected for favorable significance.

Tables [I.13](#) and [I.14](#) report every joint slope and paired 95% interval. All 80 slope percentile intervals include zero. All 48 paired changes include zero under each of the three 95% methods. The primary signed-change slope family has no BH8 rejection; its eight paired q -values are 0.994. Replacing the signed change by a decline kink gives the same qualitative limitation. In both the panel-2022 and Type-Z-inclusive sensitivities, progression-adjusted positive-destination slopes are negative but highly imprecise. These signs are retained, not interpreted as opposite effects. None of the eight primary signed-change comparisons satisfies the declared positive-reporting screen requiring a positive primary point, BH8 $q < 0.05$, and positive lower bounds under all three joint 95% methods. This screen is a reporting rule, not a separately calibrated hypothesis test. The history controls leave the primary raw point pattern nearly unchanged but do not establish equivalence, a stable population association, or a causal/perception mechanism.

Table I.13. Prior earnings and entry: complete slope intervals

Notes: All entries use 1,000 joint CPS-household/SIPP-lineage-by-cell reweighted draws and report percentage points per original raw-source CPS composition SD, unchanged across control blocks and within raw/adjusted pairs. Point-centered intervals use the original point, draw SD and t_{57} critical value. Percentile intervals use empirical 2.5th/97.5th percentiles; basic intervals reflect them about the original point. BH8 uses unadjusted two-sided point-centered reference probabilities within each sample/control-block slope family of eight raw/adjusted constructions; paired differences form separate families. These approximate positive-weight sensitivity tests do not certify complex-survey coverage. No interval is selected by its sign. Full 90/95/99 intervals, CPS-only/SIPP-only/joint engines, own-SD magnitudes and covariance arrays remain in the replication output.

All specifications retain baseline controls and their complete nuisance span. Both history blocks add two asinh annual total-earnings levels and four zero/negative indicators; the primary block adds signed arc change and the sensitivity block adds a decline kink. CPS positive-destination restrictions do not change recipient rows. Original destination weights are not history-longitudinal weights. Only positive-support destination years are listed. The full reference is not a matched no-history comparator. Processing and Type-Z sensitivities cannot substitute for the primary comparison; separate BH8 families do not authorize an omnibus claim. Neither histories nor paired differences identify beliefs, causal responses or mediation shares.

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Full-sample reference: 27,419 nonholder transitions; 19,081 people; 13,849 lineages; 58 cells. Destination years: 2020, 2021, 2022, 2023, 2024. Original baseline-nonholder sample; a selection reference, not a matched baseline.				
<i>Full-sample reference; baseline controls; raw growth</i>				
Wages, zero-inclusive	[1.105, 5.380]	[-0.329, 3.851]	[2.634, 6.814]	0.029
Total earnings, zero-inclusive	[0.543, 3.978]	[-0.562, 2.972]	[1.549, 5.083]	0.038
Wages, positive	[0.166, 1.781]	[-0.262, 1.357]	[0.590, 2.210]	0.038
Total earnings, positive	[0.203, 1.855]	[-0.265, 1.420]	[0.638, 2.323]	0.038
<i>Full-sample reference; baseline controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.292, 3.877]	[-1.109, 3.025]	[0.560, 4.694]	0.145
Total earnings, zero-inclusive	[-0.705, 3.222]	[-1.296, 2.508]	[0.009, 3.814]	0.273
Wages, positive	[-0.620, 1.379]	[-0.727, 1.256]	[-0.497, 1.486]	0.451
Total earnings, positive	[-0.646, 1.435]	[-0.768, 1.321]	[-0.532, 1.557]	0.451
Primary histories: 6,639 nonholder transitions; 6,639 people; 5,291 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Self/proxy interview status in all 24 prior monthly records; not a guarantee of unimputed earnings.				
<i>Primary histories; baseline controls; raw growth</i>				
Wages, zero-inclusive	[0.256, 8.282]	[-1.601, 6.562]	[1.976, 10.139]	0.216
Total earnings, zero-inclusive	[-0.691, 5.751]	[-1.645, 4.797]	[0.263, 6.705]	0.216
Wages, positive	[-0.309, 2.680]	[-0.788, 2.232]	[0.138, 3.159]	0.216
Total earnings, positive	[-0.372, 2.694]	[-0.864, 2.282]	[0.039, 3.185]	0.216
<i>Primary histories; baseline controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.826, 6.587]	[-1.855, 5.491]	[0.270, 7.616]	0.216
Total earnings, zero-inclusive	[-1.483, 5.344]	[-2.169, 4.467]	[-0.606, 6.030]	0.350
Wages, positive	[-1.675, 1.875]	[-1.578, 1.926]	[-1.725, 1.779]	0.910
Total earnings, positive	[-1.480, 2.199]	[-1.527, 2.123]	[-1.404, 2.245]	0.797
<i>Primary histories; signed-change history controls; raw growth</i>				
Wages, zero-inclusive	[0.271, 8.270]	[-1.641, 6.524]	[2.016, 10.181]	0.217
Total earnings, zero-inclusive	[-0.697, 5.746]	[-1.668, 4.821]	[0.228, 6.717]	0.217
Wages, positive	[-0.308, 2.677]	[-0.766, 2.216]	[0.153, 3.135]	0.217
Total earnings, positive	[-0.375, 2.693]	[-0.847, 2.255]	[0.063, 3.165]	0.217

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Continued

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 <i>q</i>
<i>Primary histories; signed-change history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.770, 6.647]	[-1.824, 5.524]	[0.354, 7.702]	0.217
Total earnings, zero-inclusive	[-1.374, 5.463]	[-2.096, 4.538]	[-0.450, 6.185]	0.315
Wages, positive	[-1.631, 1.923]	[-1.566, 1.981]	[-1.689, 1.859]	0.870
Total earnings, positive	[-1.442, 2.243]	[-1.490, 2.192]	[-1.391, 2.291]	0.760
<i>Primary histories; decline-kink history controls; raw growth</i>				
Wages, zero-inclusive	[0.265, 8.267]	[-1.602, 6.546]	[1.987, 10.134]	0.215
Total earnings, zero-inclusive	[-0.685, 5.754]	[-1.633, 4.818]	[0.251, 6.703]	0.215
Wages, positive	[-0.308, 2.674]	[-0.762, 2.221]	[0.146, 3.129]	0.215
Total earnings, positive	[-0.370, 2.692]	[-0.848, 2.244]	[0.078, 3.170]	0.215
<i>Primary histories; decline-kink history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.808, 6.639]	[-1.885, 5.524]	[0.306, 7.715]	0.215
Total earnings, zero-inclusive	[-1.415, 5.442]	[-2.142, 4.529]	[-0.502, 6.169]	0.326
Wages, positive	[-1.655, 1.906]	[-1.572, 1.954]	[-1.702, 1.824]	0.888
Total earnings, positive	[-1.465, 2.224]	[-1.502, 2.154]	[-1.395, 2.261]	0.779
Panel-2022 processing sensitivity: 1,275 nonholder transitions; 1,275 people; 1,031 lineages; 58 cells. Destination years: 2024. Primary panel-2022 histories only; no processing partition is certified clean.				
<i>Panel-2022 processing sensitivity; baseline controls; raw growth</i>				
Wages, zero-inclusive	[-2.626, 13.754]	[-5.149, 11.349]	[-0.221, 16.276]	0.911
Total earnings, zero-inclusive	[-3.956, 8.976]	[-4.793, 8.192]	[-3.172, 9.813]	0.911
Wages, positive	[-2.177, 3.915]	[-2.362, 3.623]	[-1.885, 4.101]	0.911
Total earnings, positive	[-2.514, 3.722]	[-2.634, 3.570]	[-2.363, 3.841]	0.911
<i>Panel-2022 processing sensitivity; baseline controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-3.782, 10.921]	[-4.460, 9.771]	[-2.632, 11.599]	0.911
Total earnings, zero-inclusive	[-5.747, 7.620]	[-6.332, 7.248]	[-5.375, 8.205]	0.911
Wages, positive	[-3.687, 3.294]	[-3.312, 3.483]	[-3.876, 2.919]	0.911
Total earnings, positive	[-3.899, 3.332]	[-3.380, 3.737]	[-4.303, 2.814]	0.911
<i>Panel-2022 processing sensitivity; signed-change history controls; raw growth</i>				
Wages, zero-inclusive	[-2.467, 13.915]	[-4.738, 11.250]	[0.198, 16.185]	0.862
Total earnings, zero-inclusive	[-3.796, 9.212]	[-4.745, 8.202]	[-2.786, 10.160]	0.862
Wages, positive	[-2.078, 4.023]	[-2.257, 3.675]	[-1.730, 4.202]	0.862
Total earnings, positive	[-2.394, 3.856]	[-2.466, 3.692]	[-2.230, 3.928]	0.862
<i>Panel-2022 processing sensitivity; signed-change history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-3.125, 11.469]	[-4.220, 10.070]	[-1.726, 12.564]	0.862
Total earnings, zero-inclusive	[-5.106, 8.159]	[-5.623, 7.521]	[-4.467, 8.676]	0.862
Wages, positive	[-3.592, 3.348]	[-3.242, 3.487]	[-3.731, 2.998]	0.944
Total earnings, positive	[-3.820, 3.367]	[-3.359, 3.650]	[-4.102, 2.906]	0.944
<i>Panel-2022 processing sensitivity; decline-kink history controls; raw growth</i>				
Wages, zero-inclusive	[-2.608, 13.765]	[-4.856, 11.346]	[-0.189, 16.014]	0.880
Total earnings, zero-inclusive	[-3.860, 9.132]	[-4.672, 8.261]	[-2.989, 9.944]	0.880
Wages, positive	[-2.105, 3.978]	[-2.252, 3.766]	[-1.893, 4.125]	0.880
Total earnings, positive	[-2.423, 3.812]	[-2.499, 3.549]	[-2.161, 3.887]	0.880
<i>Panel-2022 processing sensitivity; decline-kink history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-3.226, 11.429]	[-4.428, 9.996]	[-1.792, 12.632]	0.880
Total earnings, zero-inclusive	[-5.172, 8.100]	[-5.953, 7.402]	[-4.474, 8.882]	0.880
Wages, positive	[-3.594, 3.344]	[-3.239, 3.532]	[-3.782, 2.989]	0.943
Total earnings, positive	[-3.822, 3.361]	[-3.364, 3.700]	[-4.160, 2.903]	0.943

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Continued

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 <i>q</i>
Type-Z-inclusive histories: 7,406 nonholder transitions; 7,406 people; 5,672 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Domain-valid histories including legal Type-Z months; primary histories are nested within this sample.				
<i>Type-Z-inclusive histories; baseline controls; raw growth</i>				
Wages, zero-inclusive	[-0.970, 7.389]	[-2.333, 5.953]	[0.466, 8.752]	0.482
Total earnings, zero-inclusive	[-1.607, 5.108]	[-2.381, 4.363]	[-0.862, 5.881]	0.482
Wages, positive	[-0.695, 2.419]	[-1.088, 2.050]	[-0.325, 2.813]	0.482
Total earnings, positive	[-0.730, 2.456]	[-1.193, 2.075]	[-0.349, 2.919]	0.482
<i>Type-Z-inclusive histories; baseline controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.838, 6.731]	[-1.951, 5.528]	[0.364, 7.844]	0.482
Total earnings, zero-inclusive	[-2.002, 4.876]	[-2.648, 4.054]	[-1.179, 5.522]	0.541
Wages, positive	[-2.052, 1.545]	[-1.881, 1.725]	[-2.232, 1.374]	0.890
Total earnings, positive	[-1.879, 1.845]	[-1.842, 1.936]	[-1.969, 1.808]	0.986
<i>Type-Z-inclusive histories; signed-change history controls; raw growth</i>				
Wages, zero-inclusive	[-0.973, 7.380]	[-2.434, 5.884]	[0.523, 8.841]	0.487
Total earnings, zero-inclusive	[-1.621, 5.097]	[-2.419, 4.366]	[-0.891, 5.895]	0.487
Wages, positive	[-0.709, 2.402]	[-1.131, 2.038]	[-0.344, 2.825]	0.487
Total earnings, positive	[-0.742, 2.443]	[-1.222, 2.094]	[-0.393, 2.923]	0.487
<i>Type-Z-inclusive histories; signed-change history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.821, 6.740]	[-1.965, 5.642]	[0.277, 7.884]	0.487
Total earnings, zero-inclusive	[-1.935, 4.915]	[-2.597, 3.974]	[-0.994, 5.577]	0.516
Wages, positive	[-2.033, 1.563]	[-1.904, 1.725]	[-2.195, 1.435]	0.908
Total earnings, positive	[-1.864, 1.857]	[-1.882, 1.924]	[-1.931, 1.875]	0.997
<i>Type-Z-inclusive histories; decline-kink history controls; raw growth</i>				
Wages, zero-inclusive	[-0.968, 7.357]	[-2.396, 5.785]	[0.605, 8.785]	0.481
Total earnings, zero-inclusive	[-1.602, 5.092]	[-2.361, 4.310]	[-0.820, 5.851]	0.481
Wages, positive	[-0.695, 2.404]	[-1.144, 1.996]	[-0.288, 2.852]	0.481
Total earnings, positive	[-0.728, 2.443]	[-1.197, 2.074]	[-0.359, 2.912]	0.481
<i>Type-Z-inclusive histories; decline-kink history controls; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.901, 6.662]	[-2.000, 5.512]	[0.248, 7.760]	0.481
Total earnings, zero-inclusive	[-1.974, 4.880]	[-2.617, 4.003]	[-1.097, 5.523]	0.533
Wages, positive	[-2.041, 1.557]	[-1.866, 1.710]	[-2.195, 1.381]	0.901
Total earnings, positive	[-1.872, 1.849]	[-1.874, 1.936]	[-1.958, 1.851]	0.990

Table I.14. Prior earnings and entry: matched slope changes

Notes: All entries use 1,000 joint CPS-household/SIPP-lineage-by-cell reweighted draws and report percentage points per original raw-source CPS composition SD, unchanged across control blocks and within raw/adjusted pairs. Point-centered intervals use the original point, draw SD and t_{57} critical value. Percentile intervals use empirical 2.5th/97.5th percentiles; basic intervals reflect them about the original point. BH8 uses unadjusted two-sided point-centered reference probabilities within each sample/control-block slope family of eight raw/adjusted constructions; paired differences form separate families. These approximate positive-weight sensitivity tests do not certify complex-survey coverage. No interval is selected by its sign. Full 90/95/99 intervals, CPS-only/SIPP-only/joint engines, own-SD magnitudes and covariance arrays remain in the replication output.

All specifications retain baseline controls and their complete nuisance span. Both history blocks add two asinh annual total-earnings levels and four zero/negative indicators; the primary block adds signed arc change and the sensitivity block adds a decline kink. CPS positive-destination restrictions do not change recipient rows. Original destination weights are not history-longitudinal weights. Only positive-support destination years are listed. The full reference is not a matched no-history comparator. Processing and Type-Z sensitivities cannot substitute for the primary comparison; separate BH8 families do not authorize an omnibus claim. Neither histories nor paired differences identify beliefs, causal responses or mediation shares.

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 q
Primary histories: 6,639 nonholder transitions; 6,639 people; 5,291 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Self/proxy interview status in all 24 prior monthly records; not a guarantee of unimputed earnings.				
<i>Primary histories; signed-change history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-0.292, 0.294]	[-0.296, 0.310]	[-0.308, 0.298]	0.994
Total earnings, zero-inclusive	[-0.242, 0.232]	[-0.248, 0.239]	[-0.249, 0.237]	0.994
Wages, positive	[-0.110, 0.108]	[-0.117, 0.107]	[-0.109, 0.115]	0.994
Total earnings, positive	[-0.115, 0.111]	[-0.118, 0.110]	[-0.114, 0.114]	0.994
<i>Primary histories; signed-change history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.229, 0.346]	[-0.236, 0.353]	[-0.236, 0.352]	0.994
Total earnings, zero-inclusive	[-0.146, 0.374]	[-0.172, 0.343]	[-0.115, 0.400]	0.994
Wages, positive	[-0.092, 0.184]	[-0.102, 0.178]	[-0.086, 0.194]	0.994
Total earnings, positive	[-0.102, 0.184]	[-0.104, 0.182]	[-0.100, 0.186]	0.994
<i>Primary histories; decline-kink history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-0.297, 0.291]	[-0.298, 0.319]	[-0.324, 0.292]	0.997
Total earnings, zero-inclusive	[-0.235, 0.245]	[-0.244, 0.260]	[-0.250, 0.254]	0.997
Wages, positive	[-0.111, 0.107]	[-0.115, 0.111]	[-0.114, 0.112]	0.997
Total earnings, positive	[-0.114, 0.114]	[-0.112, 0.114]	[-0.114, 0.113]	0.997
<i>Primary histories; decline-kink history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.264, 0.333]	[-0.281, 0.351]	[-0.282, 0.351]	0.997
Total earnings, zero-inclusive	[-0.187, 0.354]	[-0.217, 0.340]	[-0.174, 0.384]	0.997
Wages, positive	[-0.117, 0.168]	[-0.122, 0.168]	[-0.117, 0.173]	0.997
Total earnings, positive	[-0.130, 0.170]	[-0.126, 0.178]	[-0.138, 0.166]	0.997
Panel-2022 processing sensitivity: 1,275 nonholder transitions; 1,275 people; 1,031 lineages; 58 cells. Destination years: 2024. Primary panel-2022 histories only; no processing partition is certified clean.				
<i>Panel-2022 processing sensitivity; signed-change history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-0.961, 1.281]	[-0.836, 1.377]	[-1.056, 1.156]	0.827
Total earnings, zero-inclusive	[-0.726, 1.121]	[-0.733, 1.155]	[-0.760, 1.128]	0.827
Wages, positive	[-0.319, 0.526]	[-0.366, 0.490]	[-0.284, 0.572]	0.827
Total earnings, positive	[-0.316, 0.570]	[-0.369, 0.532]	[-0.278, 0.623]	0.827

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Continued

CPS earnings construction	Point-centered 95%	Percentile 95%	Basic 95%	BH8 <i>q</i>
<i>Panel-2022 processing sensitivity; signed-change history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.392, 1.597]	[-0.538, 1.489]	[-0.284, 1.743]	0.827
Total earnings, zero-inclusive	[-0.315, 1.496]	[-0.471, 1.282]	[-0.102, 1.651]	0.827
Wages, positive	[-0.424, 0.573]	[-0.388, 0.593]	[-0.443, 0.537]	0.827
Total earnings, positive	[-0.464, 0.578]	[-0.386, 0.610]	[-0.496, 0.500]	0.827
<i>Panel-2022 processing sensitivity; decline-kink history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-1.042, 1.072]	[-0.874, 1.217]	[-1.188, 0.904]	0.978
Total earnings, zero-inclusive	[-0.768, 1.019]	[-0.739, 1.054]	[-0.803, 0.990]	0.952
Wages, positive	[-0.337, 0.472]	[-0.338, 0.473]	[-0.338, 0.472]	0.952
Total earnings, positive	[-0.338, 0.519]	[-0.356, 0.515]	[-0.334, 0.537]	0.952
<i>Panel-2022 processing sensitivity; decline-kink history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.458, 1.523]	[-0.626, 1.418]	[-0.353, 1.691]	0.952
Total earnings, zero-inclusive	[-0.352, 1.408]	[-0.523, 1.156]	[-0.101, 1.579]	0.952
Wages, positive	[-0.412, 0.555]	[-0.387, 0.565]	[-0.422, 0.530]	0.952
Total earnings, positive	[-0.449, 0.555]	[-0.421, 0.580]	[-0.474, 0.526]	0.952
Type-Z-inclusive histories: 7,406 nonholder transitions; 7,406 people; 5,672 lineages; 58 cells. Destination years: 2020, 2022, 2023, 2024. Domain-valid histories including legal Type-Z months; primary histories are nested within this sample.				
<i>Type-Z-inclusive histories; signed-change history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-0.293, 0.280]	[-0.275, 0.287]	[-0.300, 0.263]	0.965
Total earnings, zero-inclusive	[-0.256, 0.231]	[-0.247, 0.248]	[-0.272, 0.222]	0.965
Wages, positive	[-0.124, 0.094]	[-0.114, 0.099]	[-0.129, 0.083]	0.965
Total earnings, positive	[-0.126, 0.100]	[-0.115, 0.101]	[-0.126, 0.089]	0.965
<i>Type-Z-inclusive histories; signed-change history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.262, 0.288]	[-0.282, 0.295]	[-0.269, 0.308]	0.965
Total earnings, zero-inclusive	[-0.202, 0.307]	[-0.214, 0.295]	[-0.189, 0.319]	0.965
Wages, positive	[-0.120, 0.158]	[-0.115, 0.167]	[-0.128, 0.154]	0.965
Total earnings, positive	[-0.130, 0.156]	[-0.128, 0.161]	[-0.135, 0.154]	0.965
<i>Type-Z-inclusive histories; decline-kink history controls minus matched baseline; raw growth</i>				
Wages, zero-inclusive	[-0.330, 0.300]	[-0.334, 0.313]	[-0.343, 0.304]	0.969
Total earnings, zero-inclusive	[-0.262, 0.252]	[-0.260, 0.266]	[-0.277, 0.250]	0.969
Wages, positive	[-0.124, 0.108]	[-0.117, 0.118]	[-0.134, 0.101]	0.969
Total earnings, positive	[-0.127, 0.115]	[-0.120, 0.118]	[-0.129, 0.108]	0.969
<i>Type-Z-inclusive histories; decline-kink history controls minus matched baseline; progression-adjusted growth</i>				
Wages, zero-inclusive	[-0.371, 0.239]	[-0.337, 0.276]	[-0.408, 0.204]	0.969
Total earnings, zero-inclusive	[-0.263, 0.294]	[-0.271, 0.286]	[-0.255, 0.302]	0.969
Wages, positive	[-0.136, 0.158]	[-0.134, 0.160]	[-0.137, 0.157]	0.969
Total earnings, positive	[-0.149, 0.160]	[-0.151, 0.161]	[-0.150, 0.162]	0.969

J Age-Adjusted Earnings: Complete Construction Comparisons

This appendix reports the complete families underlying the main age-adjusted earnings comparisons. The main source contains 83,372 positive-endpoint histories with validated longitudinal identifiers, grouped into 58 industry–education–age cells within 30 industry–education groups. Real log growth removes CPI-U inflation; nominal arc growth uses the same histories. Survey-weighted origin-age cubics are fitted separately by industry–education group, following the profile adjustment in [Catherine, Sodini, and Zhang \(2024\)](#). These are U.S. pretax earnings rather than CSZ’s Swedish disposable income, and the historical composition measure is not their stock-return comovement measure.

J.1 Entry: source, transformation, and profile adjustment

Table [J.1](#) retains all eight combinations of wage versus total earnings, real log versus nominal arc growth, and raw versus age-adjusted growth. Observations and source scaling are common within the raw–adjusted pairs. Conventional coefficients and standard errors use 30-group CR1 inference with t_{29} tests. The finer 58-cell results and 499-draw joint remeasurement results are reported separately. The joint calculation retains its saved t_{57} convention; it is not silently relabeled as 30-group inference. All four adjusted slopes have joint eight-test q -values of 0.073, and their percentile intervals include zero. Table [J.2](#) reports all four paired adjustment contrasts using their own shared-draw covariance.

J.2 Standing shares and pooled allocation

Table [J.3](#) reports all eight standing-holder regressions: two earnings sources, two growth transformations, and two financial denominators. It also reports the four direct broad-minus-nonretirement denominator contrasts. The current-holder sample is not the prospective initial-holder sample in the main paper. Age-adjusted real-log composition has a negative nonretirement holder association for both earnings sources; the corresponding arc estimates are imprecise.

Table [J.4](#) retains all eight pooled Tobit specifications, with latent coefficients and average marginal effects distinguished and each using its own uncertainty calculation. The marginal effects are imprecise throughout. These results are supporting outcome comparisons, not replacements for the main prospective estimates.

J.3 Relationship to the coarser CSZ benchmark

The 58-cell main construction pools historical residual earnings within age bands. An earlier benchmark instead builds annual moments within coarser industry–education groups and averages them over time, more closely matching the annual aggregation in [Catherine, Sodini, and Zhang \(2024\)](#). All eight conditional composition tests in that 29-supported-group benchmark are imprecise (the smallest p -value is 0.370). The replication package retains its complete outputs. Hence the main results should not be described as invariant to aggregation or as a replication of CSZ’s systematic-risk

Table J.1. Raw and age-adjusted earnings: complete matched entry comparison

Earnings Growth	(1) Total Raw	(2) Total Age adjusted	(3) Wages Raw	(4) Wages Age adjusted
<i>A. Real log growth</i>				
Upside composition	1.096** (0.443)	0.875*** (0.279)	0.732 (0.495)	0.782** (0.307)
58-cell sensitivity p	0.0112	0.0014	0.1142	0.0062
Conditional coefficient BH8 q	0.0223	0.0168	0.1498	0.0220
Conditional-SD magnitude (pp)	0.523	0.573	0.374	0.524
Joint-refit coefficient SE	0.555	0.421	0.548	0.413
Joint original-point p	0.0533	0.0421	0.1869	0.0635
Joint coefficient BH8 q	0.0726	0.0726	0.1869	0.0726
Joint percentile 95% interval	[−0.522, 1.622]	[−0.353, 1.299]	[−0.614, 1.590]	[−0.348, 1.251]
Joint point-centered 95% interval	[−0.016, 2.208]	[0.032, 1.718]	[−0.365, 1.830]	[−0.045, 1.610]
Mean coefficient across joint draws	0.575	0.486	0.435	0.445
Nonholder transitions / joint draws	27,419 / 499	27,419 / 499	27,419 / 499	27,419 / 499

Notes: Coefficients and intervals are entry percentage points per fixed pre-adjustment source SD. The conditional-SD magnitude instead uses residual composition after all other regressors. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . All eight regressions across both panels use the same observations and controls as Table 2. Raw versus age-adjusted refers only to the group-specific cubic age fit, not the richer earnings-rank adjustment. Joint draws rebuild CPS age fits and moments with household multipliers and independently reweight SIPP sample lineages and assignment cells. Joint original-point tests retain their saved t_{57} reference and use the coefficient SD across 499 draws; their point-centered intervals use that same reference. Percentile intervals use the draw quantiles. BH8 covers all eight raw/adjusted, earnings-definition and growth-transformation slopes within each inference convention. The methods are not interchangeable when the refit distribution is displaced. Every percentile interval includes zero. Saved joint magnitude tests, distinct from slope tests, give $q = 0.1038$ for all four adjusted conditional-SD magnitudes.

measure. The matched 58-cell comparison holds the public-data assignment and earnings histories fixed while importing their age-profile adjustment.

Table J.1. Raw and age-adjusted earnings: complete matched entry comparison

Earnings Growth	(1) Total Raw	(2) Total Age adjusted	(3) Wages Raw	(4) Wages Age adjusted
<i>B. Nominal arc growth</i>				
Upside composition	1.029*** (0.350)	0.695*** (0.248)	0.974** (0.356)	0.740*** (0.263)
58-cell sensitivity p	0.0031	0.0042	0.0053	0.0044
Conditional coefficient BH8 q	0.0168	0.0168	0.0168	0.0168
Conditional-SD magnitude (pp)	0.660	0.534	0.649	0.582
Joint-refit coefficient SE	0.413	0.359	0.406	0.349
Joint original-point p	0.0157	0.0579	0.0197	0.0382
Joint coefficient BH8 q	0.0726	0.0726	0.0726	0.0726
Joint percentile 95% interval	[−0.162, 1.336]	[−0.263, 1.123]	[−0.144, 1.352]	[−0.166, 1.146]
Joint point-centered 95% interval	[0.201, 1.856]	[−0.024, 1.413]	[0.161, 1.786]	[0.042, 1.438]
Mean coefficient across joint draws	0.596	0.452	0.574	0.471
Nonholder transitions / joint draws	27,419 / 499	27,419 / 499	27,419 / 499	27,419 / 499

Notes: Coefficients and intervals are entry percentage points per fixed pre-adjustment source SD. The conditional-SD magnitude instead uses residual composition after all other regressors. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . All eight regressions across both panels use the same observations and controls as Table 2. Raw versus age-adjusted refers only to the group-specific cubic age fit, not the richer earnings-rank adjustment. Joint draws rebuild CPS age fits and moments with household multipliers and independently reweight SIPP sample lineages and assignment cells. Joint original-point tests retain their saved t_{57} reference and use the coefficient SD across 499 draws; their point-centered intervals use that same reference. Percentile intervals use the draw quantiles. BH8 covers all eight raw/adjusted, earnings-definition and growth-transformation slopes within each inference convention. The methods are not interchangeable when the refit distribution is displaced. Every percentile interval includes zero. Saved joint magnitude tests, distinct from slope tests, give $q = 0.1038$ for all four adjusted conditional-SD magnitudes.

Table J.2. Matched changes from removing group-specific age profiles

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
Adjusted minus raw composition slope	−0.221 (0.278)	0.050 (0.304)	−0.334* (0.183)	−0.234 (0.180)
58-cell sensitivity p	0.4349	0.8651	0.0674	0.1640
Conditional BH4 q	0.5787	0.8703	0.3138	0.4076
Joint original-point p	0.5655	0.8897	0.2412	0.3921
Joint BH4 q	0.7540	0.8897	0.7540	0.7540
Joint percentile 95% interval	[−0.839, 0.690]	[−0.674, 0.717]	[−0.696, 0.478]	[−0.658, 0.475]
Joint point-centered 95% interval	[−0.985, 0.544]	[−0.668, 0.768]	[−0.900, 0.231]	[−0.777, 0.309]

Notes: Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . All four slope differences use identical observations and common fixed source-SD units within each raw/adjusted pair. Their SEs retain covariance between the paired fits. Joint results use the same 499 matched draws and saved t_{57} original-point convention as Table J.1. BH4 covers all four paired differences within each convention. These are differences between descriptive specifications, not mediation effects. All joint paired intervals include zero.

Table J.3. Age-adjusted earnings composition and standing holder shares

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
<i>A. Nonretirement financial denominator</i>				
Composition	-1.317** (0.513)	-1.070** (0.504)	-0.349 (0.394)	-0.384 (0.372)
58-cell sensitivity p	0.0066	0.0200	0.4461	0.3585
Full-family BH16 q (coefficient)	0.2372	0.2372	0.4712	0.4712
Conditional-SD magnitude (pp)	-0.762	-0.705	-0.223	-0.261
Worker-year observations	17,455	17,455	17,455	17,455
People	11,773	11,773	11,773	11,773
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted OLS among current positive stock-and-mutual-fund holders; these are standing associations, not within-person reallocations. Panel C uses paired coefficient covariance. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . Age adjustment, composition, risk controls and personal controls follow Table 2, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.3. Age-adjusted earnings composition and standing holder shares (continued)

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
<i>B. Broad financial denominator</i>				
Composition	−0.817* (0.465)	−0.826** (0.393)	−0.638 (0.438)	−0.600 (0.398)
58-cell sensitivity p	0.0778	0.0400	0.1947	0.1808
Full-family BH16 q (coefficient)	0.3573	0.2372	0.4160	0.4160
Conditional-SD magnitude (pp)	−0.472	−0.544	−0.407	−0.407
<i>C. Paired broad-minus-nonretirement slope</i>				
Difference	0.500 (0.349)	0.244 (0.369)	−0.288 (0.260)	−0.216 (0.242)
Paired-difference BH4 q	0.5075	0.5126	0.5075	0.5075
Worker-year observations	17,455	17,455	17,455	17,455
People	11,773	11,773	11,773	11,773
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted OLS among current positive stock-and-mutual-fund holders; these are standing associations, not within-person reallocations. Panel C uses paired coefficient covariance. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . Age adjustment, composition, risk controls and personal controls follow Table 2, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.4. Age-adjusted earnings composition and censored standing shares

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
<i>A. Nonretirement financial denominator</i>				
Composition: latent Tobit coefficient	−0.872 (0.835)	−0.934 (0.810)	0.643 (0.710)	0.412 (0.776)
Observed-share average marginal effect	−0.187 (0.179)	−0.200 (0.174)	0.138 (0.152)	0.088 (0.166)
58-cell sensitivity p (AME)	0.3767	0.2939	0.4269	0.6188
Full-family BH16 q (coefficient)	0.4712	0.4712	0.4712	0.6849
Full-family BH16 q (AME)	0.4712	0.4712	0.4712	0.6851
Conditional-SD magnitude (pp)	−0.120	−0.135	0.103	0.068
Worker-year observations	81,603	81,603	81,603	81,603
People	48,016	48,016	48,016	48,016
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted Gaussian Tobit includes holders and genuine zero positions, with lower censoring at zero and no upper censor at one. Latent coefficients and observed-share AMEs are separate estimands; each is shown with its own SE and stars from its own zero test. AME SEs propagate the full coefficient/log-scale covariance. The AME is not a purely intensive-margin effect. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . Age adjustment, composition, risk controls and personal controls follow Table 2, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

Table J.4. Age-adjusted earnings composition and censored standing shares (continued)

Earnings Growth	(1) Total Real log	(2) Wages Real log	(3) Total Nominal arc	(4) Wages Nominal arc
<i>B. Broad financial denominator</i>				
Composition: latent Tobit coefficient	−0.529 (0.537)	−0.657 (0.529)	0.198 (0.490)	0.051 (0.551)
Observed-share average marginal effect	−0.107 (0.109)	−0.133 (0.107)	0.040 (0.099)	0.010 (0.111)
58-cell sensitivity p (AME)	0.3838	0.2345	0.7068	0.9259
Full-family BH16 q (coefficient)	0.4712	0.4712	0.7347	0.9263
Full-family BH16 q (AME)	0.4712	0.4712	0.7348	0.9263
Conditional-SD magnitude (pp)	−0.068	−0.090	0.030	0.008
Worker-year observations	81,603	81,603	81,603	81,603
People	48,016	48,016	48,016	48,016
Assignment cells / inference groups	58 / 30	58 / 30	58 / 30	58 / 30
Risk controls, personal controls, and FE	Yes	Yes	Yes	Yes

Notes: Weighted Gaussian Tobit includes holders and genuine zero positions, with lower censoring at zero and no upper censor at one. Latent coefficients and observed-share AMEs are separate estimands; each is shown with its own SE and stars from its own zero test. AME SEs propagate the full coefficient/log-scale covariance. The AME is not a purely intensive-margin effect. Both panels use identical valid 2019–2024 person-years. Shares divide stocks/mutual funds by financial assets net of retirement balances (A) or broad financial assets (B). Both denominators are positive and ratios in $[0, 1]$; invalid values are not recoded or clipped. Coefficients/AMEs are percentage points per fixed pre-adjustment source SD; the conditional-SD magnitude uses residual composition after the other controls. Stars use unadjusted two-sided zero-coefficient tests with CR1 standard errors clustered by 30 industry–education groups and t_{29} : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Fine-cell rows instead use 58-cell CR1 and t_{57} . Age adjustment, composition, risk controls and personal controls follow Table 2, using current rather than lagged values and current-year person weights. The full family spans two earnings definitions, two growth measures, two denominators and two estimators. Coefficient and AME BH16 families are separate. Inference conditions on CPS moments; no first-stage bootstrap is supplied for these models.

K Portfolio-Change Accounting and Balance Diagnostics

The balance-sheet family fixes the same 31,569 prospective observations, earnings constructions, weights, and fully interacted control design as the main allocation comparison. Before fitting the diagnostics, the analysis plan fixed two constructions and five diagnostic or accounting outcomes: midpoint risky-balance contribution, midpoint denominator contribution, growth in financial assets normalized by initial assets, similarly normalized risky-balance growth, and future normalized risky balances conditional on future normalized financial assets. Together with one share-change identity replay per construction, this gives twelve fitted models. Unconditional future normalized risky balances have the same composition coefficient as risky-balance growth because baseline share is controlled; they are not an additional model. Two further arithmetic replays verify the conditional-outcome equivalence. No observations are clipped or winsorized in response to the results.

Equation (9) is an exact finite-change identity, not a first-order approximation. Both share-change components are bounded on the retained positive-denominator, valid-share sample. Because all regressions use identical designs, the estimated composition contributions add exactly to the original share slope, including each ownership contrast. Covariance between components is retained for the arithmetic checks. Table K.1 gives the complete accounting family, and Table K.2 reports the normalized-balance diagnostics. Conventional stars use 30-group CR1/ t_{29} inference; 58-cell p -values are sensitivities.

Normalizing by initial financial assets avoids controlling for the same future denominator in every outcome but makes very small starting balances influential. Holder financial-asset growth reaches approximately 1,893 times initial assets; the corresponding nonholder maximum is approximately 464,061. Ranking observations by the absolute normalized risky-growth outcome, the largest 1% account for 66.44% of absolute weighted risky growth among holders and 99.08% among nonholders. These diagnostics must therefore be distinguished from the bounded midpoint accounting result.

The future-asset-conditioned specification is an optional comparison of a different estimand, not a robustness requirement for the main accounting result. Future assets are an endogenous conditioning variable and include risky assets themselves. Adding them is not an identified direct effect or a causal mediation design. The real-log holder coefficient is positive and imprecise ($p = 0.508$); it does not establish that greater labor income reduces demand for equity income at given financial wealth. An income-target explanation would require an additional preference or saving channel. Unconditioned normalized risky-balance growth is also imprecise and sensitive to small initial denominators. The midpoint risky-balance component combines transactions, returns, and changes in reporting; it is not net purchases or sales.

Table K.3 reports all family corrections separately: twelve accounting-component tests, eighteen normalized-balance tests, and the combined thirty-test family, each within its clustering convention. The share replays are arithmetic checks, not additional hypotheses. The real-log holder numerator

Table K.1. Portfolio-share accounting: full status contrasts and clustering sensitivity

	Real log			Nominal arc		
	(1) Nonholder	(2) Holder	(3) H–N	(4) Nonholder	(5) Holder	(6) H–N
Baseline ownership						
Risky-share change, Δs	0.360** (0.142)	–1.621** (0.715)	–1.980** (0.737)	0.264** (0.128)	–0.409 (0.642)	–0.673 (0.683)
<i>p</i> , 30 groups	0.0172	0.0311	0.0118	0.0490	0.5288	0.3325
<i>p</i> , 58 cells	0.0058	0.0150	0.0044	0.0178	0.4902	0.2887
Risky-balance term, $\Delta R/\bar{F}$	0.589** (0.226)	–3.679** (1.595)	–4.268** (1.619)	0.362 (0.227)	–1.244 (1.339)	–1.606 (1.387)
<i>p</i> , 30 groups	0.0141	0.0284	0.0133	0.1214	0.3608	0.2562
<i>p</i> , 58 cells	0.0044	0.0207	0.0087	0.0566	0.3114	0.2092
Asset term, $-\bar{s}\Delta F/\bar{F}$	–0.229** (0.101)	2.058* (1.050)	2.288** (1.058)	–0.099 (0.111)	0.835 (0.847)	0.933 (0.851)
<i>p</i> , 30 groups	0.0301	0.0595	0.0389	0.3813	0.3324	0.2820
<i>p</i> , 58 cells	0.0139	0.0567	0.0360	0.2743	0.2922	0.2421
Nonholder / holder observations	24,235 / 7,334					
Consecutive person-year pairs	31,569					

Notes: All entries are composition slopes in percentage points per common fixed pre-adjustment source SD. H–N is the directly estimated holder-minus-nonholder contrast, with shared cluster covariance. The rows decompose changes in stocks and mutual funds divided by nonretirement financial assets; their point estimates sum exactly. The share-change row reproduces the retained prospective-share result, because initial share is controlled for holders and is zero for nonholders. The sample retains all 31,569 valid consecutive pairs, including 2,165 holder-to-zero transitions. Each risk and baseline-control coefficient varies by initial ownership. Controls are composition, variance, mean growth, mean growth squared, SD, age and age squared, sex, transformed first-listed-job earnings, log gross assets, a nonpositive first-listed-job-earnings indicator, and outcome-year and baseline industry, education and age-band effects; initial share also enters the holder block. Future-year person weights are used. Parentheses and stars use primary 30-group CR1 inference and unadjusted two-sided t_{29} tests; the 58-cell sensitivity uses t_{57} . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Table K.3 reports the complete family adjustments. Inference conditions on estimated earnings moments, without a first-stage bootstrap. The accounting terms are not net trades or identified causal mechanisms.

coefficient has a twelve-test q -value of 0.090 and a thirty-test q -value of 0.113. Neither the main table nor the appendix interprets its marginal significance as a separately identified mechanism.

Table K.2. Prospective balance diagnostics: full exploratory family

	(1) Nonholder	(2) Holder	(3) H–N
Baseline ownership			
<i>Panel A: Real log</i>			
Financial-asset growth, $\Delta F/F_0$	–2458.081 (3005.350)	–252.542* (134.976)	2205.539 (3049.184)
<i>p</i> , 30 groups	0.4201	0.0715	0.4753
<i>p</i> , 58 cells	0.4099	0.1023	0.4633
Risky-balance growth, $\Delta R/F_0$	2507.971* (1356.996)	–125.930 (108.983)	–2633.901* (1389.596)
<i>p</i> , 30 groups	0.0748	0.2573	0.0680
<i>p</i> , 58 cells	0.1075	0.3098	0.0962
Future risky balance, R_1/F_0	3762.382*** (1323.286)	24.813 (36.991)	–3737.569*** (1327.363)
<i>p</i> , 30 groups	0.0081	0.5077	0.0087
<i>p</i> , 58 cells	0.0065	0.5070	0.0070
Future financial assets controlled		Final outcome only	
Nonholder / holder observations		24,235 / 7,334	
Consecutive person-year pairs		31,569	

Notes: Coefficients are nominal dollars per \$100 of initial nonretirement financial assets, per common fixed source SD of composition. All three diagnostic outcomes, both transformations, and all direct status contrasts are retained. The sample, weighting, baseline controls, and inference are those in Table K.1. The final outcome additionally conditions on future assets F_1/F_0 , separately by baseline ownership; this descriptive conditioning is not causal mediation, and future assets include risky balances. All balances and exits are untrimmed. Initial assets can be as low as \$1 for nonholders and \$7 for holders, producing very large normalized growth. Within baseline status and each normalized growth outcome, observations are ranked by the absolute unweighted outcome. The top 1% then accounts for 57.8% of the weighted absolute sum of holder asset growth and 66.4% of risky growth; corresponding nonholder shares are 91.2% and 99.1%. Parentheses and stars use 30-group CR1 and unadjusted two-sided t_{29} tests; the 58-cell p rows use t_{57} . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The complete 18-test diagnostic and 30-test combined BH adjustments appear in Table K.3. Balance changes do not identify net purchases or sales.

Table K.2. Prospective balance diagnostics: full exploratory family (continued)

	(1) Nonholder	(2) Holder	(3) H–N
Baseline ownership			
<i>Panel B: Nominal arc</i>			
Financial-asset growth, $\Delta F/F_0$	–2107.772 (1334.091)	–169.689* (92.835)	1938.083 (1359.102)
<i>p</i> , 30 groups	0.1250	0.0779	0.1645
<i>p</i> , 58 cells	0.2332	0.0863	0.2754
Risky-balance growth, $\Delta R/F_0$	803.290 (772.808)	–31.321 (58.424)	–834.610 (769.597)
<i>p</i> , 30 groups	0.3072	0.5960	0.2871
<i>p</i> , 58 cells	0.4346	0.6253	0.4138
Future risky balance, R_1/F_0	1878.875** (776.996)	69.989* (35.339)	–1808.886** (781.568)
<i>p</i> , 30 groups	0.0221	0.0572	0.0279
<i>p</i> , 58 cells	0.0285	0.0615	0.0352
Future financial assets controlled		Final outcome only	
Nonholder / holder observations		24,235 / 7,334	
Consecutive person-year pairs		31,569	

Notes: Coefficients are nominal dollars per \$100 of initial nonretirement financial assets, per common fixed source SD of composition. All three diagnostic outcomes, both transformations, and all direct status contrasts are retained. The sample, weighting, baseline controls, and inference are those in Table K.1. The final outcome additionally conditions on future assets F_1/F_0 , separately by baseline ownership; this descriptive conditioning is not causal mediation, and future assets include risky balances. All balances and exits are untrimmed. Initial assets can be as low as \$1 for nonholders and \$7 for holders, producing very large normalized growth. Within baseline status and each normalized growth outcome, observations are ranked by the absolute unweighted outcome. The top 1% then accounts for 57.8% of the weighted absolute sum of holder asset growth and 66.4% of risky growth; corresponding nonholder shares are 91.2% and 99.1%. Parentheses and stars use 30-group CR1 and unadjusted two-sided t_{29} tests; the 58-cell p rows use t_{57} . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The complete 18-test diagnostic and 30-test combined BH adjustments appear in Table K.3. Balance changes do not identify net purchases or sales.

Table K.3. Portfolio accounting and balance diagnostics: multiple-testing adjustments

	Real log			Nominal arc		
	(1) Nonholder	(2) Holder	(3) H–N	(4) Nonholder	(5) Holder	(6) H–N
Baseline ownership						
<i>Risky-balance term, $\Delta R/\bar{F}$</i>						
Family q , 30 groups	0.0848	0.0902	0.0848	0.2080	0.3813	0.3760
Family q , 58 cells	0.0521	0.0621	0.0521	0.0972	0.3114	0.3114
Combined q , 30 groups	0.1060	0.1127	0.1060	0.2205	0.4330	0.3860
Combined q , 58 cells	0.0651	0.1034	0.0651	0.1537	0.3892	0.3691
<i>Asset term, $-\bar{s}\Delta F/\bar{F}$</i>						
Family q , 30 groups	0.0902	0.1191	0.0935	0.3813	0.3813	0.3760
Family q , 58 cells	0.0554	0.0972	0.0865	0.3114	0.3114	0.3114
Combined q , 30 groups	0.1127	0.1558	0.1298	0.4399	0.4155	0.3915
Combined q , 58 cells	0.0831	0.1537	0.1202	0.3892	0.3892	0.3823
<i>Financial-asset growth, $\Delta F/F_0$</i>						
Family q , 30 groups	0.5041	0.1558	0.5347	0.2249	0.1558	0.2693
Family q , 58 cells	0.5212	0.2150	0.5212	0.4198	0.2150	0.4507
Combined q , 30 groups	0.4668	0.1558	0.5092	0.2205	0.1558	0.2742
Combined q , 58 cells	0.4775	0.2016	0.4964	0.3823	0.1991	0.3892
<i>Risky-balance growth, $\Delta R/F_0$</i>						
Family q , 30 groups	0.1558	0.3860	0.1558	0.3950	0.5960	0.3950
Family q , 58 cells	0.2150	0.4647	0.2150	0.5212	0.6253	0.5212
Combined q , 30 groups	0.1558	0.3860	0.1558	0.4007	0.5960	0.3915
Combined q , 58 cells	0.2016	0.3892	0.2016	0.4829	0.6253	0.4775
<i>Future risky balance, R_1/F_0</i>						
Family q , 30 groups	0.0779	0.5375	0.0779	0.1257	0.1558	0.1257
Family q , 58 cells	0.0627	0.5369	0.0627	0.1584	0.2150	0.1584
Combined q , 30 groups	0.1060	0.5252	0.1060	0.1127	0.1558	0.1127
Combined q , 58 cells	0.0651	0.5245	0.0651	0.1202	0.1537	0.1202

Notes: Benjamini–Hochberg q -values use the complete recorded exploratory families, separately for each clustering convention. The accounting family has 12 tests: two components $\times$ two transformations $\times$ three status contrasts. The diagnostic family has 18 tests: three balance outcomes $\times$ two transformations $\times$ three status contrasts. Combined rows pool these 30 tests. The six share-change contrasts are identity replays of existing prospective results, not additional discoveries, and are excluded from these new families. All tests are two-sided CR1, with t_{29} for the primary 30-group inference and t_{57} for the 58-cell sensitivity. Table coefficient stars use unadjusted tests, not these q -values. No first-stage bootstrap is added.

L Additional Survey Definitions

This section retains the detailed survey definitions and broader samples used in the supplementary constructions. The main age-adjusted earnings and prospective portfolio samples are defined in Section 3.

L.1 Public earnings histories

The CPS Annual Social and Economic Supplement supplies annual wage-and-salary income, adjacent-survey person identifiers, industry, education, age, and survey weights. The extract covers interview years 2014–2023. I follow workers aged 25–65 with positive baseline wage earnings to the next ASEC observation, retaining destination earnings of zero. Recorded age must rise by zero, one, or two years. The CPS-to-SIPP crosswalk yields 183,809 linked transitions in 72 candidate industry-by-education-by-age cells. It combines 12 harmonized nonmilitary industries, three education groups, and two age bands, split at age 45. Appendix C provides the mapping, support thresholds, and sample construction.

The choice to retain destination zeros is economically consequential. Positive-to-positive earnings growth describes workers with measured earnings at both endpoints; it excludes the transition to no measured earnings. Yet a zero can reflect nonemployment, a change in work intensity or arrangement, or the definition of earnings in the survey. It is not automatically a job-loss event. I therefore retain zeros in the baseline and examine their composition explicitly in Section O.

With positive baseline earnings, the bounded arc growth rate accommodates zero destination earnings without an arbitrary offset (Törnqvist, Vartia, and Vartia, 1985):

$$g_{i,t+1} = 2 \frac{y_{i,t+1} - y_{i,t}}{y_{i,t+1} + y_{i,t}}. \quad (\text{L.1})$$

It lies in $[-2, 2]$ and maps a transition to zero to -2 . Positive-to-positive arc and log growth are reported as separate conditional-earnings objects. Adding a dollar before taking logs would instead impose a unit-dependent distance between zero and small positive earnings.

SIPP provides an independent measurement comparison. The post-redesign survey combines retrospective monthly earnings with annual household balance-sheet information. I use earnings from the first-listed job (job 1, TJB1_MSUM), not total earnings or the highest-paying job. Since the 2014 redesign, job slots are ordered chronologically within the reference year, with reporting order breaking ties. The same slot need not identify the same job across waves.⁵ I match monthly job-1 earnings to the same month 12 months later, requiring positive origin earnings and allowing a destination zero. The primary reliability comparison retains one fixed transition per person; an overlapping-month construction shows what repeated observations add. Each worker always remains in the same half of a reliability split. Job-or-business class and no-job-spell fields distinguish

⁵See the Census Bureau's *2024 SIPP Users' Guide*, Section 4.2.3, pp. 71–73, and p. 76. A separate main-employer question for retirement coverage in later surveys does not redefine the earnings slots.

arrangements associated with a destination zero. Job-1 industry also determines the SIPP industry assignment. These measures do not summarize every job held by a worker.

Both surveys' distributions use official person weights. Cell support is measured by the Kish effective sample size, $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, rather than the number of records (Kish, 1965). Appendix B reports the support rules and alternative estimators. The reliability calculations and the independent CPS resampling used for portfolio inference serve different purposes and are described separately below.

The original pooled CPS assignment retains both 2014 questionnaire subsamples with their supplied weights. Because each subsample independently represents approximately the full population, this convention gives that origin year extra weight; it is not an equal-population-weighted mixture of years. I retain it as the original benchmark and explicitly examine the 2014 survey design in Appendix I.8. The new same-source wage and total-labor-earnings comparison uses only the redesigned 2014 origin subsample, with later years unchanged, following the official guidance for comparison with 2015 onward.

The observable-progression comparison additionally requires equal positive IPUMS validated person identifiers (CPSIDV) at the two earnings endpoints. It rebuilds the matched wage and total-earnings distributions on 88,862 eligible histories, including 83,372 with positive destinations in both measures. This restriction checks longitudinal demographic consistency beyond the original age rule. It is not proof that every excluded history is a false match. The original quantile, variance, and local-inference estimates remain original-linkage benchmarks; the validated-link comparison is identified separately in Section N.1 and Appendix I.13.

L.2 Standing portfolios and transitions

SIPP reports the value of directly held stocks and mutual funds and total financial assets. I aggregate available eligible monthly records within each calendar year to form person-year observations. Participation equals one when any eligible monthly stock-and-mutual-fund report is positive. Where financial assets are positive and the numerator is observed, the share is 100 times the stock-and-mutual-fund balance divided by financial assets. The all-worker measure sets the share to zero when this ratio is unavailable. A second share retains only observations with a usable positive-financial-asset ratio, and a third also requires current positive holdings. Keeping these samples separate distinguishes a population portfolio measure from allocation among holders.

These outcomes measure observed stock-and-mutual-fund exposure, not total equity exposure. Mutual funds may hold non-equity securities, SIPP does not observe the allocation inside retirement accounts, and employer-stock compensation cannot be cleanly removed. Changes in a reported balance need not be verified trades. Appendix C.3 documents financial-asset denominators, ratio bounds, eligible-month coverage, and the household-record structure.

The transition sample links consecutive annual observations. It contains 35,112 person-year pairs on 23,170 people: 27,752 begin with zero holdings, and 7,360 begin with positive holdings. The weighted nonholder-to-holder transition rate is 8.79%. Entry is defined using the lagged nonholder

risk set, rather than selecting people who hold stocks at some point in the panel. A transition into positive holdings may nevertheless be a return to holding rather than a first-ever entry.

I also consider sustained entry, which requires positive holdings for a further year, and crossings above nominal balances of \$1,000 and \$5,000. These distinguish the baseline transition from the appearance of a small or transitory positive balance. Requiring the same assignment cell in both years is a separate migration check, not a condition imposed on the primary sample.

L.3 Separating earnings measurement from portfolio outcomes

The primary design measures cell earnings distributions using CPS origin years through 2017 and destination earnings through 2018. These historical distributions are then assigned to SIPP worker-years in 2019–2024. The resulting level sample has 90,110 observations on 52,115 people in 60 supported cells. Both observations in a portfolio transition must fall in 2019–2024, so transition outcomes occur in 2020–2024. All assigned earnings histories precede both portfolio observations.

This timing eliminates overlap between the earnings-measurement window and the portfolio-outcome window. Using separate surveys also removes mechanical own-observation contamination. Neither feature removes selection into industry, education, or the lagged portfolio state. Historical cell distributions may differ from households’ expectations, and annual growth does not identify permanent and transitory risk separately. The broader 2018–2024 portfolio window is retained as a sensitivity analysis, not combined with the strictly later baseline.

M Additional Estimands and Earlier Constructions

The definitions below accompany the earlier quantile and zero-inclusive comparisons retained in this appendix. Section 3.3 defines the age-adjusted main regressions.

M.1 A comparison at fixed variance

The main comparison allocates a given earnings variance between favorable and adverse outcomes. For earnings growth g with mean μ_c in cell c , define the lower and upper centered second partial moments:

$$V_c^- = \mathbb{E}_c[(g - \mu_c)^2 \mathbf{1}\{g < \mu_c\}], \quad V_c^+ = \mathbb{E}_c[(g - \mu_c)^2 \mathbf{1}\{g > \mu_c\}]. \quad (\text{M.1})$$

These components sum exactly to total variance, $V_c = V_c^- + V_c^+$. Define upside variance composition as

$$C_c^{\text{up}} = V_c^+ - V_c^-, \quad V_c^+ = \frac{V_c + C_c^{\text{up}}}{2}, \quad V_c^- = \frac{V_c - C_c^{\text{up}}}{2}. \quad (\text{M.2})$$

The partial moments use every realization on the corresponding side of the mean. The distinction between favorable and adverse components follows the broader good- and bad-uncertainty literature (Segal, Shaliastovich, and Yaron, 2015); the application here is to individual earnings histories and subsequent household portfolio entry.

The entry regression includes (V_c, C_c^{up}) alongside mean growth and its square, the standard deviation, zero incidence, household controls, and fixed effects. Including both V_c and $\sqrt{V_c}$ preserves the original quadratic-in-standard-deviation scale specification. The coefficient on C_c^{up} compares earnings distributions with different allocations of a given variance between favorable and adverse outcomes. Its zero restriction is equivalent to equal *raw-unit* coefficients on V_c^- and V_c^+ ; an exact change of coordinates leaves fitted values unchanged. It is not a test that variance alone fully describes the distribution, and the separate partial-moment coefficients are not total derivatives when the standard-deviation control also changes.

Upside and downside composition describe the same contrast from opposite directions. If $C_c^{\text{down}} = V_c^- - V_c^+$, then $C_c^{\text{up}} = -C_c^{\text{down}}$. Reversing the orientation reverses the coefficient and interval endpoints but leaves standard errors and two-sided tests unchanged. At fixed total variance, a larger upper component necessarily entails a smaller lower component. This is not the effect of increasing upside risk while holding downside risk fixed, and the two orientations are not independent findings.

I report this comparison on the original 60 cells, on the 59 cells with adequate positive-destination support, and using only positive earnings destinations within those same 59 cells. The last design changes the conditioning event of the earnings distribution, not the SIPP entry sample. Each construction uses its own original cross-cell standard deviation, held fixed in resampling. To describe a comparison within the observed conditional support, I also multiply the coefficient by the weighted standard deviation of composition after projecting out every other regressor in its SIPP specification.

These wage-based estimates are benchmarks for the matched comparison between wages and total labor earnings. The latter retains 58 common cells and 27,419 SIPP entry transitions, compares zero-inclusive with positive destinations, and reconstructs both raw and observable-progression-adjusted earnings growth using validated longitudinal links. Section N.1 presents this rebuilt comparison. It supplies the total-earnings magnitudes highlighted in the abstract; tests from the original wage or original-linkage designs are not transferred to it.

M.2 Separating the two tails

The quantile comparison describes the two sides without squaring earnings deviations. Let G_c be the distribution of one-year earnings growth in cell c , with quantile $q_{p,c}$. Define the upper and lower tail distances as

$$U_c = q_{0.90,c} - q_{0.50,c}, \quad L_c = q_{0.50,c} - q_{0.10,c}. \quad (\text{M.3})$$

The upper distance measures the gap between the median and favorable earnings growth; the lower distance measures the gap to adverse growth. They are quantile distances, not probabilities of a gain or loss, and neither is by itself a structural shock.

Their sum, $S_c = L_c + U_c$, is the 90–10 span, a measure of dispersion distinct from variance. Their difference, $D_c = L_c - U_c$, measures the composition of that span toward the downside. Entering

(S_c, D_c) instead of (L_c, U_c) is an exact linear reparameterization. In raw units,

$$b_L L_c + b_U U_c = \frac{b_L + b_U}{2} S_c + \frac{b_L - b_U}{2} D_c.$$

Thus the composition coefficient asks whether shifting distance from the upper to the lower tail is associated with a different portfolio outcome at a given total span. It does not create a second source of identifying variation. Because the empirical regressors are standardized separately, displayed coefficients need not satisfy this identity without first restoring their raw units.

Downside-oriented Kelley shape normalizes the difference by the span:

$$K_c = \frac{L_c - U_c}{L_c + U_c} = \frac{2q_{0.50,c} - q_{0.10,c} - q_{0.90,c}}{q_{0.90,c} - q_{0.10,c}}. \quad (\text{M.4})$$

A positive value denotes a longer downside. Kelley answers a different question from the unscaled composition D_c : it discards the magnitude of the interquantile range. Appendix A defines the alternative Bowley, L-skewness, moment-skewness, and tail-probability measures.

M.3 What it means to hold other risk features fixed

The zero-inclusive distribution combines wage growth with transitions to zero measured earnings. A larger zero mass can move quantiles, the mean, and the variance even if the positive-earnings distribution is unchanged. Once the zero mass reaches ten percent, the lower quantile is at the arc-growth boundary. Lemma 1 in Appendix A.1 states this relationship formally. The resulting tail statistic remains a valid description of the distribution, but it cannot by itself isolate a change in shape holding other distributional features fixed.

I therefore distinguish raw orientation from its incremental component. Across supported cells, the latter is the residual from

$$K_c = a + \gamma p_{0c} + f_2(\mu_c) + h_2(\sigma_c) + \lambda_{e(c)} + \lambda_{a(c)} + u_c, \quad (\text{M.5})$$

where p_{0c} is the probability of zero destination earnings, μ_c and σ_c are mean growth and its standard deviation, and f_2 and h_2 are quadratic functions. Education and age components are included, and the projection is weighted by effective cell support capped at 5,000. Sequential versions remove the zero share and demographic components first, then add mean and dispersion. Each projection is re-estimated within every reliability half and CPS resample.

Controlling mean growth is important because location and Kelley shape can be closely related (Harmenberg, 2021). The residual defines an observable conditional contrast; it does not identify “true skewness” or imply that the removed variation is economically irrelevant. The portfolio application similarly conditions the two tail distances on zero incidence, mean growth, and dispersion. Its question is whether the tails contain additional information within the support of those controls.

M.4 Portfolio comparisons

Write $H_{it} = 1\{A_{it} > 0\}$ for positive reported stock-and-mutual-fund holdings. The entry regressions are estimated among lagged nonholders. The quantile specification is

$$H_{it} = \beta_L \tilde{L}_{c(i,t-1)}^{\text{CPS}} + \beta_U \tilde{U}_{c(i,t-1)}^{\text{CPS}} + \rho \hat{p}_{0,c(i,t-1)} + f_2(\hat{\mu}_{c(i,t-1)}) + h_2(\hat{\sigma}_{c(i,t-1)}) \\ + X'_{i,t-1} \gamma + \delta_t + \delta_{j(i,t-1)} + \delta_{e(i,t-1)} + \delta_{a(i,t-1)} + \varepsilon_{it}, \quad H_{i,t-1} = 0. \quad (\text{M.6})$$

Tildes denote standardization across the supported early-CPS cells, using Kish effective-N weights capped at 1,000. $X_{i,t-1}$ includes a transformed average of monthly job-1 earnings, log total assets, age and its square, sex, and an indicator for any eligible month with nonpositive decoded job-1 earnings. For mean monthly job-1 earnings $\bar{y}^{(1)}$ in unshifted dollars over the retained observations, the income control is $\log(\max\{\bar{y}^{(1)} + 1, 1\})$. Total assets are floored at one dollar before taking logs. The income-control offset is not used in the arc-growth risk measures. The nonpositive-earnings indicator is not a separately observed unemployment spell or a zero of total labor income. Fixed effects cover outcome year and lagged industry, education, and age band. Observations are weighted by the current SIPP person-year weight. Replacing the two tails with span and signed composition leaves fitted values unchanged; a further specification pairs span with incremental Kelley orientation.

I multiply the estimated coefficients in equation (M.6) by 100 to report percentage-point differences in entry per one cross-cell standard deviation of the assigned measure. They are conditional associations, not household responses to exogenous changes in risk. Risk tolerance, lifetime resources, or other characteristics can jointly influence occupation and portfolios (Ranish, 2013; d'Astous and Shore, 2024). Lagging the assignment does not solve this selection problem.

The remaining outcomes distinguish acquisition, retention, and allocation. I estimate exit, $1 - H_{it}$, among $H_{i,t-1} = 1$, and signed participation change, $H_{it} - H_{i,t-1}$, in the full transition sample, controlling lagged participation in the latter. Standing participation and the three share measures use contemporaneous household controls and assignment-cell components with the same pre-period CPS measures. The holder-only share is the relevant standing intensive margin; all-worker and positive-financial-asset shares also incorporate the extensive margin.

The prospective allocation specification instead conditions on *initial* ownership and retains future zero holdings. Let s_{it} denote stocks and mutual funds divided by nonretirement financial assets, and let $H_{i,t-1}$ indicate baseline ownership. I estimate

$$s_{it} = (1 - H_{i,t-1})(\beta_0 \tilde{C}_{c(i,t-1)}^{\text{up}} + Z'_{i,t-1} \gamma_0) \\ + H_{i,t-1}(\beta_1 \tilde{C}_{c(i,t-1)}^{\text{up}} + Z'_{i,t-1} \gamma_1 + \phi s_{i,t-1}) + \varepsilon_{it}. \quad (\text{M.7})$$

Here Z includes a constant, total variance, the other earnings-distribution controls, household controls, and fixed effects. All coefficients are allowed to differ by initial ownership. The common cross-cell scaling of composition makes $\beta_1 - \beta_0$ a direct test of the portfolio-state difference. Section 4.2

reports the age-adjusted real-log and nominal-arc constructions on the same sample. Baseline holders can subsequently exit; their outcome is not selected on continued participation.

M.5 Inference and scope of the tests

In the original wage and quantile designs, conditional standard errors cluster by SIPP person and CPS assignment cell. The large number of household records does not create a correspondingly large number of independent earnings-risk environments: these designs use at most 60 cells. A null-imposed cell-level wild bootstrap addresses this finite-cell dimension while retaining the two-way covariance structure (Roodman et al., 2019). A separate exercise resamples CPS people and reconstructs the tail measures, adding the resulting coefficient variance to the conditional variance. I call the latter a CPS plug-in sensitivity because it holds SIPP fixed and is not a joint CPS–SIPP bootstrap.

I then allow both samples to vary. In each of 2,000 draws, CPS people are resampled within their original cells, and SIPP observations receive the product of independent mean-one exponential weights for person and assignment cell, multiplied by their survey weights. This crossed reweighting follows Owen and Eckles (2012). Matched CPS-only and SIPP-only draws identify which stage changes the estimates. The sample and coefficient units remain fixed. Percentile, basic, and original-point normal-reference intervals are reported together because their disagreement is informative about nonlinear remeasurement; none is selected for favorable significance. This is a sensitivity to dependence and estimated regressors, not an exact finite-cell test or a full survey-design bootstrap. Section N.6 and Appendix I report the comparison.

The validated-link wage–total comparison uses a separate 1,000-draw reconstruction. CPS multipliers are shared within households; SIPP weights combine sample-unit-lineage and assignment-cell multipliers. Each draw refits the earnings ranks and household-disjoint predictions before rebuilding the distributions and portfolio regressions. Original-point reference tests use t_{57} , alongside percentile and basic intervals and the declared family adjustments. This design carries estimation uncertainty through the observable-progression adjustment and permits household dependence. It does not inherit the earlier 2,000-draw tests or establish full complex-survey coverage.

For the prospective allocation comparison in equation (M.7), the primary covariance clusters at the 58 assignment cells with a t_{57} reference. A coarser sensitivity clusters at 30 industry–education groups with a t_{29} reference. These tests condition on the estimated CPS moments; they do not include a new first-stage reconstruction. The complete pair of real-log and nominal-arc specifications is reported, with Benjamini–Hochberg adjustments for the two between-state contrasts. The replication output also retains the six-test family covering both state slopes and their difference in each construction.

The four standing outcomes and three transition outcomes are separately displayed testing families. Benjamini–Hochberg q -values adjust within each family and specification (Benjamini and Hochberg, 1995); the two individual tail coefficients also have a separately reported wild-bootstrap adjustment. The three variance-composition comparisons form a separate family, as do the twelve scalar coefficients in the earnings-state exercise. Every table identifies the test used for its stars. These analyses

developed through exploration rather than preregistration. The displayed adjustments do not account for every specification considered over the project's history.

N Rank-Adjusted and Quantile Portfolio Comparisons

This section preserves the richer progression, own-history, and quantile comparisons from the earlier constructions. These specifications are distinct from the CSZ-style age-profile adjustment used in Section 4; their original coefficient units and inference remain unchanged.

N.1 Variance composition and subsequent entry

At a given total earnings variance, distributions tilted toward favorable rather than adverse outcomes are associated with greater subsequent portfolio entry. Table N.1 compares wages with total labor earnings, the sum of wages and net business and farm earnings, on identical CPS histories with validated longitudinal identifiers. The population remains workers with positive origin wages. Both measures have positive origins and nonnegative destinations; a second comparison additionally requires positive destinations in both measures. All four constructions use the same 27,419 subsequent SIPP entry transitions in 58 cells. The total-earnings extension broadens the measured income components without changing the initial worker population or recoding business losses as zero.

At fixed variance, more upside composition necessarily means less downside composition. Reversing the orientation gives equal-magnitude negative coefficients, identical standard errors, and identical two-sided tests; it is not a second empirical confirmation. Nor is this an independent positive upper-tail effect with the lower partial moment held unchanged. Both separate partial-moment coefficients are negative in the original wage benchmark, and the upper one is imprecise (Appendix Table I.1). The difference between the two sides at fixed total variance is the composition contrast.

The assigned distributions contain realized earnings changes, not necessarily unanticipated risk. Favorable changes can reflect progression associated with age and prior earnings, as well as recoveries from earlier losses. I therefore also remove growth predicted by a fixed set of observed origin characteristics before reconstructing the distribution. Prediction uses five household-disjoint CPS folds and does not use SIPP portfolio outcomes. The raw and adjusted comparisons retain validated earnings links, the same 58 cells, identical portfolio observations, and common coefficient units within each pair.

Table N.1 shows that the adjustment weakens the point pattern. The residual-standard-deviation magnitude falls from 0.792 to 0.510 percentage point for zero-inclusive wages and from 0.703 to 0.370 for total earnings. In the positive-destination comparisons it falls from 0.649 to 0.207 and from 0.660 to 0.208 point, respectively. All four adjusted slopes remain positive, but none survives its complete four-test family adjustment: joint-remeasurement q -values range from 0.362 to 0.451. Thus these estimates do not establish an entry association for earnings composition after observable progression is removed.

Table N.1. Observable earnings progression and portfolio entry

Earnings Destinations	(1) Wages Zero-inclusive	(2) Total	(3) Wages Both positive	(4) Total
<i>Panel A. Observable-progression adjusted</i>				
Upside composition, C^{up}	1.793* (1.041)	1.259 (0.981)	0.380 (0.499)	0.394 (0.520)
Slope: four-model BH q	0.362	0.409	0.451	0.451
Residual-S.D. magnitude (pp)	0.510 (0.315)	0.370 (0.320)	0.207 (0.303)	0.208 (0.304)
Magnitude: four-model BH q	0.445	0.498	0.498	0.498
<i>Panel B. Matched raw earnings changes</i>				
Upside composition, C^{up}	3.242*** (1.067)	2.260** (0.858)	0.974** (0.403)	1.029** (0.412)
Slope: four-model BH q	0.014	0.019	0.019	0.019
Residual-S.D. magnitude (pp)	0.792** (0.304)	0.703** (0.299)	0.649** (0.307)	0.660** (0.302)
Magnitude: four-model BH q	0.039	0.039	0.039	0.039
Risk and household controls	Yes	Yes	Yes	Yes
Year and group effects	Yes	Yes	Yes	Yes
Observations	27,419	27,419	27,419	27,419
SIPP people	19,081	19,081	19,081	19,081
SIPP sample-unit lineages	13,849	13,849	13,849	13,849
CPS earnings histories	88,862	88,862	83,372	83,372
Active CPS households	61,480	61,480	58,798	58,798
CPS assignment cells	58	58	58	58
Joint full-pipeline draws	1,000	1,000	1,000	1,000

Notes: Outcome: subsequent entry into reported stock-and-mutual-fund holdings. Equal-positive endpoint CPSIDV restricts this repair to matched 58-cell partial moments; original 60/59-cell estimates and original-link local inference are unchanged. Slopes are percentage points per construction-specific original raw-composition cross-cell S.D. (CPS person-Kish weights capped at 1,000), common across panels. Panel A uses five household-disjoint prediction folds. Magnitudes use each model's conditional composition S.D. Parentheses report 1,000 joint CPS-household and SIPP-lineage-by-cell full-pipeline draws, rebuilding ranks, predictions, moments, regressions, and magnitude S.D.s. Stars use unrounded, unadjusted, approximate two-sided t_{57} tests against zero: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each raw/adjusted slope/magnitude family has four BH-adjusted tests. Appendix I.13 details controls and weighting; Tables I.10 and I.11 retain conflicting intervals and paired source comparisons. This exploratory sensitivity is not complete survey inference and does not identify unanticipated risk.

Exactly 1,000 matched draws reconstruct the earnings predictions, moments, and portfolio regressions while allowing dependence within CPS households and within SIPP sample-unit lineages and assignment cells. Every adjusted 95% percentile interval includes zero; basic slope intervals exclude it for both zero-inclusive constructions, only barely for total earnings. The raw references likewise have percentile intervals containing zero despite favorable original-point tests and basic intervals. Their draw means fall substantially below the original estimates. The raw-reference stars therefore do not establish inference-insensitive evidence. Appendix Table I.10 reports all three interval constructions and displacement.

The adjusted-minus-raw slope differences are also imprecise, with four-test $q = 0.281$ throughout. Repeating the complete raw-adjusted comparison on the original earnings links gives the same

qualitative conclusion; the paired changes due to link validation are imprecise as well (Appendix Table I.11). The smaller adjusted point estimates therefore do not establish that predictable progression explains the original association. Moreover, the adjustment removes only projection on observed covariates. It can absorb measurement-error mean reversion and common realized growth as well as predictable progression; the residuals are not identified earnings shocks. The exercise limits a risk interpretation without identifying the source of the original pattern.

Appendix Tables I.1 and I.2 retain the original-linkage wage and matched wage–total benchmarks. Both show similar positive point patterns. In the original wage benchmark, omitting the 2014 ASEC attenuates the associations without reversing their signs. Broader dependence-sensitive inference is inconclusive, and wage–total slope differences are imprecise. These comparisons establish neither equality across earnings definitions nor a stronger total-earnings response. Their separate local and product-reweighting uncertainty calculations are discussed in Section N.6; those tests do not supply inference for the validated estimates.

N.2 Conditioning on workers’ own prior earnings

Recent earnings levels and changes could help explain why workers assigned to different historical distributions enter the stock market at different rates. I examine this possibility using two annual totals of each worker’s all-job earnings, including signed business income, strictly before the baseline portfolio observation. Requiring complete prior histories with self or proxy interview status leaves 6,639 nonholder transitions in 58 cells, with a destination-weighted entry rate of 7.96%. This is a selected subset of the 27,419-transition reference sample, not a new representative cohort. The comparison without history controls is therefore re-estimated on exactly the same 6,639 observations.

Table N.2 adds both annual earnings levels in inverse-hyperbolic-sine form, a signed arc change, and four indicators for zero and negative annual totals. The dollar unit is fixed at \$10,000 in 2022 prices. Every specification retains the original household and earnings-distribution controls, including total variance. The raw and observable-progression-adjusted CPS distributions remain distinct constructions; two personal earnings levels do not estimate a personal risk distribution or elicit beliefs.

Adding the history controls leaves the raw point estimates almost unchanged. For total earnings, the residual-standard-deviation magnitude moves from 0.777 to 0.775 percentage point; with positive CPS destinations in both earnings measures, it moves from 0.729 to 0.727 point. Columns (3)–(4) and (7)–(8) use identical workers within each comparison. These magnitudes use each specification’s own residual spread; the table separately reports changes in slopes expressed in common original CPS units. The complete eight-test family of primary matched slope changes has $q = 0.994$. Neither the small point changes nor the imprecise paired differences establish equivalence or rule out sorting.

The restricted sample does not deliver an inference-stable association beyond recent earnings. All eight history-augmented slopes are positive, but none survives BH8 adjustment; q -values range from 0.217 to 0.870. Every primary 95% percentile interval includes zero. The same-sample baseline coefficients also fail their eight-test family adjustment, so this loss of significance cannot be attributed

Table N.2. Earnings composition and entry conditional on prior earnings

	CPS earnings: zero-inclusive				CPS earnings: positive			
	Wages		Total earnings		Wages		Total earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prior-history controls	No	Yes	No	Yes	No	Yes	No	Yes
<i>Panel A: Raw earnings growth</i>								
Upside composition	4.269**	4.270**	2.530	2.525	1.185	1.184	1.161	1.159
	(2.004)	(1.997)	(1.609)	(1.609)	(0.746)	(0.745)	(0.765)	(0.766)
Composition: BH8 q	0.216	0.217	0.216	0.217	0.216	0.217	0.216	0.217
All 95% CIs positive	–	No	–	No	–	No	–	No
Per own residual SD (pp)	1.024*	1.024*	0.777	0.775	0.772	0.771	0.729	0.727
	(0.544)	(0.542)	(0.537)	(0.536)	(0.537)	(0.535)	(0.530)	(0.530)
Magnitude: BH8 q	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
Matched slope change	–	0.001	–	–0.005	–	–0.001	–	–0.002
		(0.146)		(0.118)		(0.054)		(0.056)
Paired change: BH8 q	–	0.994	–	0.994	–	0.994	–	0.994
Baseline controls and FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nonholder transitions	6,639; 6,639 people; 5,291 lineages; 58 cells							
Destination years	2020, 2022, 2023, 2024							
Weighted entry rate (%)	7.96							

Outcome: later positive stock-and-mutual-fund holdings among baseline nonholders. All columns use the same self/proxy histories covering 24 prior months; earnings recodes may be imputed. CPS origins require positive wages and totals; negative total destinations are excluded. Positive headers require both CPS destinations positive, without changing recipients. Slopes are percentage points per original raw-source composition SD, common within raw/adjusted pairs and control blocks.

All models retain variance, mean and square, SD, applicable zero incidence, baseline personal controls and fixed effects. Histories add two annual totals strictly before baseline: $\text{asinh}(Y/10,000)$ in 2022 dollars, signed arc change and four zero/negative indicators. They are not identified shocks or beliefs. Weights are original destination weights, not history-longitudinal weights.

Parentheses: SDs of 1,000 joint CPS-household/SIPP-lineage-by-cell draws. Stars: unadjusted two-sided original-point t_{57} tests, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. This approximation does not certify survey coverage. BH8 covers eight raw/adjusted constructions separately by block and target; paired changes have a separate eight-test family. Magnitudes use their own SEs/tests and reflect changing residual spread. Slope changes are augmented minus matched baseline, not mediation shares. “Yes”: all three joint point-centered, percentile and basic 95% slope intervals are positive at unrounded bounds, not a new p-value. The positive-reporting screen additionally requires a positive primary augmented point and BH8 $q < 0.05$. Both interval companions retain every method.

to adding the controls. The decline-kink and alternative interview/processing samples provide no family-adjusted slope rejection either. Appendix 1.16 reports the complete comparisons. The exercise shows that observed recent earnings do not substantially change the estimated point pattern on the matched sample; it does not identify a causal response to risk, establish unanticipated shocks, or recover perceived risk.

N.3 The comparison with conventional skewness

The allocation of variance between its two sides is not the same statistic as a third moment. It nevertheless need not contain separately measurable information beyond conventional skewness. In the original-linkage matched earnings constructions, the conditional correlation between upside

Table N.2. Earnings composition and entry conditional on prior earnings (continued)

	CPS earnings: zero-inclusive				CPS earnings: positive			
	Wages (1) No	(2) Yes	Total earnings (3) No	(4) Yes	Wages (5) No	(6) Yes	Total earnings (7) No	(8) Yes
Prior-history controls								
<i>Panel B: Observable-progression-adjusted growth</i>								
Upside composition	2.880 (1.851)	2.939 (1.852)	1.930 (1.705)	2.044 (1.707)	0.100 (0.886)	0.146 (0.887)	0.359 (0.919)	0.400 (0.920)
Composition: BH8 q	0.216	0.217	0.350	0.315	0.910	0.870	0.797	0.760
All 95% CIs positive	–	No	–	No	–	No	–	No
Per own residual SD (pp)	0.802 (0.548)	0.817 (0.547)	0.553 (0.540)	0.585 (0.539)	0.054 (0.534)	0.079 (0.533)	0.187 (0.530)	0.208 (0.530)
Magnitude: BH8 q	0.280	0.280	0.413	0.376	0.920	0.883	0.830	0.796
Matched slope change	–	0.058 (0.144)	–	0.114 (0.130)	–	0.046 (0.069)	–	0.041 (0.072)
Paired change: BH8 q	–	0.994	–	0.994	–	0.994	–	0.994
Baseline controls and FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nonholder transitions	6,639; 6,639 people; 5,291 lineages; 58 cells							
Destination years	2020, 2022, 2023, 2024							
Weighted entry rate (%)	7.96							

Outcome: later positive stock-and-mutual-fund holdings among baseline nonholders. All columns use the same self/proxy histories covering 24 prior months; earnings recodes may be imputed. CPS origins require positive wages and totals; negative total destinations are excluded. Positive headers require both CPS destinations positive, without changing recipients. Slopes are percentage points per original raw-source composition SD, common within raw/adjusted pairs and control blocks.

All models retain variance, mean and square, SD, applicable zero incidence, baseline personal controls and fixed effects. Histories add two annual totals strictly before baseline: $\text{asinh}(Y/10,000)$ in 2022 dollars, signed arc change and four zero/negative indicators. They are not identified shocks or beliefs. Weights are original destination weights, not history-longitudinal weights.

Parentheses: SDs of 1,000 joint CPS-household/SIPP-lineage-by-cell draws. Stars: unadjusted two-sided original-point t_{57} tests, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. This approximation does not certify survey coverage. BH8 covers eight raw/adjusted constructions separately by block and target; paired changes have a separate eight-test family. Magnitudes use their own SEs/tests and reflect changing residual spread. Slope changes are augmented minus matched baseline, not mediation shares. “Yes”: all three joint point-centered, percentile and basic 95% slope intervals are positive at unrounded bounds, not a new p -value. The positive-reporting screen additionally requires a positive primary augmented point and BH8 $q < 0.05$. Both interval companions retain every method.

composition and standardized skewness ranges from 0.90 to 0.97. Only zero-inclusive total earnings retains positive incremental-composition reliability under both the CPS support metric and the SIPP regression’s exact control-profiled metric; the other three constructions fail this outcome-free information requirement.

In that one eligible construction, adding standardized skewness leaves a positive composition coefficient but does not establish an incremental association. Its two-survey p -value is 0.100 and its two-test q -value is 0.200; the corresponding skewness q -value is 0.472. The pair is jointly informative under the local covariance ($p = 0.038$), but not under the broader dependence sensitivity ($p = 0.145$). A joint rejection tests whether both coefficients vanish, not whether composition adds information beyond skewness. Appendix Table I.12 reports both covariance conventions and the complete two-test

family. The supported interpretation remains a comparison beyond variance, not a distinct empirical premium for semivariance composition over conventional skewness.

N.4 Lower and upper quantile distances

Table N.3 provides a complementary quantile comparison. In column (1), the lower and upper tail distances enter jointly. A one-standard-deviation wider lower tail is associated with a 2.712-percentage-point lower probability of positive stock-and-mutual-fund holdings one year later. The corresponding upper-tail coefficient is +1.917 points. The sample contains 27,752 worker transitions beginning with no reported holdings, with a weighted entry rate of 8.79%. The opposite signs distinguish the favorable and adverse sides of the assigned earnings distribution; the uncertainty is appreciably larger for the upper side.

Column (2) expresses the same comparison in terms of total span and signed composition. The span coefficient is imprecise, while a one-standard-deviation shift toward the downside is associated with 4.456 points less entry. The CPS plug-in p -values are 0.232 for span and 0.014 for composition. The fitted values from columns (1) and (2) agree to 1.1×10^{-8} : the rotation reveals the economic contrast already present in the joint two-tail regression. It does not show that total dispersion is irrelevant, or provide a second independent rejection.

Column (3) pairs total span with incremental Kelley orientation. The orientation coefficient is negative but less precise once uncertainty in the estimated CPS distribution is included (-0.870 point, $p = 0.068$). Normalizing composition by span therefore does not strengthen the result. Figure N.1 shows how adding the CPS estimation component changes the uncertainty around all three representations. In particular, the wider upper-tail and Kelley intervals cross zero.

This contrast is not a recovered variance sign reversal. In the same entry sample, the standardized linear term in cell standard deviation, evaluated at mean standardized dispersion and included with its square, is -0.262 point (SE 1.120; $p = 0.816$). Adding Kelley orientation or the separate tails does not uncover a positive scale coefficient. The empirical content is the difference between the tail associations conditional on dispersion, not an explanation of a positive variance–portfolio-share coefficient.

N.5 Quantile orientation across portfolio margins

Appendix Table C.4 compares the standing outcomes with portfolio transitions using raw and incremental Kelley orientation. Among lagged nonholders, incremental downside orientation is associated with 1.047 points less entry, or 11.9% of the baseline transition rate. Its conditional three-outcome q -value is below 0.001. Raw Kelley has a similar point estimate but less precision: its unadjusted $p = 0.042$ becomes $q = 0.126$ within the transition family. Separating zero incidence and lower moments changes the precision and interpretation of the comparison rather than revealing a larger raw effect.

Table N.3. Lower and upper earnings tails and portfolio entry

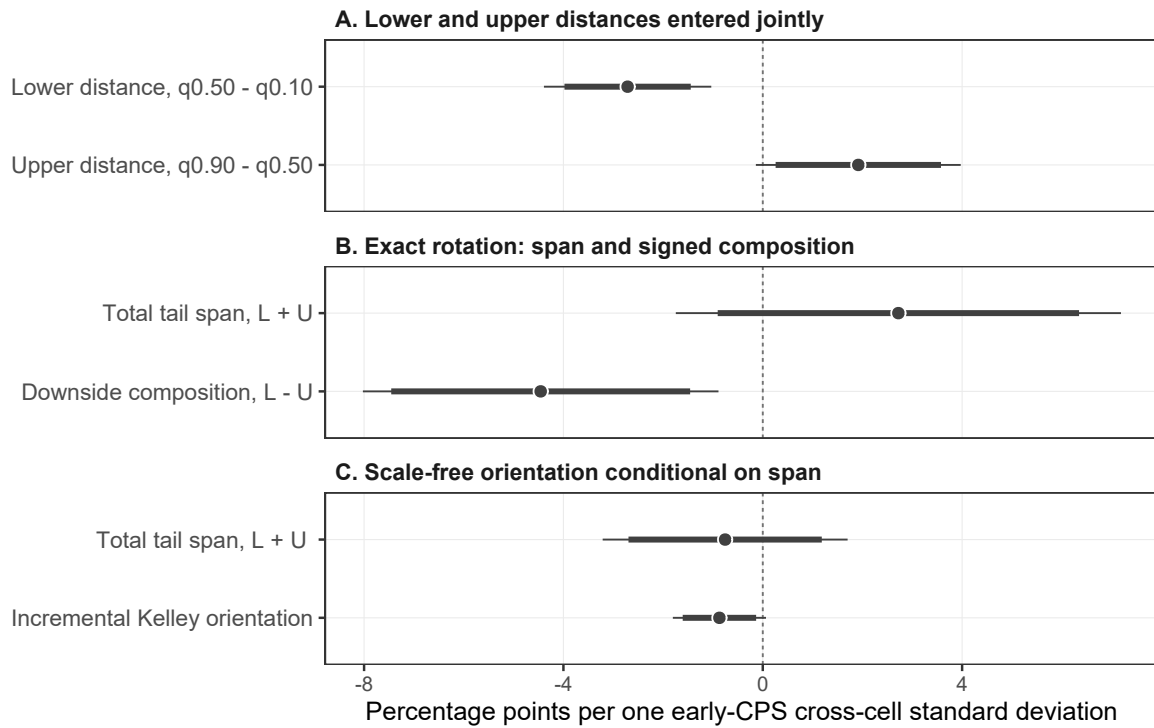
	(1) Lower and upper	(2) Span and composition	(3) Span and orientation
Lower distance, $L = q_{0.50} - q_{0.10}$	-2.712*** (0.857)		
Upper distance, $U = q_{0.90} - q_{0.50}$	1.917* (1.049)		
Total tail span, $S = L + U$		2.721 (2.279)	-0.755 (1.254)
Downside composition, $D = L - U$		-4.456** (1.820)	
Incremental Kelley orientation			-0.870* (0.477)
Diagnostic component order	L / U	S / D	S / Kelley
Conditional S.E.	0.646 / 0.846	1.850 / 1.530	0.989 / 0.376
CPS coefficient S.D.	0.562 / 0.619	1.331 / 0.986	0.770 / 0.293
Plug-in p	0.002 / 0.068	0.232 / 0.014	0.547 / 0.068
Joint two-component plug-in p	0.002	0.002	0.011
Lagged controls and fixed effects	Yes	Yes	Yes
Observations	27,752	27,752	27,752
People	19,300	19,300	19,300
CPS signal cells	60	60	60

Notes: The dependent variable is a transition from zero to positive reported stock-and-mutual-fund holdings; both annual observations are in 2019–2024. Coefficients are percentage points per one cross-cell standard deviation of the named early-CPS measure. Each column is a separate regression. All columns include lagged zero share, quadratic mean and dispersion, household controls, and year, lagged industry, education, and age-band fixed effects, with current SIPP weights. Parentheses report plug-in standard errors: the conditional variance, clustered by person and lagged CPS cell, plus the coefficient variance from 200 CPS resamples. This is a first-stage sensitivity with SIPP held fixed, not joint inference. Stars use two-sided standard-normal plug-in tests: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$. Paired diagnostic entries follow the stated component order. In a separate 99,999-draw null-imposed wild-cluster check, the lower and upper distances have two-component BH q -values of 0.019 and 0.087; signed composition has $p = 0.033$. Assignment is endogenous.

The standing-share results for Kelley orientation do not establish an intensive-margin response. Incremental Kelley is negatively associated with the all-worker and positive-financial-asset shares, but both samples include nonholders. The all-worker coefficient is -0.234 point, with conditional $q = 0.039$; after rebuilding the CPS first stage, its p -value rises to 0.109. Among current holders, the coefficient is -0.153 point with SE 0.424 and $q = 0.721$. Exit is also imprecise. Within this quantile construction, the evidence is strongest for acquisition from a zero position. These standing-outcome estimates concern a different earnings statistic and conditioning event from the prospective baseline-status comparison in Table 3.

Several checks show that the entry association is not confined to a fleeting or very small positive balance. Sustained entry has an incremental-Kelley coefficient of -0.697 point (SE 0.293; six-test $q = 0.041$). Crossings above \$1,000 and \$5,000 have coefficients of -1.104 and -0.453 points, with $q < 0.001$ and $q = 0.045$ in the same family. Requiring the same earnings-risk cell in both years yields -0.958 point, and omitting each outcome year in turn leaves five negative estimates between -1.203

Figure N.1. Tail composition and uncertainty about portfolio entry



Notes: Points report the coefficients in columns (1)–(3) of Table N.3, in percentage points per one early-CPS cross-cell standard deviation. Thick intervals are the estimate plus or minus 1.96 conditional two-way-clustered standard errors. Thin intervals use the corresponding CPS plug-in standard error, which adds the coefficient variance across 200 CPS resamples. These are normal-approximation sensitivity intervals, not joint-bootstrap confidence intervals. Components are standardized separately within each panel and enter jointly; the dashed line marks zero.

and -0.881 points. These tests concern incremental orientation; they are not separate confirmations that both individual tail coefficients are robust.

Household composition also matters because the estimation unit is a worker transition involving household-reported assets. The coefficient is -1.057 point in the stable-household sample and -0.869 point when one eligible worker, ranked by mean lagged job-1 earnings, is retained per household transition. Every leave-one-cell and leave-one-industry coefficient is negative. Appendix Tables C.6, C.9, and C.3 report the full families, alternative covariance estimates, and decision-unit checks. The broader portfolio-level window appears in Appendix Table E.1.

N.6 Uncertainty in the assigned earnings distributions

The following uncertainty calculations concern the original wage and quantile designs; they do not supply inference for the age-adjusted total-earnings or validated-linkage comparisons. Reconstructing the CPS distributions leaves incremental Kelley entry as the strongest plug-in result ($p = 0.0052$; three-outcome $q = 0.0157$). The lower and upper quantile distances are jointly informative, but the upper distance alone is imprecisely estimated ($p = 0.068$). Thus the evidence for the overall orientation contrast is stronger than that for an independent upper-tail association.

Jointly resampling both surveys exposes sensitivity that the plug-in calculation does not resolve. The remeasured coefficient distributions shift toward zero. Their 95% percentile intervals contain zero, whereas basic intervals exclude it for the lower tail, upper tail, and signed composition. The same disagreement appears for smooth variance components. Neither interval construction is treated as definitive, and a narrower spread after remeasurement is not evidence of greater precision.

For the original wage variance-composition benchmarks, local propagation of moment uncertainty gives family-adjusted $q = 0.015$ and $q = 0.049$ for zero-inclusive and positive-destination earnings. Replacing the SIPP component with the broader crossed-reweighting covariance raises these to 0.066 and 0.105. Appendix I.4 preserves the complete outcome family, remeasurement diagnostics, and covariance comparisons. The conditional associations therefore require a distinction between sampling precision and stability to reconstructing the assigned earnings environment.

N.7 The lower tail and zero earnings

The lower-tail coefficient depends substantially on the support boundary. In 11 of the 60 cells, more than ten percent of CPS destinations have zero wage earnings. The 10th percentile is then -2 , and the lower distance equals $2 + q_{0.50,c}$. Across cells, standardized lower distance correlates 0.89 with the zero share and 0.77 with an indicator that the quantile is at the bound. This makes the composition of zero wage earnings central to the disaster interpretation.

Appendix Table C.5 varies the zero-endpoint construction while retaining the baseline portfolio design. Excluding destinations classified as self-employed removes 6.2% of early transitions. Counting a zero only when no weeks were worked removes the 2.3% of transitions that report work without wage earnings. These changes reduce the cross-cell standard deviation of the lower distance from 0.40 to 0.25 and 0.23 arc-growth units. Its portfolio coefficient falls from -2.71 to -0.91 and -0.76 points, respectively, and neither alternative is distinguishable from zero. Restricting the baseline to the 49 cells with an interior 10th percentile yields -1.22 points (SE 1.35).

Signed composition is more stable than either a standalone lower-tail effect or normalized Kelley orientation across these constructions. Its coefficients are -1.73 , -1.77 , and -4.46 points in the three alternatives, with conditional p -values of 0.048, 0.023, and 0.020. The upper coefficient remains positive but is imprecise after excluding self-employed destinations ($p = 0.13$). Incremental Kelley is imprecise in two of the alternatives ($p = 0.16$ and $p = 0.080$). These comparisons preserve the direction of the signed-composition relation within the baseline assignment, but they do not propagate CPS first-stage uncertainty and do not establish two construction-invariant tail effects.

An additional business-asset comparison does not support an entrepreneurial-substitution interpretation. Among transitions starting with neither business equity nor stock-and-mutual-fund holdings, the lower tail predicts less stock-and-mutual-fund-only entry (-2.99 points; eight-coefficient plug-in $q = 0.0098$). The upper estimate has $q = 0.131$, and neither business-only entry nor the broader business-entry outcome survives the adjustment. These results also disappear under the self-employment-sensitive tail constructions. The distinction between a CPS self-employment classifi-

cation and a household's business ownership does not resolve the baseline lower-tail dependence on wage zeros.

A separate earnings-state exercise distinguishes zeros with no reported weeks of work from zeros despite some work. On a common 59-cell sample, allowing the two probabilities to enter separately leaves the unconditional lower-upper pair jointly informative under the CPS plug-in sensitivity (three-model $q = 0.019$). None of the twelve scalar coefficients survives the complete family adjustment. When quantiles are instead estimated only among positive earnings destinations, signed composition is essentially zero, but its incremental reliability is negative. That estimate is uninformative about the absence of an economically relevant positive-earnings contrast. Appendix Table I.5 reports the full family. The exercise rules out neither nonlinear zero-boundary mechanics nor endogenous sorting; it shows that a simple linear distinction between the two zero states does not absorb the baseline joint comparison.

N.8 Sorting, assignment, and predictive scope

Within-person comparisons do not deliver a stable alternative to the cross-cell association. Among 5,789 workers with repeated entry-risk observations, person fixed effects give lower and upper coefficients of -0.58 (SE 1.30) and 4.27 (SE 1.99) under lagged assignment. Assigning the current cell instead gives -0.05 (SE 0.96) and -1.17 (SE 0.75). Changes in assigned tails are themselves imprecise. The timing sensitivity of the upper estimate prevents these results from ruling out time-invariant sorting.

The grouping used to assign risk is consequential as well. Adding occupation to the industry-based grouping reduces the upper estimate to 0.32 and 0.55 point in 69- and 96-cell designs, with incremental reliability of 0.32 and 0.18. Replacing industry with occupation in an 18-cell design reverses it to -1.74 points (SE 0.98). The baseline upper-tail sign is not invariant to the assignment scheme.

A single pre-pandemic comparison is inconclusive. The 6,033 transitions from 2018 to 2019 give lower and upper coefficients of -0.34 and -0.46 point (SE 1.08 and 0.99). This window includes a baseline portfolio observation overlapping the final CPS destination year, and is kept outside the strictly later sample. Its estimates are too imprecise to corroborate or refute the baseline; pooling the windows gives -2.42 and 1.57 points. Split-sample results by liquid assets also do not establish heterogeneous coefficients: formal interaction tests give $p = 0.38$ for the lower tail and $p = 0.89$ for the upper tail. Appendix D reports these comparisons in full.

The decomposition has not demonstrated superior out-of-cell forecasting. Leaving one assignment cell out at a time, the separate tails provide no detectable improvement over total span ($p = 0.571$, $q = 0.695$). Nor do comparisons with entry into bank, bond, retirement, home-equity, and vehicle-equity positions establish stock-market specificity. A joint bond contrast has $q = 0.017$ in the all-worker sample but $q = 0.102$ after selecting one worker per household; neither individual tail contrast survives the full ten-coefficient family. These tests limit forecast and mechanism claims without changing the in-sample conditional comparison.

Within the baseline historical industry-based assignment, composition favoring the upper over the lower side of earnings growth is associated with greater subsequent entry. The quantile and centered-variance representations express this comparison using different features of the distribution. The age-adjusted real-log comparison additionally distinguishes subsequent allocation by initial ownership, with opposite slopes for nonholders and holders. Construction, assignment, and inference determine the scope of these patterns. The next section examines the features of the earnings distributions that make the assigned measures reproducible.

O Measurement Evidence for the Earlier Constructions

O.1 Reproducibility and the earnings support

A portfolio association is more informative when different workers generate a similar ranking of earnings environments. I assess this reproducibility by assigning every person globally to one of two halves and estimating the cell statistics separately. Global assignment keeps a worker in the same half after a change in industry or education. If the half-sample correlation is $r_{1/2}$, the full-sample reliability projection is the Spearman–Brown transform (Spearman, 1910; Brown, 1910):

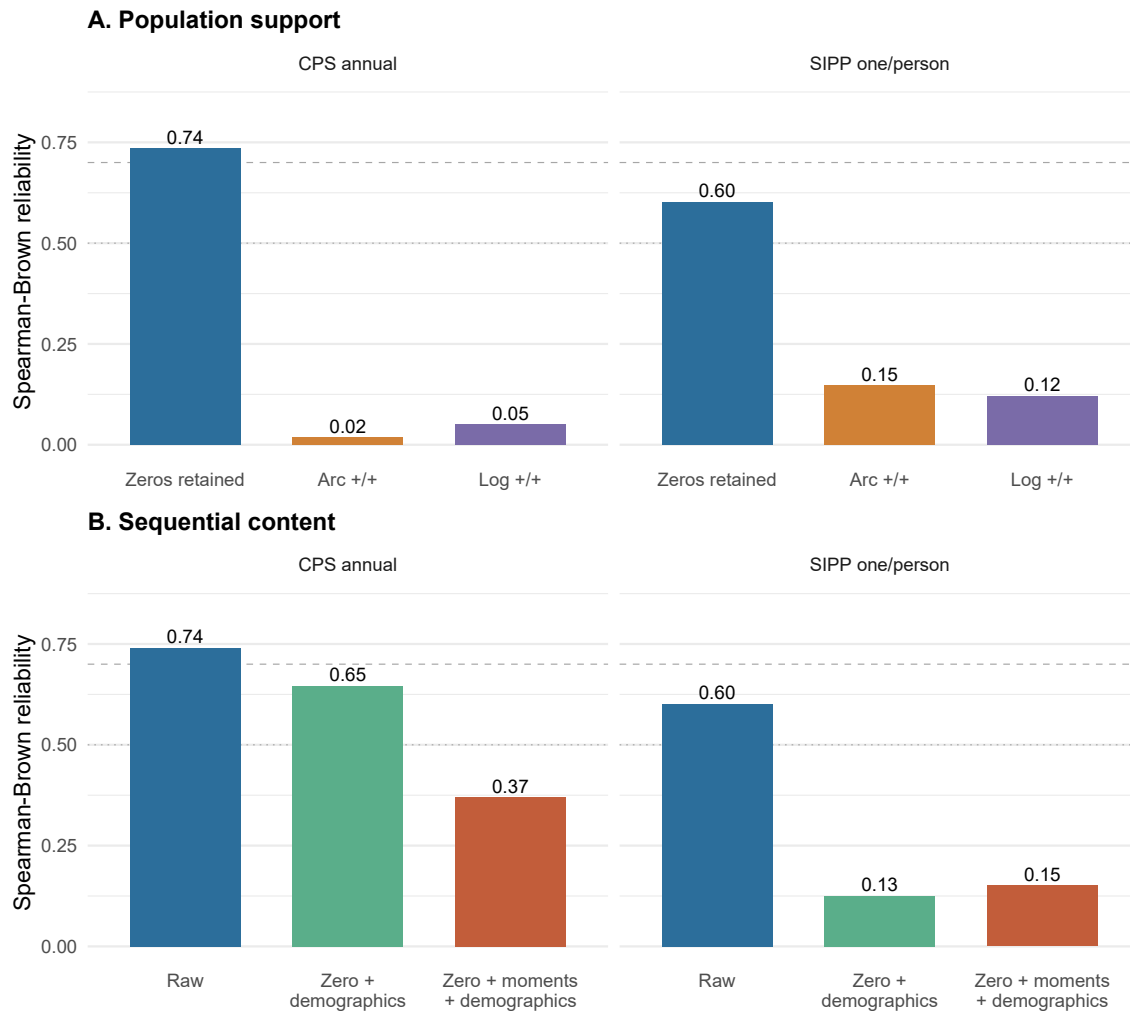
$$\rho_{SB} = \frac{2r_{1/2}}{1 + r_{1/2}}. \quad (\text{O.1})$$

I apply this transform to the mean correlation across 30 independent-person splits. Rank correlations and early-versus-late correlations provide separate measures of ordering and temporal stability. A high split reliability means that the statistic reproduces a cross-cell ranking; it does not reveal which economic component produces that ranking. Lemma 2 formalizes this distinction.

Retaining destination zeros changes the result sharply. In the companion CPS construction comparison, raw Kelley reliability is 0.018 for positive-to-positive arc growth and 0.735 when destination zeros are retained. The exact crosswalk used for the portfolio application gives 0.740. In the one-transition-per-person SIPP comparison, the corresponding arc-growth reliabilities are 0.147 and 0.602; positive-to-positive log growth has reliability 0.120. Figure O.1 shows that this ordering does not depend on treating overlapping SIPP months as independent evidence: zero-inclusive reliability is 0.622 with repeated months and 0.602 with one transition per person.

The difference is not simply a gain from a larger sample. A zero destination maps to the lower arc-growth boundary and changes the population distribution. Conditioning it away changes the question to earnings growth among workers with positive measured earnings at both endpoints. Independent worker samples reproduce the zero-inclusive downside ranking within each survey. The economic meaning of that ranking depends on what a zero records.

Figure O.1. Earnings support and the reliability of tail orientation



Notes: Panel A compares raw downside-oriented Kelley reliability when destination zeros are retained with positive-to-positive constructions. Panel B uses the pooled exact CPS crosswalk and matched SIPP. The second stage removes the linear zero share and demographic components; the third also removes quadratic mean and dispersion. Reliability is the Spearman–Brown transform of the mean independent-person half-sample correlation across 30 deterministic splits. Each projection is re-estimated inside each half.

O.2 The economic content of a recorded zero

Table O.1 separates nonwork from work without wage earnings. Among 11,828 CPS annual wage-zero destinations, 64.50% report no work, 35.50% report some work, and 26.02% report work in all 52 weeks. Each component is reproducible across independent worker samples. No-work incidence has reliability 0.840, while worked-without-wage incidence has reliability 0.940, the same as total wage-zero incidence. The raw lower-tail ranking is therefore not simply a ranking of full-year nonemployment.

The SIPP zero refers to a different earnings concept and time interval. Of 25,310 workers in the strict one-transition-per-person anatomy sample, 914 reach zero monthly job-1 earnings. Among

Table O.1. Work arrangements at the zero-earnings endpoint

Endpoint or component	Weighted share (%)	Zero rows	CPS incidence reliability
<i>Panel A. CPS annual wage-and-salary income</i>			
Destination annual wage zero	6.63	11,828	0.940
No work in destination year	64.50		0.840
Worked in destination year	35.50		0.940
Worked all 52 weeks	26.02		0.914
<i>Panel B. SIPP monthly job-1 earnings, one transition per person</i>			
Destination monthly job-1 zero	3.51	914	
Self-employed work arrangement	64.47		
Employer work arrangement	34.62		
Any no-job spell in destination month	2.06		

Overall zero rates condition on positive origin earnings; indented shares condition on a destination zero. Official person weights are used. CPS shares use 183,809 linked annual transitions in all 72 candidate cells; incidence reliabilities use 181,946 transitions in 65 supported cells. Reliability is the Spearman–Brown transform of the mean independent-person half-sample correlation across 30 splits. The strict SIPP anatomy sample selects one eligible 2018–2023 origin transition for each of 25,310 people without conditioning on destination state, giving outcomes in 2019–2024. Arrangement class and no-job indicators refer to the destination month and job 1. Arrangement classes are mutually exclusive, but a no-job spell can coexist with a work arrangement during the month.

these zeros, 64.47% have a self-employment arrangement and 34.62% an employer arrangement in that month. A zero in the first group can represent zero business earnings rather than lost wages. Only 19 zeros, a weighted 2.06%, coincide with any recorded no-job spell. The corresponding weighted share is 0.99% when all eligible overlapping transitions are retained. A work arrangement and a no-job spell can occur within the same month, so the latter is not a mutually exclusive class.

An out-of-time comparison asks whether the CPS ranking also predicts subsequent SIPP job-1 zeros. Across 30 choices of one eligible transition per person, raw Kelley gives a positive coefficient in 29 cases, with a median of 0.297 percentage point. Incremental Kelley gives a positive coefficient in 26 cases, with a median of 0.154 point. No selection passes the four-outcome $q < 0.05$ threshold. Appendix Table B.5 reports the ranges and complete families. The signs are compatible with some shared information about earnings support, but the exercise does not establish a reliable cross-survey prediction.

O.3 Incremental orientation and persistence

Table O.2 separates the stable components of Kelley orientation. In pooled CPS, removing the zero share and demographic components reduces reliability from 0.740 to 0.646; adding quadratic mean and dispersion reduces it to 0.370. The exact early-CPS measure assigned to portfolios has raw reliability of 0.687 and incremental reliability of 0.397. The matched SIPP sequence is 0.602, 0.125, and 0.151. The highly reproducible zero incidence—0.940 in pooled CPS and 0.809 in matched SIPP—is one component of a broader downside environment, rather than an estimate of shape alone.

Time stability is weaker than within-sample reproducibility. The CPS early-versus-late correlation falls from 0.631 for raw Kelley to 0.230 for incremental Kelley; in matched SIPP, it falls from 0.414

Table O.2. Reliability and persistence of earnings-tail orientation

Design	Supported records	Cells	Raw	Zero adjusted	Moment adjusted	Zero share
<i>Panel A. Independent-person reliability</i>						
CPS, pooled	181,946	65	0.740	0.646	0.370	0.940
CPS, early assignment	106,404	60	0.687	0.564	0.397	0.909
SIPP, one per person	50,137	58	0.602	0.125	0.151	0.809
SIPP, repeated months	953,809	60	0.622	0.178	0.070	0.831
<i>Panel B. Early-versus-late Pearson correlation</i>						
CPS, pooled			0.631	0.591	0.230	0.921
SIPP, one per person			0.414	0.153	−0.100	0.691
SIPP, repeated months			0.279	0.066	0.054	0.614

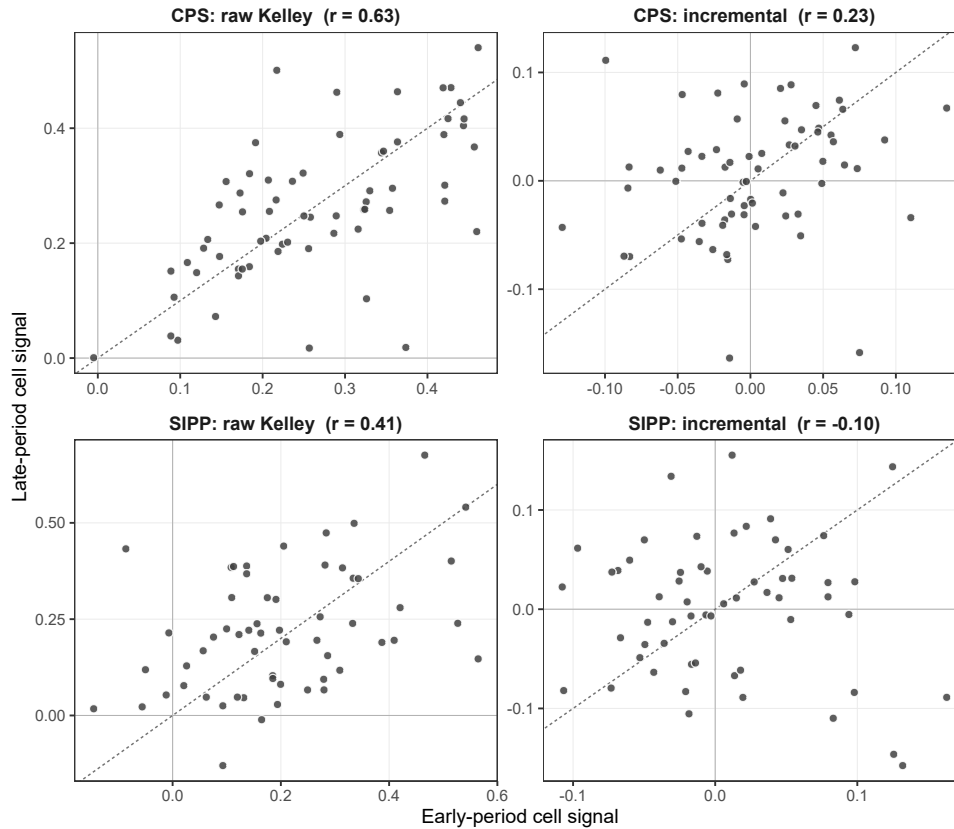
Panel A applies the Spearman–Brown transform to the mean global-person split correlation. Zero-adjusted Kelley removes the linear zero-transition share and education and age components; moment-adjusted Kelley also removes quadratic mean and standard deviation, as in equation (M.5). The zero-share column reports the corresponding statistic for zero incidence itself. Records and cells are post-support counts. Candidate samples contain 183,809 pooled CPS transitions, 108,707 early CPS transitions, 51,152 matched SIPP transitions, and 967,563 overlapping SIPP transitions. Pooled CPS requires Kish support of 200 overall and 100 in each period; the early row requires 200 in the early period. SIPP requires effective-person support of 100 overall and 50 in each period. The one-per-person transition is selected before any destination-positive restriction.

to −0.100. The corresponding rank correlations are 0.634 and 0.326 in CPS and 0.396 and −0.025 in SIPP. Figure O.2 makes the loss of linear persistence visible. Repeated monthly observations do not produce a more stable incremental ranking.

Direct cross-survey transport is weaker still for the incremental component. Over common years, the pooled CPS–SIPP correlations are 0.482 for raw Kelley and 0.053 for incremental Kelley. Alternative splits, one-transition selections, and annual job-1 constructions are reported in Appendix Table B.4. In the broader 2018–2024 portfolio comparison, alternative centered estimators—Bowley, moment skewness, and L-skewness—do not provide a stronger adjusted result; the smallest family q -value is 0.244.

These results support a distinction between measuring a survey-specific earnings environment and isolating a portable shape component. Raw downside orientation contains stable information, much of it shared with zero incidence and lower moments. Incremental orientation is less reliable and less persistent. This does not make the removed variation spurious or imply that latent shape is unimportant. It limits how precisely an assigned public-data measure can represent that latent object, and prevents reliability alone from licensing a structural interpretation of the portfolio coefficient.

Figure O.2. Persistence of raw and incremental tail orientation



Notes: Each point is one supported cell. The horizontal axis reports the early-period signal and the vertical axis the late-period signal; the dashed line marks equality. Panel titles report Pearson correlations. Raw Kelley uses the zero-inclusive construction. Incremental Kelley is the residual in equation (M.5), re-estimated within each period. CPS uses the exact crosswalk with 65 cells; SIPP uses one transition per person and 58 cells.

P Summary of the Earlier Validated-Link Comparison

This section records the interpretation of the earlier validated-link, zero-inclusive and rank-adjusted comparisons. Their original inference conventions remain visible; the main conclusions using age-adjusted earnings are in Section 6.

At a given total earnings variance, distributions tilted toward favorable rather than adverse outcomes are associated with greater subsequent stock-and-mutual-fund entry in the historical CPS-to-SIPP assignment. In the validated total-earnings comparison, a one-standard-deviation difference in composition remaining after the other regressors corresponds to 0.70 percentage point more entry, or 0.66 when both wage and total earnings destinations remain positive. More upside and less downside describe the same comparison, not independent tail effects.

The positive pattern also extends from wages to a matched measure including net business and farm earnings and persists under explicit treatment of the 2014 survey design. This broadens the earnings construct behind the household-finance comparison. It does not establish a wage-total difference or extend the comparison to negative total-earnings destinations.

In the original-linkage benchmarks, support from local CPS-plus-two-way-SIPP inference is sensitive to broader person-by-cell reweighting. The validated-link reconstruction uses a separate household-and-lineage dependence design. Its raw slopes survive family-adjusted original-point tests, but all 95% percentile intervals include zero, whereas basic intervals exclude it. Wage-zero construction, assignment, and sorting further limit interpretation. The contribution is therefore a conditional within-variance comparison with measured uncertainty, not identified tail effects or a precise life-cycle calibration target.

In the richer rank-based progression-adjusted entry comparisons, removing observable earnings progression reduces the point magnitudes, and none of the adjusted slopes survives its joint-remeasurement family adjustment. The raw-adjusted differences are imprecise, including when the full comparison is rebuilt using validated longitudinal earnings links. The evidence also fails to establish an incremental composition association beyond conventional skewness. These findings limit the claim to historical earnings variation; they do not identify a portfolio response to unanticipated risk.

Conditioning on workers' own recent annual earnings leaves the raw point estimates almost unchanged on the smaller matched-history sample. Neither the baseline nor the history-controlled coefficients survive their full testing families there, and paired differences remain imprecise. This comparison addresses a limited observed-history objection; it does not establish equivalence, eliminate sorting, or measure perceived risk.

Earlier headline units. For the earlier validated-link comparison, the total-earnings entry magnitude was 0.70 percentage point per conditional standard deviation, or 0.66 with positive earnings destinations. These figures refer to the earlier raw arc-growth construction and its own residual scaling, not the age-adjusted real-log coefficients that lead the current paper. The accompanying rank-adjusted, linkage, and remeasurement comparisons are retained above and in Appendix I.